31/7 equals

**A.** 4 + 1/7

**B.** 4 + 3/7

**C.** 3 + 4/7

**D.** 4 + 3/31

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q4

Grade 4 • Number & Operations–Fractions • 4.NF.B.3 • Set 22 • BACKWARD 1 • Q4/5

801

**QUESTION ID**

## S22-B1-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. 4 + 3/7**

**GRADING NOTE** $31 = 7 \cdot 4 + 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q4

Grade 4 • Number & Operations–Fractions • 4.NF.B.3 • Set 22 • BACKWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Write 100/9 in the form q + r/9.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q5

Grade 4 • Number & Operations–Fractions • 4.NF.B.3 • Set 22 • BACKWARD 1 • Q5/5

802

QUESTION ID

**S22-B1-Q5**

TYPE

Short answer

CORRECT ANSWER

**11 + 1/9**

GRADING NOTE Form required: q + r/9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q5

Grade 4 • Number & Operations–Fractions • 4.NF.B.3 • Set 22 • BACKWARD 1 • Q5/5

SET 22 • HSA-APR.D.6

# BACKWARD 2

Multiply to check a division • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Compute $(x - 1)(x^2 + 3x - 2) + 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 22 • BACKWARD 2 • Q1/5

804

**QUESTION ID**

## S22-B2-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**$x^3 + 2x^2 - 5x + 3$**

**GRADING NOTE** Any equivalent form.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S22-B2-Q1**

**High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 22 • BACKWARD 2 • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Compute $(x^2 + 1)(2x - 3) + (2x - 4)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 22 • BACKWARD 2 • Q2/5

805

QUESTION ID

**S22-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**$2x^3 - 3x^2 + 4x - 7$**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 22 • BACKWARD 2 • Q2/5

**True or false?**

$(2x + 1)(3x - 1) - 3 = 6x^2 + x - 4.$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 22 • BACKWARD 2 • Q3/5

806

QUESTION ID

## S22-B2-Q3

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** $6x^2 + x - 1 - 3.$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 22 • BACKWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

$(x + 3)(x^2 - 3x + 5)$ equals

**A.** $x^3 - 4x + 15$

**B.** $x^3 + 15$

**C.** $x^3 - 4x + 5$

**D.** $x^3 + 6x^2 - 4x + 15$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 22 • BACKWARD 2 • Q4/5

807

**QUESTION ID**

## S22-B2-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. $x^3 - 4x + 15$**

**GRADING NOTE** $x^3 - 3x^2 + 5x + 3x^2 - 9x + 15$.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S22-B2-Q4**

**High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 22 • BACKWARD 2 • Q4/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Compute $(x - 2)(x^3 + 2x^2 + 4x + 8)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 22 • BACKWARD 2 • Q5/5

808

QUESTION ID

**S22-B2-Q5**

TYPE

Short answer

CORRECT ANSWER

**$x^4 - 16$**

**GRADING NOTE** Any equivalent form.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S22-B2-Q5**

**High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 22 • BACKWARD 2 • Q5/5**

SET 22 • HSA-APR.D.6

# BACKWARD 3

Use the Remainder Theorem to check the remainder • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the remainder of $x^3 + 2x^2 - 5x + 3$ divided by $x - 1$ by evaluating.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 22 • BACKWARD 3 • Q1/5

810

**QUESTION ID**
**S22-B3-Q1**

**TYPE**
Short answer

**CORRECT ANSWER**
**1**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 22 • BACKWARD 3 • Q1/5

Find the remainder of $x^3 - 4x + 5$ divided by $x + 3$ by evaluating.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 22 • BACKWARD 3 • Q2/5

811

**QUESTION ID**

**S22-B3-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**−10**

**GRADING NOTE** Any equivalent form.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S22-B3-Q2**

**High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 22 • BACKWARD 3 • Q2/5**

**True or false?**

The remainder of $x^4 - 16$ divided by $x - 2$ is 16.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 22 • BACKWARD 3 • Q3/5

812

QUESTION ID
**S22-B3-Q3**

TYPE
True or false

CORRECT ANSWER
**B. False**

**GRADING NOTE** $2^4 - 16 = 0$, so the remainder is 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 22 • BACKWARD 3 • Q3/5

The remainder of $6x^2 + x - 4$ divided by $2x + 1$ is

**A.** 3

**B.** $-4$

**C.** 1

**D.** $-3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 22 • BACKWARD 3 • Q4/5



QUESTION ID

**S22-B3-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**D. −3**

**GRADING NOTE** Evaluate at $x = -1/2$: $1.5 - 0.5 - 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 22 • BACKWARD 3 • Q4/5

Find the remainder of $2x^3 + x - 1$ divided by $x - 2$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 22 • BACKWARD 3 • Q5/5

814

**QUESTION ID**

**S22-B3-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**17**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B3-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 22 • BACKWARD 3 • Q5/5

SET 22 • HSA-APR.D.6

# FORWARD 1

Divide with unknown coefficients and higher-degree divisors • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6

When $x^3 + ax^2 + 2x + 4$ is divided by $x^2 - x + 1$, the remainder is a constant. Find a and the remainder.

a = _____

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • FORWARD 1 • Q1/5

816

**QUESTION ID**

**S22-F1-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**−2; 5**

**GRADING NOTE** Form required: both. Quotient x + (a + 1); remainder (2 + a)x + (3 − a), constant when a = −2, giving 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • FORWARD 1 • Q1/5

Divide $x^4 + x^2 + 1$ by $x^2 + x + 1$. Give the quotient q(x) and the remainder.

q(x) = _____

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • FORWARD 1 • Q2/5

817

**QUESTION ID**

## S22-F1-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**$x^2 - x + 1$; 0**

**GRADING NOTE** Form required: both. $(x^2 + x + 1)(x^2 - x + 1) = x^4 + x^2 + 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • FORWARD 1 • Q2/5

**True or false?**

Dividing a degree-5 polynomial by a degree-2 polynomial gives a quotient of degree 2.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • FORWARD 1 • Q3/5

818

QUESTION ID

## S22-F1-Q3

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** The quotient has degree $5 - 2 = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • FORWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

The remainder of a division by $x^2 + 4$ can have degree at most

**A.** 0

**B.** 1

**C.** 2

**D.** 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • FORWARD 1 • Q4/5

819

QUESTION ID

**S22-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**B. 1**

GRADING NOTE Lower than the divisor.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • FORWARD 1 • Q4/5

Find the polynomial a(x) for which
$a(x)/(x - 2) = x^2 + 1 + 3/(x - 2)$.

a(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • FORWARD 1 • Q5/5

820

**QUESTION ID**
## S22-F1-Q5

**TYPE**
Short answer

**CORRECT ANSWER**
**$x^3 - 2x^2 + x + 1$**

**GRADING NOTE** Any equivalent form. $(x - 2)(x^2 + 1) + 3$.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S22-F1-Q5**

**High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • FORWARD 1 • Q5/5**

SET 22 • HSA-APR.D.6

# FORWARD 2

Use division to find slant asymptotes • High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d

Use division to find the slant asymptote of $f(x) = (x^2 + 3x + 1)/(x + 1)$.

Asymptote: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 22 • FORWARD 2 • Q1/5

822

QUESTION ID

**S22-F2-Q1**

TYPE

Short answer

CORRECT ANSWER

**$y = x + 2$**

**GRADING NOTE** Form required: an equation. $(x^2 + 3x + 1)/(x + 1) = x + 2 - 1/(x + 1)$.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S22-F2-Q1**

**High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 22 • FORWARD 2 • Q1/5**

Find the slant asymptote of $f(x) = (2x^2 - x - 4)/(x - 1)$.

Asymptote: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 22 • FORWARD 2 • Q2/5

823

**QUESTION ID**

## S22-F2-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**y = 2x + 1**

**GRADING NOTE** Form required: an equation. Remainder −3.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S22-F2-Q2**

**High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 22 • FORWARD 2 • Q2/5**

**True or false?**

A rational function with numerator degree exactly one more than the denominator degree has a slant asymptote.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 22 • FORWARD 2 • Q3/5

824

QUESTION ID
**S22-F2-Q3**

TYPE
True or false

CORRECT ANSWER
**A. True**

**GRADING NOTE** The quotient is linear (unless the division is exact, when the graph is the line with a hole).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 22 • FORWARD 2 • Q3/5

The slant asymptote of $y = (x^2 + 1)/x$ is

**A.** $y = x$

**B.** $y = 1$

**C.** $x = 0$

**D.** $y = x + 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 22 • FORWARD 2 • Q4/5

825

QUESTION ID
**S22-F2-Q4**

TYPE
Multiple choice

CORRECT ANSWER
**A. y = x**

GRADING NOTE $x + 1/x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 22 • FORWARD 2 • Q4/5

Why does the graph of $y = x + 2 - 1/(x + 1)$ get close to the line $y = x + 2$ for large $|x|$?

**A.** $1/(x + 1)$ approaches 0 as $|x|$ grows, so the vertical distance $y - (x + 2) = -1/(x + 1)$ approaches 0.

**B.** $1/(x + 1)$ grows as $|x|$ grows, so the graph gets pushed toward the line $y = x + 2$ for large $|x|$.

**C.** The graph and the line $y = x + 2$ meet at $x = -1$, so they stay close for large $|x|$.

**D.** At $x = 0$, $y = x + 2$ exactly.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 22 • FORWARD 2 • Q5/5

826

QUESTION ID

**S22-F2-Q5**

TYPE

Multiple choice

CORRECT ANSWER

**A. $1/(x + 1)$ approaches 0 as $|x|$ grows, so the vertical distance $y - (x + 2) = -1/(x + 1)$ approaches 0.**

**GRADING NOTE** The difference is $-1/(x + 1)$, which tends to 0. The others use one point ($y(0) = 1 \neq 2$), a false claim, or the excluded value $x = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 22 • FORWARD 2 • Q5/5

# Rewrite simple rational expressions by inspection

| | | |
|---|---|---|
| **MAIN** | Algebra II main question | **HSA-APR.D.6** |
| **BACKWARD 1** | Split a fraction with a sum in the numerator | **7.EE.A.1** |
| **BACKWARD 2** | Rewrite a numerator as a multiple of the denominator plus a cons… | **6.EE.A.3** |
| **BACKWARD 3** | Divide by long division when inspection fails | **HSA-APR.D.6** |
| **FORWARD 1** | Use the rewritten form to reason about values | **HSA-APR.D.6** |
| **FORWARD 2** | Graph rational functions as transformations of y = 1/x | **HSF-BF.B.3** |

SET 23 • HSA-APR.D.6

# MAIN

Algebra II main question • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6

Rewrite (x + 5)/(x + 2) by inspection in the form q + r/(x + 2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • MAIN • Q1/5

829

**QUESTION ID**
**S23-M-Q1**

**TYPE**
Short answer

**CORRECT ANSWER**
**1 + 3/(x + 2)**

**GRADING NOTE** Form required: q + r/(x + 2). x + 5 = (x + 2) + 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Rewrite $(2x - 1)/(x - 3)$ by inspection in the form $q + r/(x - 3)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • MAIN • Q2/5

830

**QUESTION ID**

**S23-M-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**$2 + 5/(x - 3)$**

**GRADING NOTE** Form required. $2x - 1 = 2(x - 3) + 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • MAIN • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Rewrite $(x^2 + 1)/x$ by inspection in the form $q(x) + r/x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • MAIN • Q3/5

831

QUESTION ID
**S23-M-Q3**

TYPE
Short answer

CORRECT ANSWER
**x + 1/x**

GRADING NOTE Form required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • MAIN • Q3/5

Rewrite (3x + 7)/(x + 1) by inspection in the form q + r/(x + 1).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • MAIN • Q4/5

832

**QUESTION ID**

## S23-M-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**3 + 4/(x + 1)**

**GRADING NOTE** Form required. 3x + 7 = 3(x + 1) + 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • MAIN • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Rewrite $x^2/(x - 1)$ by inspection in the form $q(x) + r/(x - 1)$. (Hint: $x^2 = (x^2 - 1) + 1$.)

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • MAIN • Q5/5

833

**QUESTION ID**

## S23-M-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**$x + 1 + 1/(x - 1)$**

**GRADING NOTE** Form required.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S23-M-Q5**

**High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • MAIN • Q5/5**

SET 23 • HSA-APR.D.6

# BACKWARD 1

Split a fraction with a sum in the numerator • Grade 7 • Expressions & Equations • 7.EE.A.1

Which writes (x + 3)/x as a sum of a whole number and a fraction?

**A.** 1/x + 3/x

**B.** x + 3/x

**C.** 1 + 3/x

**D.** 1 + 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • BACKWARD 1 • Q1/5

835

QUESTION ID

**S23-B1-Q1**

TYPE

Multiple choice

CORRECT ANSWER

**C. 1 + 3/x**

**GRADING NOTE** x/x + 3/x = 1 + 3/x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Simplify (2(x − 3) + 5)/(x − 3) as a sum.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • BACKWARD 1 • Q2/5

836

QUESTION ID

**S23-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**2 + 5/(x − 3)**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • BACKWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

$(a + b)/c = a/c + b/c.$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • BACKWARD 1 • Q3/5

837

QUESTION ID

## S23-B1-Q3

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Division distributes over a sum in the numerator.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • BACKWARD 1 • Q3/5

$(x^2 + 1)/x$ equals

**A.** $x + 1/x$

**B.** $1 + 1/x$

**C.** $x + 1$

**D.** $x^2 + 1/x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • BACKWARD 1 • Q4/5

838

QUESTION ID

**S23-B1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**A. $x + 1/x$**

**GRADING NOTE** $x^2/x + 1/x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • BACKWARD 1 • Q4/5

**True or false?**

c/(a + b) = c/a + c/b.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • BACKWARD 1 • Q5/5

839

QUESTION ID

**S23-B1-Q5**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Splitting works only for the numerator.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • BACKWARD 1 • Q5/5

SET 23 • HSA-APR.D.6

# BACKWARD 2

Rewrite a numerator as a multiple of the denominator plus a constant • Grade 6 • Expressions & Equations • 6.EE.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

Find c so that x + 5 = (x + 2) + c.

c = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 2 • Q1/5

841

QUESTION ID

**S23-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**3**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Find c so that $2x - 1 = 2(x - 3) + c$.

c = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 2 • Q2/5

842

QUESTION ID

**S23-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**5**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 2 • Q2/5

**True or false?**

$3x + 7 = 3(x + 1) + 7.$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 2 • Q3/5

843

QUESTION ID

**S23-B2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE $3(x + 1) + 7 = 3x + 10$; the constant is 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 2 • Q3/5

$x^2 - 1$ equals

**A.** $x(x - 1)$

**B.** $(x + 1)^2$

**C.** $(x - 1)^2$

**D.** $(x - 1)(x + 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 2 • Q4/5

844

QUESTION ID

**S23-B2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**D. $(x - 1)(x + 1)$**

**GRADING NOTE** Difference of squares.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Find c so that $4x + 1 = 4(x - 2) + c$.

c = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 2 • Q5/5

845

**QUESTION ID**

**S23-B2-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**9**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 2 • Q5/5

SET 23 • HSA-APR.D.6

# BACKWARD 3

Divide by long division when inspection fails • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6

Divide $x^2$ by $x - 1$ with long division. Give the quotient and the remainder.

Quotient: _____

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • BACKWARD 3 • Q1/5

847

QUESTION ID

**S23-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**x + 1; 1**

**GRADING NOTE** Form required: both. $x^2 = (x - 1)(x + 1) + 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • BACKWARD 3 • Q1/5

Divide 3x + 7 by x + 1. Give the quotient and the remainder.

Quotient: _____

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • BACKWARD 3 • Q2/5

848

**QUESTION ID**

**S23-B3-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**3; 4**

**GRADING NOTE** Form required: both. 3x + 7 = 3(x + 1) + 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • BACKWARD 3 • Q2/5

**True or false?**

Long division and inspection give the same result.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • BACKWARD 3 • Q3/5

849

QUESTION ID

## S23-B3-Q3

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** The form q + r/b is unique.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • BACKWARD 3 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

The remainder of $(2x - 1) \div (x - 3)$ is

**A.** $-5$

**B.** 2

**C.** $-1$

**D.** 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • BACKWARD 3 • Q4/5

850

**QUESTION ID**

**S23-B3-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. 5**

**GRADING NOTE** $2 \cdot 3 - 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • BACKWARD 3 • Q4/5

Divide x + 5 by x + 2. Give the quotient and the remainder.

Quotient: _____

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B3-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • BACKWARD 3 • Q5/5

851

**QUESTION ID**

**S23-B3-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**1; 3**

**GRADING NOTE** Form required: both. x + 5 = 1(x + 2) + 3.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S23-B3-Q5**

**High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • BACKWARD 3 • Q5/5**

SET 23 • HSA-APR.D.6

# FORWARD 1

Use the rewritten form to reason about values • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6

**ANSWER KEY** Reduced copy of the question page, with its answer

For which integers x is (x + 5)/(x + 2) an integer? (Use (x + 5)/(x + 2) = 1 + 3/(x + 2).)

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • FORWARD 1 • Q1/5

853

QUESTION ID

**S23-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**−5 or −3 or −1 or 1**

**GRADING NOTE** Form required: all four, in any order. x + 2 must divide 3: x + 2 = ±1 or ±3.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S23-F1-Q1**

**High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • FORWARD 1 • Q1/5**

Why can (2x − 1)/(x − 3) never equal 2?

**A.** It is undefined for every value of x.

**B.** Its numerator 2x − 1 is never 2.

**C.** At x = 0 it equals 1/3, which is not 2.

**D.** It equals 2 + 5/(x − 3), and 5/(x − 3) is never 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • FORWARD 1 • Q2/5

854

**QUESTION ID**

**S23-F1-Q2**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. It equals 2 + 5/(x − 3), and 5/(x − 3) is never 0.**

**GRADING NOTE** Rewrite by inspection: 2x − 1 = 2(x − 3) + 5. The others check one value, give a false reason, or a false fact (it is undefined only at x = 3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • FORWARD 1 • Q2/5

**True or false?**

$(3x + 7)/(x + 1) > 3$ for every $x > -1$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • FORWARD 1 • Q3/5

855

QUESTION ID

## S23-F1-Q3

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** $4/(x + 1) > 0$ when $x > -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • FORWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

As x grows large, (x + 5)/(x + 2) approaches

**A.** 3

**B.** 0

**C.** 5/2

**D.** 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • FORWARD 1 • Q4/5

856

**QUESTION ID**

## S23-F1-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. 1**

**GRADING NOTE** 3/(x + 2) → 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • FORWARD 1 • Q4/5

For $x > 0$, what is the least value of $x + 1/x$, and at which $x$ does it occur? (Use $x + 1/x - 2 = (x - 1)^2/x$.)

Least value: _____

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • FORWARD 1 • Q5/5

857

**QUESTION ID**

**S23-F1-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**2; 1**

**GRADING NOTE** Form required: both. $(x - 1)^2/x \geq 0$ for $x > 0$, with equality only at $x = 1$.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S23-F1-Q5**

**High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • FORWARD 1 • Q5/5**

SET 23 • HSA-APR.D.6

# FORWARD 2

Graph rational functions as transformations of y = 1/x • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

**ANSWER KEY** Reduced copy of the question page, with its answer

How is the graph of $y = 1 + 3/(x + 2)$ obtained from the graph of $y = 3/x$?

**A.** Shift it left 2 and down 1.

**B.** Shift it left 2 and up 1.

**C.** Shift it right 2 and up 1.

**D.** Shift it up 2 and left 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 23 • FORWARD 2 • Q1/5

859

QUESTION ID
**S23-F2-Q1**

TYPE
Multiple choice

CORRECT ANSWER
**B. Shift it left 2 and up 1.**

**GRADING NOTE** $x + 2$ shifts left 2; $+ 1$ shifts up 1. The asymptotes move to $x = -2$ and $y = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 23 • FORWARD 2 • Q1/5

Give the vertical and horizontal asymptotes of $y = 2 + 5/(x - 3)$ as equations.

Vertical asymptote: _____

Horizontal asymptote: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 23 • FORWARD 2 • Q2/5

860

**QUESTION ID**

**S23-F2-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**x = 3; y = 2**

**GRADING NOTE** Form required: both equations. $5/(x - 3)$ is undefined at $x = 3$ and approaches 0 for large $|x|$.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S23-F2-Q2**

**High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 23 • FORWARD 2 • Q2/5**

**True or false?**

y = (3x + 7)/(x + 1) has horizontal asymptote y = 7.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 23 • FORWARD 2 • Q3/5

861

QUESTION ID

**S23-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** (3x + 7)/(x + 1) = 3 + 4/(x + 1), so the asymptote is y = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 23 • FORWARD 2 • Q3/5

$y = (x + 5)/(x + 2)$ is a translation of $y = 3/x$ by

**A.** left 2, up 1

**B.** right 2, up 1

**C.** left 2, up 5

**D.** left 5, up 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 23 • FORWARD 2 • Q4/5

862

**QUESTION ID**

**S23-F2-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. left 2, up 1**

**GRADING NOTE** $1 + 3/(x + 2)$.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S23-F2-Q4**

**High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 23 • FORWARD 2 • Q4/5**

Write (4x − 3)/(x − 1) in the form q + r/(x − 1) with constants q and r. Then give its vertical and horizontal asymptotes as equations.

q = _____

r = _____

Vertical asymptote: _____

Horizontal asymptote: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 23 • FORWARD 2 • Q5/5

863

**QUESTION ID**

**S23-F2-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**4; 1; x = 1; y = 4**

**GRADING NOTE** Form required: all four. 4x − 3 = 4(x − 1) + 1.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S23-F2-Q5**

**High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 23 • FORWARD 2 • Q5/5**

# Multiply and divide rational expressions

| | | |
|---|---|---|
| **MAIN** | Algebra II main question | **HSA-APR.D.7** |
| **BACKWARD 1** | Multiply and divide fractions | **6.NS.A.1** |
| **BACKWARD 2** | Factor numerators and denominators | **HSA-SSE.A.2** |
| **BACKWARD 3** | Division by zero is undefined | **7.NS.A.2.b** |
| **FORWARD 1** | Simplify complex fractions | **HSA-APR.D.7** |
| **FORWARD 2** | Explain closure of rational expressions | **HSA-APR.D.7** |

SET 24 • HSA-APR.D.7

# MAIN

Algebra II main question • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.7

Simplify $(x^2 - 9)/(x^2 + x - 6) \cdot (x - 2)/(x - 3)$. Give the result and all excluded values.

Result: _____

Excluded values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • MAIN • Q1/5

866

QUESTION ID
**S24-M-Q1**

TYPE
Short answer

CORRECT ANSWER
**1; {−3, 2, 3}**

**GRADING NOTE** Form required: both. $(x - 3)(x + 3)/((x + 3)(x - 2)) \cdot (x - 2)/(x - 3) = 1$; the original denominators are zero at −3, 2, and 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • MAIN • Q1/5

Simplify $(x + 2)/(3x^2) \div (x^2 - 4)/x$. Write the result as one fraction with its denominator multiplied out, and give all excluded values.

Result: _____

Excluded values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q2

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • MAIN • Q2/5

867

QUESTION ID
**S24-M-Q2**

TYPE
Short answer

CORRECT ANSWER
**$1/(3x^2 - 6x)$; {−2, 0, 2}**

**GRADING NOTE** Form required: both. $(x + 2)/(3x^2) \cdot x/((x - 2)(x + 2)) = 1/(3x(x - 2)) = 1/(3x^2 - 6x)$. Exclusions: 0 (denominators), and −2 and 2 (zeros of the divisor $x^2 - 4$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q2

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • MAIN • Q2/5

Simplify $(2x - 6)/(x + 1) \cdot (x + 1)/(x^2 - 9)$. Give the result and all excluded values.

Result: _____

Excluded values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • MAIN • Q3/5

868

**QUESTION ID**
**S24-M-Q3**

**TYPE**
Short answer

**CORRECT ANSWER**
**2/(x + 3); {−3, −1, 3}**

**GRADING NOTE** Form required: both. $2(x - 3)/(x + 1) \cdot (x + 1)/((x - 3)(x + 3))$.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S24-M-Q3**
**High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions**
**HSA-APR.D.7 • Set 24 • MAIN • Q3/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Simplify (3x + 3)/(x − 4) ÷ (x + 1)/(x − 4). Give the result and all excluded values.

Result: _____

Excluded values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q4

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • MAIN • Q4/5

869

QUESTION ID
**S24-M-Q4**

TYPE
Short answer

CORRECT ANSWER
**3; {−1, 4}**

**GRADING NOTE** Form required: both. 3(x + 1)/(x − 4) · (x − 4)/(x + 1) = 3; x = 4 (denominators) and x = −1 (zero of the divisor) are excluded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q4

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • MAIN • Q4/5

Simplify $(5x^2 + 5x)/(x^2 - 4) \div (x + 1)/(x - 2)$. Give the result and all excluded values.

Result: _____

Excluded values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • MAIN • Q5/5

870

**QUESTION ID**
**S24-M-Q5**

**TYPE**
Short answer

**CORRECT ANSWER**
**5x/(x + 2); {−2, −1, 2}**

**GRADING NOTE** Form required: both. $5x(x + 1)/((x - 2)(x + 2)) \cdot (x - 2)/(x + 1)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • MAIN • Q5/5

SET 24 • HSA-APR.D.7

# BACKWARD 1

Multiply and divide fractions • Grade 6 • The Number System • 6.NS.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

Compute 2/3 · 9/4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 24 • BACKWARD 1 • Q1/5

872

QUESTION ID

**S24-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**3/2**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 24 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Compute 5/6 ÷ 10/9.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 24 • BACKWARD 1 • Q2/5

873

QUESTION ID

**S24-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**3/4**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 24 • BACKWARD 1 • Q2/5

**True or false?**

$a/b \div c/d = a/b \cdot c/d$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 24 • BACKWARD 1 • Q3/5

874

QUESTION ID

**S24-B1-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Dividing multiplies by the reciprocal d/c.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 24 • BACKWARD 1 • Q3/5

3/8 ÷ 3/4 equals

**A.** 9/32

**B.** 2

**C.** 3/4

**D.** 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 24 • BACKWARD 1 • Q4/5

875

QUESTION ID

**S24-B1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**D. 1/2**

**GRADING NOTE** 3/8 · 4/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 24 • BACKWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Compute 14/15 · 5/7.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 24 • BACKWARD 1 • Q5/5

876

QUESTION ID

**S24-B1-Q5**

TYPE

Short answer

CORRECT ANSWER

**2/3**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 24 • BACKWARD 1 • Q5/5

SET 24 • HSA-APR.D.7

# BACKWARD 2

Factor numerators and denominators • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

Factor $x^2 + x - 6$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 24 • BACKWARD 2 • Q1/5

878

QUESTION ID

**S24-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**(x + 3)(x − 2)**

**GRADING NOTE** Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 24 • BACKWARD 2 • Q1/5

Factor $5x^2 + 5x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 24 • BACKWARD 2 • Q2/5

879

**QUESTION ID**

**S24-B2-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**5x(x + 1)**

**GRADING NOTE** Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 24 • BACKWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

$x^2 - 4$ equals

**A.** $(x - 4)(x + 1)$

**B.** $(x - 2)^2$

**C.** $(x - 2)(x + 2)$

**D.** $x(x - 4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 24 • BACKWARD 2 • Q3/5

880

QUESTION ID

**S24-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**C. (x − 2)(x + 2)**

GRADING NOTE Difference of squares.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 24 • BACKWARD 2 • Q3/5

**True or false?**

$2x - 6 = 2(x - 3)$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 24 • BACKWARD 2 • Q4/5

881

QUESTION ID

**S24-B2-Q4**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Common factor.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 24 • BACKWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Factor $x^2 - 9$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 24 • BACKWARD 2 • Q5/5

882

QUESTION ID

**S24-B2-Q5**

TYPE

Short answer

CORRECT ANSWER

**(x − 3)(x + 3)**

**GRADING NOTE** Form required: factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 24 • BACKWARD 2 • Q5/5

SET 24 • HSA-APR.D.7

# BACKWARD 3

Division by zero is undefined • Grade 7 • The Number System • 7.NS.A.2.b

**ANSWER KEY** Reduced copy of the question page, with its answer

For which x is 1/(x − 4) undefined?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 24 • BACKWARD 3 • Q1/5

884

QUESTION ID

**S24-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**4**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 24 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

For which x is $(x + 2)/(x^2 - 9)$ undefined?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 24 • BACKWARD 3 • Q2/5

885

QUESTION ID

**S24-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**3 or −3**

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 24 • BACKWARD 3 • Q2/5

**True or false?**

After canceling (x − 3), x = 3 is allowed in the original expression.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 24 • BACKWARD 3 • Q3/5

886

QUESTION ID

**S24-B3-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** The original expression is undefined at x = 3, so it stays excluded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 24 • BACKWARD 3 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Dividing by (x + 1)/(x − 2) is impossible when

**A.** x = −1 only

**B.** x = 0

**C.** x = −1 or x = 2

**D.** x = 2 only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 24 • BACKWARD 3 • Q4/5

887

**QUESTION ID**

**S24-B3-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. x = −1 or x = 2**

**GRADING NOTE** Divisor undefined or equal to 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 24 • BACKWARD 3 • Q4/5

Why is $x = 0$ excluded from $(x + 2)/(3x^2)$?

**A.** Every expression excludes $x = 0$.

**B.** The value of the expression is 0 when $x = 0$.

**C.** The denominator $3x^2$ equals 0 when $x = 0$.

**D.** The numerator $x + 2$ equals 0 when $x = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 24 • BACKWARD 3 • Q5/5

888

QUESTION ID

**S24-B3-Q5**

TYPE

Multiple choice

CORRECT ANSWER

**C. The denominator $3x^2$ equals 0 when $x = 0$.**

GRADING NOTE Division by 0 is undefined. The others claim every expression excludes 0, or look at where the value or the numerator is 0, which does not make the expression undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B3-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 24 • BACKWARD 3 • Q5/5

SET 24 • HSA-APR.D.7

# FORWARD 1

Simplify complex fractions • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.7

**ANSWER KEY** Reduced copy of the question page, with its answer

Which is (1/x + 1)/(1/x − 1) simplified, with all excluded values?

**A.** (1 + x)/(1 − x); x ≠ 0, 1

**B.** −1; x ≠ 0

**C.** (x + 1)/(x − 1); x ≠ 0, 1

**D.** (1 + x)/(1 − x); x ≠ 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q1

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 1 • Q1/5

890

QUESTION ID

**S24-F1-Q1**

TYPE

Multiple choice

CORRECT ANSWER

**A. (1 + x)/(1 − x); x ≠ 0, 1**

GRADING NOTE Multiply the top and bottom by x. x = 0 stays excluded (the original contains 1/x), and x = 1 makes the denominator zero. The others drop x = 0, cancel incorrectly, or flip a sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q1

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 1 • Q1/5

Which is $(1 - 4/x^2)/(1 + 2/x)$ simplified, with all excluded values?

**A.** $(x - 2)/x;\ x \neq 0, -2$

**B.** $x - 2;\ x \neq -2$

**C.** $(x + 2)/x;\ x \neq 0, -2$

**D.** $(x - 2)/x;\ x \neq 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 1 • Q2/5

891

QUESTION ID
**S24-F1-Q2**

TYPE
Multiple choice

CORRECT ANSWER
**A. $(x - 2)/x;\ x \neq 0, -2$**

**GRADING NOTE** Multiply the top and bottom by $x^2$: $(x^2 - 4)/(x^2 + 2x) = (x - 2)(x + 2)/(x(x + 2))$. $x = -2$ stays excluded. The others drop it, cancel the wrong factor, or lose the x in the denominator.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 1 • Q2/5

**True or false?**

$(a/b)/(c/d) = ad/(bc)$ when $b, c, d \neq 0$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 1 • Q3/5

892

QUESTION ID
**S24-F1-Q3**

TYPE
True or false

CORRECT ANSWER
**A. True**

**GRADING NOTE** Multiply by the reciprocal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

(x/2)/(x/4) for x ≠ 0 equals

**A.** 2

**B.** 1/2

**C.** $x^2/8$

**D.** $8/x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q4

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 1 • Q4/5

893

QUESTION ID

**S24-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**A. 2**

GRADING NOTE x/2 · 4/x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q4

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 1 • Q4/5

Which is $(x - 1/x)/(x + 1)$ simplified, with all excluded values?

**A.** $(x + 1)/x;\ x \neq 0, -1$

**B.** $(x - 1)/x;\ x \neq 0, -1$

**C.** $x - 1;\ x \neq -1$

**D.** $(x - 1)/x;\ x \neq 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 1 • Q5/5

894

QUESTION ID
**S24-F1-Q5**

TYPE
Multiple choice

CORRECT ANSWER
**B. $(x - 1)/x;\ x \neq 0, -1$**

**GRADING NOTE** $(x^2 - 1)/(x(x + 1)) = (x - 1)/x$, with $x = 0$ and $x = -1$ excluded from the original. The others drop an exclusion, cancel the wrong factor, or lose the denominator x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 1 • Q5/5

SET 24 • HSA-APR.D.7

# FORWARD 2

Explain closure of rational expressions • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.7

Why is the product of two rational expressions a rational expression?

**A.** Rational expressions are always polynomials, and a product of polynomials is a polynomial.

**B.** (a/b)(c/d) = (a + c)/(b + d), and a + c and b + d are polynomials, so the product is a ratio of polynomials.

**C.** The product of any two expressions is a polynomial.

**D.** (a/b)(c/d) = ac/(bd), and ac and bd are polynomials (bd is not the zero polynomial), so the product is a ratio of polynomials.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 2 • Q1/5

896

QUESTION ID
**S24-F2-Q1**

TYPE
Multiple choice

CORRECT ANSWER
**D. (a/b)(c/d) = ac/(bd), and ac and bd are polynomials (bd is not the zero polynomial), so the product is a ratio of polynomials.**

**GRADING NOTE** Polynomials are closed under multiplication. The others use an invalid rule or false statements (1/x is not a polynomial).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 2 • Q1/5

Why is the quotient of a rational expression by a nonzero rational expression a rational expression?

**A.** Any quotient is undefined, so the question does not apply.

**B.** (a/b) ÷ (c/d) = (a ÷ c)/(b ÷ d), and quotients of polynomials are polynomials.

**C.** (a/b) ÷ (c/d) = ad/(bc), a ratio of polynomials, and bc is not the zero polynomial because c is not.

**D.** Dividing never changes an expression.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 2 • Q2/5

897

**QUESTION ID**
**S24-F2-Q2**

**TYPE**
Multiple choice

**CORRECT ANSWER**
**C.**
**(a/b) ÷ (c/d) = ad/(bc), a ratio of polynomials, and bc is not the zero polynomial because c is not.**

**GRADING NOTE** Multiply by the reciprocal. The others claim polynomials are closed under division (they are not) or give non-reasons.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 2 • Q2/5

**True or false?**

Rational expressions are closed under division by the zero expression.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 2 • Q3/5

898

QUESTION ID
**S24-F2-Q3**

TYPE
True or false

CORRECT ANSWER
**B. False**

**GRADING NOTE** Division by 0 is undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 2 • Q3/5

Which system is closed under addition, subtraction, multiplication, and division by nonzero elements?

**A.** Whole numbers

**B.** Rational expressions

**C.** Integers

**D.** Polynomials

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 2 • Q4/5

899

QUESTION ID
**S24-F2-Q4**

TYPE
Multiple choice

CORRECT ANSWER
**B. Rational expressions**

**GRADING NOTE** Like the rational numbers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Which example shows that polynomials are NOT closed under division?

**A.** $6x^2 \div 3x = 2x$ for $x \neq 0$, which is a polynomial

**B.** $(x^3 + x) \div x = x^2 + 1$ for $x \neq 0$, a polynomial

**C.** $x \div (x + 1) = x/(x + 1)$, which is not a polynomial

**D.** $(x^2 - 1) \div (x - 1) = x + 1$ for $x \neq 1$, a polynomial

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q5

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 2 • Q5/5

900

QUESTION ID
**S24-F2-Q5**

TYPE
Multiple choice

CORRECT ANSWER
**C. $x \div (x + 1) = x/(x + 1)$, which is not a polynomial**

**GRADING NOTE** One quotient that is not a polynomial is enough. The other quotients are polynomials, so they do not show the failure of closure.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q5

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 2 • Q5/5

# Add and subtract rational expressions

| | | |
|---|---|---|
| **MAIN** | Algebra II main question | **HSA-APR.D.7** |
| **BACKWARD 1** | Add fractions with unlike denominators | **5.NF.A.1** |
| **BACKWARD 2** | Find a least common denominator by factoring | **HSA-SSE.A.2** |
| **BACKWARD 3** | Subtract a whole numerator | **6.EE.A.3** |
| **FORWARD 1** | Decompose into partial fractions | **HSA-APR.D.7** |
| **FORWARD 2** | Solve rational equations | **HSA-REI.A.2** |

SET 25 • HSA-APR.D.7

# MAIN

Algebra II main question • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.7

**ANSWER KEY** Reduced copy of the question page, with its answer

Simplify $(x + 1)/(x^2 + x) + 1/x$. Give the result and all excluded values.

Result: _____

Excluded values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • MAIN • Q1/5

903

QUESTION ID
**S25-M-Q1**

TYPE
Short answer

CORRECT ANSWER
**2/x; {0, −1}**

**GRADING NOTE** Form required: both. $(x + 1)/(x(x + 1)) = 1/x$ for $x \neq -1$, so the sum is $2/x$; $x = -1$ stays excluded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • MAIN • Q1/5

Write 2/(x + 1) + 3/(x − 2) as one fraction with denominator (x + 1)(x − 2). Give the numerator and all excluded values.

Numerator: _____

Excluded values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • MAIN • Q2/5

904

QUESTION ID
**S25-M-Q2**

TYPE
Short answer

CORRECT ANSWER
**5x − 1; {−1, 2}**

**GRADING NOTE** Form required: both. 2(x − 2) + 3(x + 1) = 5x − 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • MAIN • Q2/5

Simplify x/(x − 3) − 3/(x − 3). Give the result and all excluded values.

Result: _____

Excluded values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S25-M-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • MAIN • Q3/5

905

**QUESTION ID**

## S25-M-Q3

**TYPE**

Short answer

**CORRECT ANSWER**

**1; {3}**

**GRADING NOTE** Form required: both. (x − 3)/(x − 3) = 1 for x ≠ 3.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S25-M-Q3**

**High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions**
**HSA-APR.D.7 • Set 25 • MAIN • Q3/5**

Write $1/(x^2 - 1) - 1/(x - 1)$ as one fraction with denominator $(x - 1)(x + 1)$. Give the numerator and all excluded values.

Numerator: _____

Excluded values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • MAIN • Q4/5

906

QUESTION ID
**S25-M-Q4**

TYPE
Short answer

CORRECT ANSWER
**$-x$; {1, −1}**

**GRADING NOTE** Form required: both. $1 - (x + 1) = -x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • MAIN • Q4/5

Write $4/(x^2 - 4x) - 1/x$ as one fraction with denominator $x(x - 4)$. Give the numerator and all excluded values.

Numerator: _____

Excluded values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • MAIN • Q5/5

907

**QUESTION ID**
**S25-M-Q5**

**TYPE**
Short answer

**CORRECT ANSWER**
**8 − x; {0, 4}**

**GRADING NOTE** Form required: both. $4 - (x - 4) = 8 - x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • MAIN • Q5/5

SET 25 • HSA-APR.D.7

# BACKWARD 1

Add fractions with unlike denominators • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

Compute 2/3 + 3/4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 25 • BACKWARD 1 • Q1/5

909

**QUESTION ID**

**S25-B1-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**17/12**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 25 • BACKWARD 1 • Q1/5

Compute 5/6 − 1/4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 25 • BACKWARD 1 • Q2/5

910

**QUESTION ID**

**S25-B1-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**7/12**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 25 • BACKWARD 1 • Q2/5

**True or false?**

1/2 + 1/3 = 2/5.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 25 • BACKWARD 1 • Q3/5

911

QUESTION ID

**S25-B1-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE 5/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 25 • BACKWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

1/4 + 1/6 equals

**A.** 5/12

**B.** 2/10

**C.** 2/12

**D.** 1/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 25 • BACKWARD 1 • Q4/5

912

**QUESTION ID**

**S25-B1-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. 5/12**

**GRADING NOTE** 3/12 + 2/12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 25 • BACKWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Compute 3/10 + 2/15.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 25 • BACKWARD 1 • Q5/5

913

QUESTION ID

**S25-B1-Q5**

TYPE

Short answer

CORRECT ANSWER

**13/30**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 25 • BACKWARD 1 • Q5/5

SET 25 • HSA-APR.D.7

# BACKWARD 2

Find a least common denominator by factoring • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $x^2 + x$ and $x^2 - 4x$.

$x^2 + x =$ _____

$x^2 - 4x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 25 • BACKWARD 2 • Q1/5

915

QUESTION ID

**S25-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**x(x + 1); x(x − 4)**

GRADING NOTE Form required: both factored.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 25 • BACKWARD 2 • Q1/5

What is the least common denominator of $1/(x^2 - 1)$ and $1/(x - 1)$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 25 • BACKWARD 2 • Q2/5

916

**QUESTION ID**

**S25-B2-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**$(x - 1)(x + 1)$**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 25 • BACKWARD 2 • Q2/5

The LCD of $\frac{1}{x}$ and $\frac{1}{x(x-4)}$ is

**A.** $x - 4$

**B.** $x^2(x - 4)$

**C.** $x(x - 4)$

**D.** $x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 25 • BACKWARD 2 • Q3/5

917

QUESTION ID

**S25-B2-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**C. $x(x - 4)$**

GRADING NOTE Use each factor to its highest power.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 25 • BACKWARD 2 • Q3/5

**True or false?**

The LCD of $\frac{2}{x+1}$ and $\frac{3}{x-2}$ is $x - 1$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 25 • BACKWARD 2 • Q4/5



QUESTION ID

**S25-B2-Q4**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** The denominators have no common factor; the LCD is $(x + 1)(x - 2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 25 • BACKWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

What is the LCD of $1/(x^2 + x)$ and $1/x$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 25 • BACKWARD 2 • Q5/5

919

QUESTION ID

**S25-B2-Q5**

TYPE

Short answer

CORRECT ANSWER

**x(x + 1)**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 25 • BACKWARD 2 • Q5/5

SET 25 • HSA-APR.D.7

# BACKWARD 3

Subtract a whole numerator • Grade 6 • Expressions & Equations • 6.EE.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

Simplify 4 − (x − 4).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 25 • BACKWARD 3 • Q1/5

921

QUESTION ID

**S25-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**8 − x**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 25 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Simplify 1 − (x + 1).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 25 • BACKWARD 3 • Q2/5

922

QUESTION ID

**S25-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**−x**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 25 • BACKWARD 3 • Q2/5

**True or false?**

$3 - (x - 3) = -x.$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B3-Q3

**Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 25 • BACKWARD 3 • Q3/5**

923

**QUESTION ID**

## S25-B3-Q3

**TYPE**

True or false

**CORRECT ANSWER**

## B. False

**GRADING NOTE** $3 - x + 3 = 6 - x.$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B3-Q3

**Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 25 • BACKWARD 3 • Q3/5**

$2(x - 2) - 3(x + 1)$ equals

**A.** $5x - 7$

**B.** $-x + 1$

**C.** $-x - 7$

**D.** $-x - 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 25 • BACKWARD 3 • Q4/5

924

QUESTION ID

**S25-B3-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**C. $-x - 7$**

**GRADING NOTE** $2x - 4 - 3x - 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 25 • BACKWARD 3 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Simplify $x - (3 - x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 25 • BACKWARD 3 • Q5/5

925

QUESTION ID

**S25-B3-Q5**

TYPE

Short answer

CORRECT ANSWER

**2x − 3**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 25 • BACKWARD 3 • Q5/5

SET 25 • HSA-APR.D.7

# FORWARD 1

Decompose into partial fractions • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.7

**ANSWER KEY** Reduced copy of the question page, with its answer

Find A and B so that
$(5x + 1)/((x - 1)(x + 2)) = A/(x - 1) + B/(x + 2)$.

A = _____

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • FORWARD 1 • Q1/5

927

QUESTION ID
**S25-F1-Q1**

TYPE
Short answer

CORRECT ANSWER
**2; 3**

GRADING NOTE Form required: both. $5x + 1 = A(x + 2) + B(x - 1)$; $x = 1$ gives $A = 2$, $x = -2$ gives $B = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • FORWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Find A and B so that $1/(x(x + 1)) = A/x + B/(x + 1)$.

A = _____

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q2

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • FORWARD 1 • Q2/5

928

QUESTION ID

**S25-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**1; −1**

GRADING NOTE Form required: both. $1 = A(x + 1) + Bx$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q2

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • FORWARD 1 • Q2/5

**True or false?**

$1/(x(x + 1)) = 1/x + 1/(x + 1)$ for $x \neq 0, -1$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • FORWARD 1 • Q3/5

929

QUESTION ID
**S25-F1-Q3**

TYPE
True or false

CORRECT ANSWER
**B. False**

**GRADING NOTE** $1/x + 1/(x + 1) = (2x + 1)/(x(x + 1))$; the difference $1/x - 1/(x + 1)$ gives $1/(x(x + 1))$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • FORWARD 1 • Q3/5

$4/(x^2 - 4)$ equals

**A.** $2/(x - 2) - 2/(x + 2)$

**B.** $1/(x - 2) + 1/(x + 2)$

**C.** $4/(x - 2) - 4/(x + 2)$

**D.** $1/(x - 2) - 1/(x + 2)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • FORWARD 1 • Q4/5

930

QUESTION ID
**S25-F1-Q4**

TYPE
Multiple choice

CORRECT ANSWER
**D.**
**$1/(x - 2) - 1/(x + 2)$**

**GRADING NOTE** Combine to check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • FORWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Use $1/(n(n+1)) = 1/n - 1/(n+1)$ to find $1/(1\cdot2) + 1/(2\cdot3) + 1/(3\cdot4) + 1/(4\cdot5)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q5

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • FORWARD 1 • Q5/5

931

QUESTION ID
**S25-F1-Q5**

TYPE
Short answer

CORRECT ANSWER
**4/5**

**GRADING NOTE** Any equivalent form. Telescoping: $1 - 1/5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q5

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • FORWARD 1 • Q5/5

SET 25 • HSA-APR.D.7

# FORWARD 2

Solve rational equations • High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve 2/(x + 1) + 3/(x − 2) = 0.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 25 • FORWARD 2 • Q1/5

933

QUESTION ID

**S25-F2-Q1**

TYPE

Short answer

CORRECT ANSWER

**1/5**

**GRADING NOTE** Any equivalent form. 5x − 1 = 0; not an excluded value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 25 • FORWARD 2 • Q1/5

Which is the solution set of x/(x − 3) − 3/(x − 3) = 1?

**A.** x = 3 only

**B.** Every real number

**C.** No solution

**D.** Every real number except 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 25 • FORWARD 2 • Q2/5

934

**QUESTION ID**

**S25-F2-Q2**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. Every real number except 3**

**GRADING NOTE** For x ≠ 3 the left side is (x − 3)/(x − 3) = 1, so every allowed x works; x = 3 is excluded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 25 • FORWARD 2 • Q2/5

**True or false?**

A solution of a rational equation that makes a denominator 0 must be rejected.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 25 • FORWARD 2 • Q3/5

935

QUESTION ID

## S25-F2-Q3

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Extraneous.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 25 • FORWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

The solution of 1/x + 1/(2x) = 3 is

**A.** x = 1/2

**B.** x = 2

**C.** x = 3/2

**D.** no solution

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 25 • FORWARD 2 • Q4/5

936

**QUESTION ID**

## S25-F2-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. x = 1/2**

**GRADING NOTE** 3/(2x) = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 25 • FORWARD 2 • Q4/5

Solve $4/(x^2 - 4x) - 1/x = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 25 • FORWARD 2 • Q5/5

937

QUESTION ID

**S25-F2-Q5**

TYPE

Short answer

CORRECT ANSWER

**8**

GRADING NOTE Any equivalent form. $(8 - x)/(x(x - 4)) = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 25 • FORWARD 2 • Q5/5

SET 26 • HSA-CED.A.1 • CREATING EQUATIONS AND REASONING WITH EQUATIONS

# Create and solve rational, radical, and exponential equations from contexts

| | | |
|---|---|---|
| **MAIN** | Algebra II main question | **HSA-CED.A.1** |
| **BACKWARD 1** | Express rates of work and unit rates | **6.RP.A.3.b** |
| **BACKWARD 2** | Write exponential growth models | **HSF-LE.A.2** |
| **BACKWARD 3** | Solve linear equations after clearing a denominator | **HSA-REI.B.3** |
| **FORWARD 1** | Create equations whose algebraic solutions must be checked aga. | **HSA-CED.A.1** |
| **FORWARD 2** | Solve exponential context equations with logarithms | **HSF-LE.A.4** |

SET 26 • HSA-CED.A.1

# MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1

Ana can paint a room in 4 hours. Working together, Ana and Ben paint it in 2.4 hours. Write an equation for Ben's time b (in hours) working alone, and use it to find b.

b (hours) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • MAIN • Q1/5

940

**QUESTION ID**

## S26-M-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**6**

**GRADING NOTE** Form required: the value. 1/4 + 1/b = 1/2.4, so 1/b = 5/12 − 3/12 = 1/6.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S26-M-Q1**

**High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • MAIN • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

A car's speed s (in mph) before braking is estimated from its skid length d (in feet) by $s = \sqrt{(22.5d)}$. Write an equation for the skid length of a car going 45 mph, and use it to find d.

d (feet) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • MAIN • Q2/5

941

**QUESTION ID**
**S26-M-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**90**

**GRADING NOTE** Form required: the value. $45 = \sqrt{(22.5d)}$, so $2025 = 22.5d$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • MAIN • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A town of 1,200 people doubles in population every 5 years. Write an equation for the time t (in years) when the population is 9,600, and use it to find t.

t (years) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • MAIN • Q3/5

942

**QUESTION ID**

**S26-M-Q3**

**TYPE**

Short answer

**CORRECT ANSWER**

**15**

**GRADING NOTE** Form required: the value. $1200 \cdot 2^{t/5} = 9600$, so $2^{t/5} = 8 = 2^3$ and $t/5 = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • MAIN • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A printer charges $500 plus $4 per shirt. Write an equation for the number of shirts x that makes the average cost per shirt $9, and use it to find x.

x (shirts) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • MAIN • Q4/5

943

**QUESTION ID**
**S26-M-Q4**

**TYPE**
Short answer

**CORRECT ANSWER**
**100**

**GRADING NOTE** Form required: the value. (500 + 4x)/x = 9, so 500 + 4x = 9x and x = 100 (x > 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • MAIN • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A tank holds 10 L of a 20% salt solution. Write an equation for the liters x of pure water to add to make an 8% solution, and use it to find x.

x (liters) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • MAIN • Q5/5

944

**QUESTION ID**

## S26-M-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**15**

**GRADING NOTE** Form required: the value. The salt stays 2 L: 2/(10 + x) = 0.08, so 10 + x = 25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • MAIN • Q5/5

SET 26 • HSA-CED.A.1

# BACKWARD 1

Express rates of work and unit rates • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b

Ana paints a room in 4 hours. What fraction of the room does she paint per hour?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 26 • BACKWARD 1 • Q1/5

946

**QUESTION ID**

**S26-B1-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**1/4**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 26 • BACKWARD 1 • Q1/5

A pump fills a pool in 6 hours. What fraction does it fill in 2 hours?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 26 • BACKWARD 1 • Q2/5

947

**QUESTION ID**

**S26-B1-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**1/3**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 26 • BACKWARD 1 • Q2/5

**True or false?**

If two people work together, their times to finish the job add.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 26 • BACKWARD 1 • Q3/5

948

QUESTION ID

## S26-B1-Q3

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Their rates (jobs per hour) add; working together takes less time, not more.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 26 • BACKWARD 1 • Q3/5

A machine makes 300 parts in 5 hours. Its rate is

**A.** 1500 parts per hour

**B.** 300 parts per hour

**C.** 60 parts per hour

**D.** 5/300 parts per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 26 • BACKWARD 1 • Q4/5

949

**QUESTION ID**

## S26-B1-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. 60 parts per hour**

**GRADING NOTE** 300/5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 26 • BACKWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Working together, two people finish a job in 2.4 hours. What fraction of the job do they finish per hour? Give a fraction.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 26 • BACKWARD 1 • Q5/5

950

**QUESTION ID**

**S26-B1-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**5/12**

**GRADING NOTE** 1/2.4 = 5/12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 26 • BACKWARD 1 • Q5/5

SET 26 • HSA-CED.A.1

# BACKWARD 2

Write exponential growth models • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

A population of 1,200 doubles every 5 years. Write the population after t years.

P(t) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 26 • BACKWARD 2 • Q1/5

952

QUESTION ID

**S26-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**$1200 \cdot 2^{t/5}$**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 26 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A $500 deposit grows 3% per year. Write its value after t years.

V(t) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 26 • BACKWARD 2 • Q2/5

953

**QUESTION ID**

**S26-B2-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**$500(1.03)^t$**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 26 • BACKWARD 2 • Q2/5

**True or false?**

A quantity that doubles every 5 years is multiplied by $2^5$ after 20 years.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 26 • BACKWARD 2 • Q3/5

954

QUESTION ID

**S26-B2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** 20/5 = 4 doublings, a factor of $2^4 = 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 26 • BACKWARD 2 • Q3/5

A quantity triples every year starting at 7. After t years it is

**A.** $7 + 3t$

**B.** $21t$

**C.** $3 \cdot 7^t$

**D.** $7 \cdot 3^t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 26 • BACKWARD 2 • Q4/5

955

**QUESTION ID**

## S26-B2-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. $7 \cdot 3^t$**

**GRADING NOTE** Repeated multiplication.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 26 • BACKWARD 2 • Q4/5

How many doublings happen in 15 years if the doubling time is 5 years?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 26 • BACKWARD 2 • Q5/5

956

**QUESTION ID**

## S26-B2-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**3**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 26 • BACKWARD 2 • Q5/5

SET 26 • HSA-CED.A.1

# BACKWARD 3

Solve linear equations after clearing a denominator • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve 500 + 4x = 9x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 26 • BACKWARD 3 • Q1/5

958

QUESTION ID

**S26-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**100**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 26 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve 2 = 0.08(10 + x).

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 26 • BACKWARD 3 • Q2/5

959

QUESTION ID

**S26-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**15**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 26 • BACKWARD 3 • Q2/5

**True or false?**

To solve (500 + 4x)/x = 9 we may multiply both sides by x when x ≠ 0.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 26 • BACKWARD 3 • Q3/5

960

QUESTION ID

## S26-B3-Q3

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Multiplying by a nonzero quantity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 26 • BACKWARD 3 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

2025 = 22.5d gives d =

**A.** 2002.5

**B.** 45

**C.** 9

**D.** 90

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 26 • BACKWARD 3 • Q4/5

961

**QUESTION ID**

## S26-B3-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. 90**

**GRADING NOTE** 2025/22.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 26 • BACKWARD 3 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve 5x/12 = 1/2.4 for x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 26 • BACKWARD 3 • Q5/5

962

QUESTION ID

**S26-B3-Q5**

TYPE

Short answer

CORRECT ANSWER

**1**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 26 • BACKWARD 3 • Q5/5

SET 26 • HSA-CED.A.1

# FORWARD 1

Create equations whose algebraic solutions must be checked against the context • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1

A boat travels 24 km downstream and 24 km back upstream in 5 hours. The current is 2 km/h. The still-water speed b satisfies 24/(b + 2) + 24/(b − 2) = 5. Solve the equation and give the speed that fits the context, in km/h.

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • FORWARD 1 • Q1/5

964

**QUESTION ID**

**S26-F1-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**10**

**GRADING NOTE** $5b^2 - 48b - 20 = 0$ gives b = 10 or b = −0.4; reject −0.4 (a speed must be greater than 2 to go upstream).

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S26-F1-Q1**

**High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • FORWARD 1 • Q1/5**

The length of a rectangle is 3 more than its width w, and its diagonal is 15. Then $w^2 + (w + 3)^2 = 225$. Find the width that fits the context.

w = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • FORWARD 1 • Q2/5

965

**QUESTION ID**

## S26-F1-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**9**

**GRADING NOTE** $2w^2 + 6w - 216 = 0$ gives w = 9 or w = −12; reject −12 (a length is positive).

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S26-F1-Q2**

**High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • FORWARD 1 • Q2/5**

**True or false?**

A negative solution of an equation for a time must be rejected.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • FORWARD 1 • Q3/5

966

QUESTION ID
**S26-F1-Q3**

TYPE
True or false

CORRECT ANSWER
**A. True**

**GRADING NOTE** Time in the context is nonnegative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • FORWARD 1 • Q3/5

A model gives t = −3 or t = 4 for the time (in seconds) of an event. The answer is

**A.** neither

**B.** both

**C.** 4 seconds

**D.** −3 seconds

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • FORWARD 1 • Q4/5

967

QUESTION ID

**S26-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**C. 4 seconds**

**GRADING NOTE** Reject the negative time.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • FORWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A pipe fills a tank in x hours and a second pipe takes 2 hours longer. Together they fill it in 2.4 hours. Write a rational equation and solve it for x (reject any value that makes no sense in the context).

x (hours) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • FORWARD 1 • Q5/5

968

QUESTION ID

**S26-F1-Q5**

TYPE

Short answer

CORRECT ANSWER

**4**

**GRADING NOTE** Form required: the value. $1/x + 1/(x + 2) = 1/2.4$; $2.4(2x + 2) = x(x + 2)$ gives $x^2 - 2.8x - 4.8 = 0$, so $x = 4$ or $x = -1.2$, and a time cannot be negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • FORWARD 1 • Q5/5

SET 26 • HSA-CED.A.1

# FORWARD 2

Solve exponential context equations with logarithms • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

A town of 1,200 people doubles every 5 years, so its population is $1200 \cdot 2^{t/5}$ after t years. Solve $1200 \cdot 2^{t/5} = 5000$ for t, to the nearest tenth of a year. A calculator may be used.

t (years) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 26 • FORWARD 2 • Q1/5

970

**QUESTION ID**

**S26-F2-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**10.3**

**GRADING NOTE** Rounding to 0.1. $2^{t/5} = 25/6$, so $t = 5 \log_2(25/6) = 5 \ln(25/6)/\ln 2 \approx 10.29$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 26 • FORWARD 2 • Q1/5

Solve $500(1.04)^t = 800$ for t, to the nearest hundredth.
A calculator may be used.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 26 • FORWARD 2 • Q2/5

971

**QUESTION ID**
**S26-F2-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**11.98**

**GRADING NOTE** Rounding to 0.01. $(1.04)^t = 1.6$, so $t = \log 1.6/\log 1.04 \approx 11.9836$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 26 • FORWARD 2 • Q2/5

**True or false?**

$2^{t/5} = 8$ can be solved without logarithms, but $2^{t/5} = 5$ cannot be solved exactly without them.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 26 • FORWARD 2 • Q3/5



**QUESTION ID**

## S26-F2-Q3

**TYPE**

True or false

**CORRECT ANSWER**

## A. True

**GRADING NOTE** 8 is a power of 2; 5 is not.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 26 • FORWARD 2 • Q3/5

The exact solution of $3^x = 20$ is

**A.** 20/3

**B.** log(20/3)

**C.** log 20/log 3

**D.** 3 log 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 26 • FORWARD 2 • Q4/5

973

QUESTION ID

**S26-F2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**C. log 20/log 3**

**GRADING NOTE** Take logarithms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 26 • FORWARD 2 • Q4/5

An amount grows 4% per year, so after t years it is multiplied by $(1.04)^t$. Why is t = log 1.6/log 1.04 the time for it to grow by 60%?

**A.** Growing by 60% means $(1.04)^t = 60$; taking logarithms gives t log 1.04 = log 60, so t = log 60/log 1.04.

**B.** Growing by 60% means $(1.04)^t = 1.6$; taking logarithms gives t log 1.04 = log 1.6, so t = log 1.6/log 1.04.

**C.** Growing by 60% means 1.04t = 1.6, because the amount is multiplied by 1.04 for each year, so t = 1.6/1.04.

**D.** log 1.6/log 1.04 = log(1.6 − 1.04), the difference of the rates.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 26 • FORWARD 2 • Q5/5

974

QUESTION ID

**S26-F2-Q5**

TYPE

Multiple choice

CORRECT ANSWER

**B. Growing by 60% means $(1.04)^t = 1.6$; taking logarithms gives t log 1.04 = log 1.6, so t = log 1.6/log 1.04.**

**GRADING NOTE** Write the growth equation, then use $\log(b^t) = t \log b$. The others treat the growth as linear, use 60 instead of the factor 1.6, or use a false logarithm rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 26 • FORWARD 2 • Q5/5

SET 27 • HSA-CED.A.1 • CREATING EQUATIONS AND REASONING WITH EQUATIONS

# Create and solve nonlinear inequalities from contexts

| | | |
|---|---|---|
| **MAIN** | Algebra II main question | **HSA-CED.A.1** |
| **BACKWARD 1** | Solve quadratic equations by factoring | **HSA-REI.B.4.b** |
| **BACKWARD 2** | Sign of a product on intervals | **7.NS.A.2** |
| **BACKWARD 3** | Reverse the inequality when multiplying by a negative | **HSA-REI.B.3** |
| **FORWARD 1** | Solve rational inequalities with sign charts and excluded values | **HSA-CED.A.1** |
| **FORWARD 2** | Write systems of constraints and judge viability | **HSA-CED.A.3** |

SET 27 • HSA-CED.A.1

# MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

A ball's height is $h = -16t^2 + 64t$ feet after t seconds ($0 \le t \le 4$). Use an inequality to find the times when the ball is at least 48 feet high. Write them as a compound inequality.

Times: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • MAIN • Q1/5

977

**QUESTION ID**

## S27-M-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**$1 \le t \le 3$**

**GRADING NOTE** Form required: the compound inequality. $-16t^2 + 64t \ge 48$ gives $t^2 - 4t + 3 \le 0$, so $(t - 1)(t - 3) \le 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A company sells x items at a price of 100 − 2x dollars each (0 ≤ x ≤ 50). Use an inequality to find the values of x that give revenue of at least $1,200. Write them as a compound inequality.

Values of x: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • MAIN • Q2/5

978

QUESTION ID
**S27-M-Q2**

TYPE
Short answer

CORRECT ANSWER
**20 ≤ x ≤ 30**

**GRADING NOTE** Form required: the compound inequality. x(100 − 2x) ≥ 1200 gives (x − 20)(x − 30) ≤ 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • MAIN • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

The average cost of making x posters (x > 0) is (300 + 2x)/x dollars. Use an inequality to find the numbers of posters with an average cost of at most $5.

Values of x: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • MAIN • Q3/5

979

QUESTION ID

**S27-M-Q3**

TYPE

Short answer

CORRECT ANSWER

**x ≥ 100**

**GRADING NOTE** Form required: the inequality. With x > 0, (300 + 2x)/x ≤ 5 gives 300 + 2x ≤ 5x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • MAIN • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A drug concentration is $C(t) = 20t/(t^2 + 4)$ mg/L after $t$ hours ($t \geq 0$). Use an inequality to find the times when $C$ is at least 4 mg/L. Write them as a compound inequality.

Times: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • MAIN • Q4/5

980

**QUESTION ID**

**S27-M-Q4**

**TYPE**

Short answer

**CORRECT ANSWER**

**$1 \leq t \leq 4$**

**GRADING NOTE** Form required: the compound inequality. $t^2 + 4 > 0$, so $20t \geq 4t^2 + 16$, which gives $t^2 - 5t + 4 \leq 0$.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S27-M-Q4**

**High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • MAIN • Q4/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

$500 is invested at 4% interest, credited once at the end of each year. Use an inequality to find the least whole number of years t after which the value is at least $800. A calculator may be used.

t (whole years) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • MAIN • Q5/5

981

QUESTION ID

**S27-M-Q5**

TYPE

Short answer

CORRECT ANSWER

**12**

**GRADING NOTE** Form required: the whole number. $500(1.04)^t \geq 800$ holds for $t \geq 11.98$, and interest is credited only at whole years: $500(1.04)^{11} \approx 769.73 < 800 \leq 500(1.04)^{12} \approx 800.52$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • MAIN • Q5/5

SET 27 • HSA-CED.A.1

# BACKWARD 1

Solve quadratic equations by factoring • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve $t^2 - 4t + 3 = 0$.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 27 • BACKWARD 1 • Q1/5

983

QUESTION ID

**S27-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**1 or 3**

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 27 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve $x^2 - 50x + 600 = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 27 • BACKWARD 1 • Q2/5

984

QUESTION ID

**S27-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**20 or 30**

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 27 • BACKWARD 1 • Q2/5

**True or false?**

The solutions of $t^2 - 5t + 4 = 0$ are $-1$ and $-4$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 27 • BACKWARD 1 • Q3/5

985

QUESTION ID
**S27-B1-Q3**

TYPE
True or false

CORRECT ANSWER
**B. False**

**GRADING NOTE** $(t - 1)(t - 4) = 0$ gives $t = 1$ or $t = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 27 • BACKWARD 1 • Q3/5

The solutions of $x^2 - x - 12 = 0$ are

**A.** 4 and −3

**B.** 6 and −2

**C.** 12 and −1

**D.** −4 and 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 27 • BACKWARD 1 • Q4/5

986

**QUESTION ID**

## S27-B1-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

## A. 4 and −3

**GRADING NOTE** $(x - 4)(x + 3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 27 • BACKWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve $2x^2 - 100x + 1200 = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 27 • BACKWARD 1 • Q5/5

987

**QUESTION ID**

## S27-B1-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**20 or 30**

**GRADING NOTE** Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 27 • BACKWARD 1 • Q5/5

SET 27 • HSA-CED.A.1

# BACKWARD 2

Sign of a product on intervals • Grade 7 • The Number System • 7.NS.A.2

Is (t − 1)(t − 3) positive or negative when t = 2? Answer Positive or Negative.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 27 • BACKWARD 2 • Q1/5

989

QUESTION ID

**S27-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**Negative**

GRADING NOTE (1)(−1) = −1 < 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 27 • BACKWARD 2 • Q1/5

Is (t − 1)(t − 3) positive or negative when t = 5? Answer Positive or Negative.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 27 • BACKWARD 2 • Q2/5

990

QUESTION ID

**S27-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**Positive**

GRADING NOTE (4)(2) = 8 > 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 27 • BACKWARD 2 • Q2/5

**True or false?**

A product of two negative numbers is negative.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 27 • BACKWARD 2 • Q3/5

991

QUESTION ID

## S27-B2-Q3

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** It is positive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 27 • BACKWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

$(x - 20)(x - 30) \leq 0$ when

**A.** $x \leq 20$ or $x \geq 30$

**B.** $x \leq 20$

**C.** $20 \leq x \leq 30$

**D.** $x \geq 30$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 27 • BACKWARD 2 • Q4/5

992

**QUESTION ID**

**S27-B2-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. $20 \leq x \leq 30$**

**GRADING NOTE** Factors have opposite signs between the roots.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 27 • BACKWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

For which t is $(t - 1)(t - 4) < 0$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 27 • BACKWARD 2 • Q5/5

993

QUESTION ID

**S27-B2-Q5**

TYPE

Short answer

CORRECT ANSWER

**$1 < t < 4$**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 27 • BACKWARD 2 • Q5/5

SET 27 • HSA-CED.A.1

# BACKWARD 3

Reverse the inequality when multiplying by a negative • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Both sides of $-16t^2 + 64t - 48 \geq 0$ are divided by $-16$. Which inequality results?

**A.** $t^2 - 4t + 3 \leq 0$

**B.** $t^2 - 4t - 3 \geq 0$

**C.** $t^2 + 4t + 3 \leq 0$

**D.** $t^2 - 4t + 3 \geq 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 27 • BACKWARD 3 • Q1/5

995

QUESTION ID

**S27-B3-Q1**

TYPE

Multiple choice

CORRECT ANSWER

**A. $t^2 - 4t + 3 \leq 0$**

**GRADING NOTE** Dividing by a negative number reverses the inequality, and every term is divided: $-16t^2/(-16) = t^2$, $64t/(-16) = -4t$, $-48/(-16) = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 27 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve $-3x \geq 12$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 27 • BACKWARD 3 • Q2/5

996

QUESTION ID

**S27-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**$x \leq -4$**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 27 • BACKWARD 3 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

Multiplying both sides of an inequality by x is always allowed.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 27 • BACKWARD 3 • Q3/5

997

QUESTION ID
**S27-B3-Q3**

TYPE
True or false

CORRECT ANSWER
**B. False**

GRADING NOTE Only when the sign of x is known.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 27 • BACKWARD 3 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

If $x > 0$, then $(300 + 2x)/x \leq 5$ is equivalent to

**A.** $300 + 2x \leq 5x$

**B.** $300 + 2x \geq 5x$

**C.** $2x \leq 5$

**D.** $300 \leq 2x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 27 • BACKWARD 3 • Q4/5

998

QUESTION ID

**S27-B3-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**A. 300 + 2x ≤ 5x**

**GRADING NOTE** Multiplying by positive x keeps the direction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 27 • BACKWARD 3 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve $2 - x < 7$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 27 • BACKWARD 3 • Q5/5

999

QUESTION ID

**S27-B3-Q5**

TYPE

Short answer

CORRECT ANSWER

**$x > -5$**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 27 • BACKWARD 3 • Q5/5

SET 27 • HSA-CED.A.1

# FORWARD 1

Solve rational inequalities with sign charts and excluded values • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve $(x - 3)/(x + 1) \leq 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • FORWARD 1 • Q1/5

1001

**QUESTION ID**

## S27-F1-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**$-1 < x \leq 3$**

**GRADING NOTE** Form required: the interval; −1 is excluded and 3 is included.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S27-F1-Q1**

**High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • FORWARD 1 • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Which is the solution set of $(x + 2)/(x - 4) > 0$?

**A.** $-2 < x < 4$

**B.** $x < -2$ or $x > 4$

**C.** $x > 4$ only

**D.** $x < -2$ or $x \geq 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • FORWARD 1 • Q2/5

1002

QUESTION ID
**S27-F1-Q2**

TYPE
Multiple choice

CORRECT ANSWER
**B. $x < -2$ or $x > 4$**

**GRADING NOTE** The quotient is positive where the numerator and denominator have the same sign: both negative ($x < -2$) or both positive ($x > 4$). $x = 4$ is excluded; between $-2$ and 4 the quotient is negative.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S27-F1-Q2**

**High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • FORWARD 1 • Q2/5**

**True or false?**

Multiplying $(x - 3)/(x + 1) \leq 0$ by $x + 1$ gives $x - 3 \leq 0$ for all $x$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • FORWARD 1 • Q3/5

1003

QUESTION ID

## S27-F1-Q3

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** The direction depends on the sign of $x + 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • FORWARD 1 • Q3/5

The solution set of $1/x > 2$ is

**A.** $x > 1/2$

**B.** $x < 1/2$

**C.** $x > 0$

**D.** $0 < x < 1/2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • FORWARD 1 • Q4/5

1004

**QUESTION ID**

## S27-F1-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. $0 < x < 1/2$**

**GRADING NOTE** For $x > 0$, $1 > 2x$; for $x < 0$ the left side is negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • FORWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A model's average cost is (400 + 3x)/x dollars. For how many items x > 0 is the average cost under $7?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • FORWARD 1 • Q5/5

1005

**QUESTION ID**

**S27-F1-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**x > 100**

**GRADING NOTE** Any equivalent form. 400 + 3x < 7x.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S27-F1-Q5**

**High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • FORWARD 1 • Q5/5**

SET 27 • HSA-CED.A.1

# FORWARD 2

Write systems of constraints and judge viability • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

A ball's height is $h(t) = -16t^2 + 64t$ feet after $t$ seconds, for $0 \le t \le 4$ (while it is in the air). Its height must be at least 48 ft and the time must be at most 2.5 s. Give the viable times as a compound inequality.

Viable times: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 27 • FORWARD 2 • Q1/5

1007

**QUESTION ID**

**S27-F2-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**$1 \le t \le 2.5$**

**GRADING NOTE** Form required: the compound inequality. $-16t^2 + 64t \ge 48$ gives $1 \le t \le 3$; with $t \le 2.5$ the viable times are $1 \le t \le 2.5$.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S27-F2-Q1**

**High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 27 • FORWARD 2 • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

A company sells x items (x a whole number) at a price of 100 − 2x dollars each. Revenue x(100 − 2x) must be at least $1,200 and at most 24 items can be made. Give the viable values of x as a compound inequality.

Viable x: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 27 • FORWARD 2 • Q2/5

1008

**QUESTION ID**
**S27-F2-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**$20 \le x \le 24$**

**GRADING NOTE** Form required: the compound inequality (x a whole number). x(100 − 2x) ≥ 1200 gives 20 ≤ x ≤ 30; with x ≤ 24 the viable values are the whole numbers 20 to 24.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S27-F2-Q2**

**High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 27 • FORWARD 2 • Q2/5**

**True or false?**

A value that solves the inequality may still not be viable in the context.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 27 • FORWARD 2 • Q3/5

1009

QUESTION ID
**S27-F2-Q3**

TYPE
True or false

CORRECT ANSWER
**A. True**

**GRADING NOTE** For example a non-whole number of items.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 27 • FORWARD 2 • Q3/5

Revenue $x(100 - 2x) \geq 1200$ with x a whole number is satisfied by

**A.** x = 19

**B.** x = 25

**C.** x = 31

**D.** x = 50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 27 • FORWARD 2 • Q4/5

1010

**QUESTION ID**

## S27-F2-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. x = 25**

**GRADING NOTE** $20 \leq x \leq 30$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 27 • FORWARD 2 • Q4/5

$500 is invested at 4% interest credited once at the end of each year. The equation $500(1.04)^t = 800$ gives $t \approx 11.98$. After how many years is the account first worth at least $800, and why?

**A.** 12 years: because 11.98 rounds to 12 when rounded to the nearest whole number of years.

**B.** 11.98 years: the balance reaches exactly $800 at that time, as the equation shows.

**C.** 11 years: round 11.98 down, because after 11 years the account is already close to $800.

**D.** 12 years: interest is credited only at year ends, so the balance is below $800 until the 12th.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 27 • FORWARD 2 • Q5/5

1011

QUESTION ID
**S27-F2-Q5**

TYPE
Multiple choice

CORRECT ANSWER
**D. 12 years: interest is credited only at year ends, so the balance is below $800 until the 12th.**

**GRADING NOTE** The model value 11.98 is a threshold for a continuous curve; the real balance changes only at whole years. $500(1.04)^{11} \approx 769.73$ and $500(1.04)^{12} \approx 800.52$. The others ignore when interest is credited, round down below the target, or give a coincidental reason (ordinary rounding).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 27 • FORWARD 2 • Q5/5

# Create and graph nonlinear two-variable models

| | | |
|---|---|---|
| **MAIN** | Algebra II main question | **HSA-CED.A.2** |
| **BACKWARD 1** | Make a table of values from an equation | **6.EE.C.9** |
| **BACKWARD 2** | Choose scales and label axes | **HSN-Q.A.1** |
| **BACKWARD 3** | Write a growth or decay model from a percent | **HSF-LE.A.2** |
| **FORWARD 1** | Create equations in more than two variables and graph a member | **HSA-CED.A.2** |
| **FORWARD 2** | Interpret key features of the created models | **HSF-IF.B.4** |

SET 28 • HSA-CED.A.2

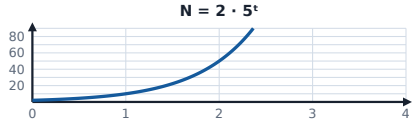
# MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2

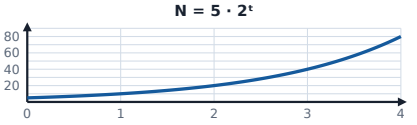
**ANSWER KEY** Reduced copy of the question page, with its answer

A rabbit population starts at 5 and doubles every year. Which graph shows a correct equation for the situation with a correct graph of that equation for $0 \le t \le 4$ (t in years, N rabbits)?

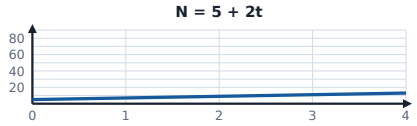
**A**



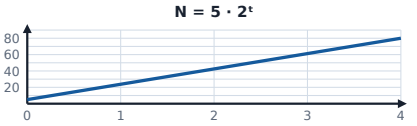

**B**




**C**




**D**




**A.** Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • MAIN • Q1/5

1014

**QUESTION ID**

## S28-M-Q1

**TYPE**

Multiple choice

**CORRECT ANSWER**

## B. Graph B

**GRADING NOTE** Doubling every year from 5 gives $N = 5 \cdot 2^t$, an exponential curve through (0, 5), (1, 10), (2, 20), (3, 40), (4, 80). The others use the wrong model ($2 \cdot 5^t$ or $5 + 2t$) or pair the right equation with a straight line, which is not its graph.

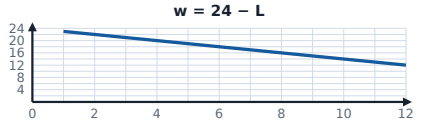
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • MAIN • Q1/5

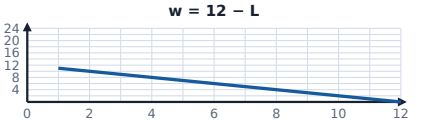
**ANSWER KEY** Reduced copy of the question page, with its answer

A rectangle has area 24 square units. Which graph shows a correct equation for the situation with a correct graph of that equation for $1 \le L \le 12$ (length L, width w)?

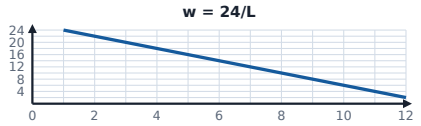
**A**



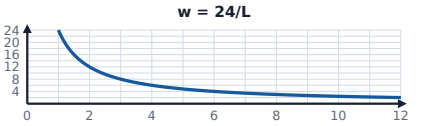

**B**




**C**




**D**




**A.** Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • MAIN • Q2/5

1015

**QUESTION ID**

## S28-M-Q2

**TYPE**

Multiple choice

**CORRECT ANSWER**

## D. Graph D

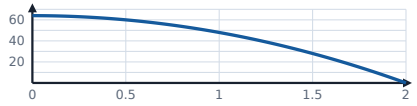
**GRADING NOTE** Area = Lw = 24 gives w = 24/L, a decreasing curve that flattens and approaches the L-axis, through (1, 24), (2, 12), (4, 6), (12, 2). The others use w = 24 − L, w = 12 − L (a perimeter idea), or draw a straight line for 24/L.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q2

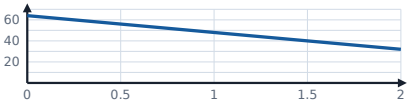
High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • MAIN • Q2/5

A stone is dropped from 64 feet. Its height after t seconds is 64 minus 16 times the square of t. Which graph shows a correct equation for the situation with a correct graph of that equation for $0 \le t \le 2$ (h in feet)?

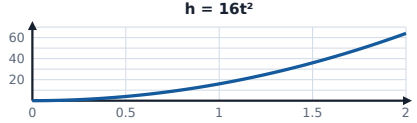
**A**



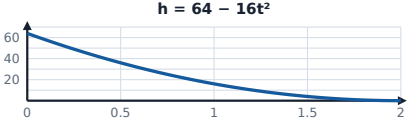

**B**




**C**




**D**




**A.** Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • MAIN • Q3/5

1016

**QUESTION ID**

## S28-M-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

## A. Graph A

**GRADING NOTE** $h = 64 - 16t^2$: the height falls slowly at first and then faster, through (0, 64), (0.5, 60), (1, 48), (1.5, 28), (2, 0). The others use $64 - 16t$ (not squared), $16t^2$ (the distance fallen), or draw a curve that does not match $64 - 16t^2$ (it gives 16 at t = 1 instead of 48).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • MAIN • Q3/5

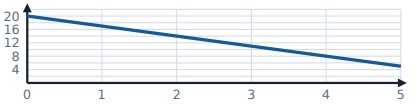
**ANSWER KEY** Reduced copy of the question page, with its answer

A $20,000 car loses 15% of its value each year. Which graph shows a correct equation for the situation with a correct graph of that equation for $0 \le t \le 5$ (V in thousands of dollars, t in years)?

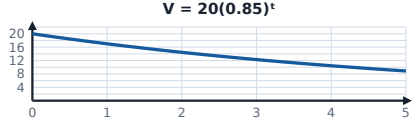
**A**



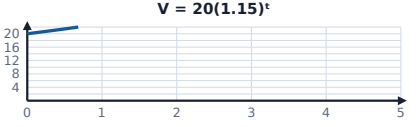

**B**




**C**




**D**




**A.** Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • MAIN • Q4/5

1017

**QUESTION ID**

## S28-M-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

## C. Graph C

**GRADING NOTE** Keeping 85% each year gives $V = 20(0.85)^t$, through (1, 17), (2, 14.45), (5, 8.87). The others keep only 15% each year, subtract a fixed $3,000 each year (15% of the first value only), or grow 15% each year.

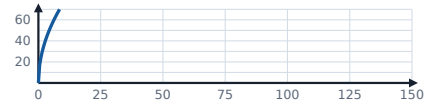
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • MAIN • Q4/5

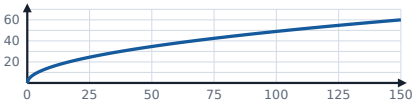
**ANSWER KEY** Reduced copy of the question page, with its answer

A car's speed s (mph) is the square root of 24 times its skid length d (ft). Which graph shows a correct equation for the situation with a correct graph of that equation for $0 \le d \le 150$?

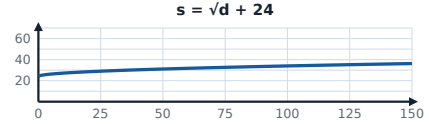
**A**



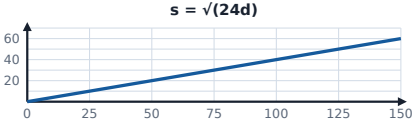

**B**




**C**




**D**




**A.** Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • MAIN • Q5/5

1018

**QUESTION ID**

## S28-M-Q5

**TYPE**

Multiple choice

**CORRECT ANSWER**

## B. Graph B

**GRADING NOTE** s = √(24d): an increasing curve that flattens, through (6, 12), (24, 24), (96, 48), (150, 60). The others use 24√d or √d + 24, or draw a straight line for √(24d) (it gives 24 at d = 60, not 37.9).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • MAIN • Q5/5

SET 28 • HSA-CED.A.2

# BACKWARD 1

Make a table of values from an equation • Grade 6 • Expressions & Equations • 6.EE.C.9

**ANSWER KEY** Reduced copy of the question page, with its answer

For $N = 200 \cdot 2^t$, find N when t = 0, 1, 2, 3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q1/5

1020

**QUESTION ID**

**S28-B1-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**200, 400, 800, 1600**

**GRADING NOTE** Form required: all four.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

For w = 24/L, find w when L = 2, 3, 4, 6.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q2/5

1021

**QUESTION ID**

**S28-B1-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**12, 8, 6, 4**

**GRADING NOTE** Form required: all four.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q2/5

**True or false?**

For $h = 64 - 16t^2$, $h = 48$ when $t = 2$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q3/5

1022

QUESTION ID

**S28-B1-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE $h(2) = 64 - 64 = 0$; $h = 48$ when $t = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

For s = √(24d), s when d = 96 is

**A.** 96

**B.** 48

**C.** 2304

**D.** 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q4/5

1023

QUESTION ID

**S28-B1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**B. 48**

GRADING NOTE √2304.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

For $V = 20000(0.85)^t$, find V when t = 1 and when t = 2.

V at t = 1: _____

V at t = 2: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q5/5

1024

**QUESTION ID**
**S28-B1-Q5**

**TYPE**
Short answer

**CORRECT ANSWER**
**17000; 14450**

**GRADING NOTE** Form required: both. 20000 · 0.85 = 17000; 17000 · 0.85 = 14450.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q5/5

SET 28 • HSA-CED.A.2

# BACKWARD 2

Choose scales and label axes • High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

Values of N go from 200 to 3,200. Which step for the vertical axis, starting at 0, shows all of them using about 7 grid lines?

**A.** 500

**B.** 50

**C.** 1,000

**D.** 100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 28 • BACKWARD 2 • Q1/5

1026

QUESTION ID

**S28-B2-Q1**

TYPE

Multiple choice

CORRECT ANSWER

**A. 500**

GRADING NOTE 0 to 3,500 in steps of 500 uses 7 steps. Steps of 100 or 50 need 32 or more lines; steps of 1,000 use only 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 28 • BACKWARD 2 • Q1/5

V depends on t, the time in years. Which quantity goes on the horizontal axis?

**A.** V, the value

**B.** Either one; it makes no difference

**C.** t, the time in years

**D.** Neither; time is not graphed

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 28 • BACKWARD 2 • Q2/5

1027

**QUESTION ID**
**S28-B2-Q2**

**TYPE**
Multiple choice

**CORRECT ANSWER**
**C. t, the time in years**

**GRADING NOTE** The input (independent variable) goes on the horizontal axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 28 • BACKWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

Axis labels should include units.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 28 • BACKWARD 2 • Q3/5

1028

QUESTION ID

**S28-B2-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Units make the graph readable.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 28 • BACKWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

The best vertical scale for values from 0 to 64 is

**A.** steps of 0.5

**B.** steps of 1 from 0 to 10

**C.** steps of 100 from 0 to 1000

**D.** steps of 10 from 0 to 70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 28 • BACKWARD 2 • Q4/5

1029

**QUESTION ID**
**S28-B2-Q4**

**TYPE**
Multiple choice

**CORRECT ANSWER**
**D. steps of 10 from 0 to 70**

**GRADING NOTE** Fits the range.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 28 • BACKWARD 2 • Q4/5

1029

**ANSWER KEY** Reduced copy of the question page, with its answer

A horizontal axis runs from d = 0 to d = 150 with tick marks every 10. How many intervals of 10 are there?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 28 • BACKWARD 2 • Q5/5

1030

QUESTION ID

**S28-B2-Q5**

TYPE

Short answer

CORRECT ANSWER

**15**

GRADING NOTE 150/10 = 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 28 • BACKWARD 2 • Q5/5

SET 28 • HSA-CED.A.2

# BACKWARD 3

Write a growth or decay model from a percent • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

Write the value of $20,000 after losing 15% per year for t years.

V = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 28 • BACKWARD 3 • Q1/5

1032

QUESTION ID

**S28-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**$20000(0.85)^t$**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 28 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

What is the growth factor for 4% growth?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 28 • BACKWARD 3 • Q2/5

1033

QUESTION ID

**S28-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**1.04**

**GRADING NOTE** Any equivalent form.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S28-B3-Q2**

**High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 28 • BACKWARD 3 • Q2/5**

**True or false?**

Losing 15% each year means multiplying by 0.15 each year.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 28 • BACKWARD 3 • Q3/5

1034

QUESTION ID

## S28-B3-Q3

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Multiply by $1 - 0.15 = 0.85$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 28 • BACKWARD 3 • Q3/5

A quantity doubles every hour starting at 200. After t hours it is

**A.** $2 \cdot 200^t$

**B.** $200 \cdot 2^t$

**C.** 200 + 2t

**D.** 400t

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 28 • BACKWARD 3 • Q4/5

1035

QUESTION ID
**S28-B3-Q4**

TYPE
Multiple choice

CORRECT ANSWER
**B. $200 \cdot 2^t$**

**GRADING NOTE** Doubling.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 28 • BACKWARD 3 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

What is the decay factor for 30% decay?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B3-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 28 • BACKWARD 3 • Q5/5

1036

QUESTION ID

**S28-B3-Q5**

TYPE

Short answer

CORRECT ANSWER

**0.7**

**GRADING NOTE** Any equivalent form.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S28-B3-Q5**

**High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 28 • BACKWARD 3 • Q5/5**

SET 28 • HSA-CED.A.2

# FORWARD 1

Create equations in more than two variables and graph a member of the family • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

The current I (amps) through a resistor is the voltage V divided by the resistance R. Write I in terms of V and R, then write I as a function of R when V = 12.

I = _____

I (when V = 12) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • FORWARD 1 • Q1/5

1038

**QUESTION ID**

**S28-F1-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**V/R; 12/R**

**GRADING NOTE** Form required: both expressions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • FORWARD 1 • Q1/5

The volume of a cylinder is $V = \pi r^2 h$. Write h in terms of r when $V = 100\pi$ ($r > 0$).

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • FORWARD 1 • Q2/5

1039

**QUESTION ID**

**S28-F1-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**$100/r^2$**

**GRADING NOTE** Any equivalent expression. $100\pi = \pi r^2 h$ gives $h = 100/r^2$.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S28-F1-Q2**

**High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • FORWARD 1 • Q2/5**

**True or false?**

Fixing one variable of a three-variable equation gives a two-variable equation that can be graphed.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • FORWARD 1 • Q3/5

1040

QUESTION ID
**S28-F1-Q3**

TYPE
True or false

CORRECT ANSWER
**A. True**

**GRADING NOTE** A family of curves.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • FORWARD 1 • Q3/5

For $A = P(1.05)^t$ with $P = 1000$, the graph of A against t is

**A.** a line

**B.** an increasing exponential curve

**C.** a parabola

**D.** a decreasing curve

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • FORWARD 1 • Q4/5

1041

**QUESTION ID**

**S28-F1-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. an increasing exponential curve**

**GRADING NOTE** Growth factor 1.05 > 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • FORWARD 1 • Q4/5

For the area formula $A = \pi r^2$ with $r \geq 0$, which describes the graph of A against r?

**A.** A full parabola opening down from its vertex

**B.** An increasing exponential curve through (0, 1)

**C.** Half of a parabola, starting at the origin and increasing

**D.** A straight line through the origin with slope π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • FORWARD 1 • Q5/5

1042

QUESTION ID

**S28-F1-Q5**

TYPE

Multiple choice

CORRECT ANSWER

**C. Half of a parabola, starting at the origin and increasing**

**GRADING NOTE** $A = \pi r^2$ is quadratic in r; only $r \geq 0$ is used, so it is the right half of a parabola with vertex at the origin.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • FORWARD 1 • Q5/5

SET 28 • HSA-CED.A.2

# FORWARD 2

Interpret key features of the created models • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4

A bacteria population is $N = 200 \cdot 2^t$ after t hours. What does the N-intercept, (0, 200), mean?

**A.** The population never goes below 200 in the past.

**B.** There are 200 bacteria at the start (t = 0).

**C.** The population doubles every 200 hours.

**D.** There are 200 bacteria after 1 hour.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 28 • FORWARD 2 • Q1/5

1044

QUESTION ID
**S28-F2-Q1**

TYPE
Multiple choice

CORRECT ANSWER
**B. There are 200 bacteria at the start (t = 0).**

**GRADING NOTE** The N-intercept is the value when t = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 28 • FORWARD 2 • Q1/5

A stone's height is $h = 64 - 16t^2$ feet after t seconds. At what time (the t-intercept, $t > 0$) does it hit the ground?

t (seconds) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 28 • FORWARD 2 • Q2/5

1045

**QUESTION ID**

**S28-F2-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**2**

**GRADING NOTE** Form required: the value. $64 - 16t^2 = 0$ gives $t^2 = 4$; the negative root has no meaning here.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 28 • FORWARD 2 • Q2/5

**True or false?**

The graph of $w = 24/L$ never reaches the L-axis because a rectangle of area 24 cannot have width 0.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 28 • FORWARD 2 • Q3/5

1046

QUESTION ID
**S28-F2-Q3**

TYPE
True or false

CORRECT ANSWER
**A. True**

**GRADING NOTE** $w > 0$ for every $L > 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 28 • FORWARD 2 • Q3/5

For $V = 20000(0.85)^t$, the graph

**A.** decreases and approaches 0

**B.** increases

**C.** is a line

**D.** decreases to 0 at t = 6.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 28 • FORWARD 2 • Q4/5

1047

**QUESTION ID**

**S28-F2-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. decreases and approaches 0**

**GRADING NOTE** Decay.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S28-F2-Q4**

**High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 28 • FORWARD 2 • Q4/5**

A car's speed is $s = \sqrt{(24d)}$ mph for a skid length of d ft. How does the rate of change of s change as d grows?

**A.** It increases: the speed grows faster and faster, so the graph gets steeper.

**B.** It becomes negative after a while: the speed starts to decrease for long skids, so the graph turns down.

**C.** It stays positive but decreases: the speed increases more and more slowly.

**D.** It is constant: s increases by the same amount for each foot of skid length.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 28 • FORWARD 2 • Q5/5

1048

**QUESTION ID**

## S28-F2-Q5

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. It stays positive but decreases: the speed increases more and more slowly.**

**GRADING NOTE** Square-root growth: increasing and concave down. For example s goes from 0 to 12 between d = 0 and 6 but from 48 to 60 between 96 and 150.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 28 • FORWARD 2 • Q5/5

# Represent nonlinear constraints and judge viable options

| | | |
|---|---|---|
| **MAIN** | Algebra II main question | **HSA-CED.A.3** |
| **BACKWARD 1** | Represent constraints with a linear system | **HSA-CED.A.3** |
| **BACKWARD 2** | Test whether a point satisfies an inequality | **6.EE.B.5** |
| **BACKWARD 3** | Solve quadratic inequalities | **HSA-CED.A.1** |
| **FORWARD 1** | Optimize within nonlinear constraints | **HSA-CED.A.3** |
| **FORWARD 2** | Locate constraint boundaries as graph intersections | **HSA-REI.D.11** |

SET 29 • HSA-CED.A.3

# MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

A garden has length L and width W (feet). The fencing gives L + W = 30, and the area must be at least 200 ft². Give the viable widths as a compound inequality, and decide whether W = 8 is viable.

Viable widths: _____

Is W = 8 viable? (Yes or No) _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • MAIN • Q1/5

1051

**QUESTION ID**

## S29-M-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**10 ≤ W ≤ 20; No**

**GRADING NOTE** Form required: both. L = 30 − W and W(30 − W) ≥ 200 give (W − 10)(W − 20) ≤ 0; W = 8 gives area 176.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S29-M-Q1**

**High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • MAIN • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

A box has a square base of side x in and a volume of 32 $in^3$; both x and the height h must be at most 4 in. Give the viable x as a compound inequality with exact values, and decide whether x = 3 is viable.

Viable x: _____

Is x = 3 viable? (Yes or No) _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • MAIN • Q2/5

1052

**QUESTION ID**

**S29-M-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**$2\sqrt{2} \le x \le 4$; Yes**

**GRADING NOTE** Form required: both. $h = 32/x^2 \le 4$ gives $x^2 \ge 8$, so $x \ge 2\sqrt{2} \approx 2.83$; with $x \le 4$. At x = 3, $h = 32/9 \approx 3.56 \le 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • MAIN • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A plan invests $1,000 at 5% compounded yearly and needs at least $1,500 within 10 years. Find the value after 10 years to the nearest cent, and decide whether the plan is viable. A calculator may be used.

Value after 10 years ($): _____

Viable? (Yes or No) _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • MAIN • Q3/5

1053

QUESTION ID

**S29-M-Q3**

TYPE

Short answer

CORRECT ANSWER

**1628.89; Yes**

**GRADING NOTE** Form required: both. $1000(1.05)^t \geq 1500$ for some $t \leq 10$; the value grows with t, and $1000(1.05)^{10} \approx 1628.89 \geq 1500$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • MAIN • Q3/5

Attendance at a show is 500 − 10p people when tickets cost p dollars (0 ≤ p ≤ 50). Revenue must be at least $6,000. Give the viable prices as a compound inequality, and decide whether $35 is viable.

Viable prices: _____

Is p = 35 viable? (Yes or No) _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • MAIN • Q4/5

1054

**QUESTION ID**

## S29-M-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**20 ≤ p ≤ 30; No**

**GRADING NOTE** Form required: both. p(500 − 10p) ≥ 6000 gives (p − 20)(p − 30) ≤ 0; at p = 35 revenue is 35 · 150 = 5250.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • MAIN • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A rectangular pen with sides x and y (m) needs area at least 50 m² and perimeter at most 40 m. Decide whether (5, 10) and (4, 12) are viable.

Is (5, 10) viable? (Yes or No) _____

Is (4, 12) viable? (Yes or No) _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • MAIN • Q5/5

1055

QUESTION ID

**S29-M-Q5**

TYPE

Short answer

CORRECT ANSWER

**Yes; No**

**GRADING NOTE** Form required: both. $xy \geq 50$ and $2x + 2y \leq 40$. (5, 10): area 50, perimeter 30. (4, 12): area 48 < 50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • MAIN • Q5/5

SET 29 • HSA-CED.A.3

# BACKWARD 1

Represent constraints with a linear system • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

Write a constraint for "the total of L and W is 30".

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • BACKWARD 1 • Q1/5

1057

QUESTION ID

**S29-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**L + W = 30**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • BACKWARD 1 • Q1/5

A rectangle has sides x and y (m), and at most 40 m of fence can be used for its perimeter. Which constraint is correct?

**A.** $2x + 2y \leq 40$

**B.** $x + y \leq 40$

**C.** $2x + 2y \geq 40$

**D.** $xy \leq 40$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • BACKWARD 1 • Q2/5

1058

**QUESTION ID**

**S29-B1-Q2**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. $2x + 2y \leq 40$**

**GRADING NOTE** The perimeter is 2x + 2y and "at most" means ≤.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • BACKWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

"At least 200" is written ≤ 200.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • BACKWARD 1 • Q3/5

1059

QUESTION ID

**S29-B1-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** At least 200 means 200 or more: ≥ 200.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • BACKWARD 1 • Q3/5

1059

**ANSWER KEY** Reduced copy of the question page, with its answer

"The height h must be at most 4" is

**A.** $h \leq 4$

**B.** $h \geq 4$

**C.** $h < 4$

**D.** $h = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • BACKWARD 1 • Q4/5

1060

QUESTION ID

**S29-B1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**A. $h \leq 4$**

**GRADING NOTE** At most: less than or equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • BACKWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Is (12, 20) a solution of $x + y \leq 30$? Find $x + y$, then answer Yes or No.

$x + y =$ _____

Solution? (Yes or No) _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • BACKWARD 1 • Q5/5

1061

**QUESTION ID**

**S29-B1-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**32; No**

**GRADING NOTE** Form required: both. 32 > 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • BACKWARD 1 • Q5/5

SET 29 • HSA-CED.A.3

# BACKWARD 2

Test whether a point satisfies an inequality • Grade 6 • Expressions & Equations • 6.EE.B.5

**ANSWER KEY** Reduced copy of the question page, with its answer

Does W = 8 satisfy W(30 − W) ≥ 200? Find W(30 − W), then answer Yes or No.

W(30 − W) = _____

Satisfies? (Yes or No) _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 29 • BACKWARD 2 • Q1/5

1063

**QUESTION ID**

**S29-B2-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**176; No**

**GRADING NOTE** Form required: both. 8 · 22 = 176 < 200.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 29 • BACKWARD 2 • Q1/5

1063

**ANSWER KEY** Reduced copy of the question page, with its answer

Does p = 25 satisfy p(500 − 10p) ≥ 6000? Find p(500 − 10p), then answer Yes or No.

p(500 − 10p) = _____

Satisfies? (Yes or No) _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 29 • BACKWARD 2 • Q2/5

1064

QUESTION ID

**S29-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**6250; Yes**

GRADING NOTE Form required: both. 25 · 250 = 6250 ≥ 6000.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 29 • BACKWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

(4, 12) satisfies $xy \geq 50$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 29 • BACKWARD 2 • Q3/5

1065

QUESTION ID

**S29-B2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE $48 < 50$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 29 • BACKWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Which x satisfies $x^2 \geq 8$?

**A.** 3

**B.** 2

**C.** 1

**D.** 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 29 • BACKWARD 2 • Q4/5

1066

QUESTION ID

**S29-B2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**A. 3**

**GRADING NOTE** $9 \geq 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 29 • BACKWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Does t = 10 satisfy $1000(1.05)^t \geq 1500$? Find $1000(1.05)^{10}$ to the nearest cent, then answer Yes or No. A calculator may be used.

$1000(1.05)^{10}$ = _____

Satisfies? (Yes or No) _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 29 • BACKWARD 2 • Q5/5

1067

**QUESTION ID**

**S29-B2-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**1628.89; Yes**

**GRADING NOTE** Form required: both. $1000(1.05)^{10} \approx 1628.89 \geq 1500$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 29 • BACKWARD 2 • Q5/5

SET 29 • HSA-CED.A.3

# BACKWARD 3

Solve quadratic inequalities • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve $W^2 - 30W + 200 \leq 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 29 • BACKWARD 3 • Q1/5

1069

**QUESTION ID**

**S29-B3-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**$10 \leq W \leq 20$**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 29 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve $p^2 - 50p + 600 \leq 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 29 • BACKWARD 3 • Q2/5

1070

**QUESTION ID**

**S29-B3-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**$20 \leq p \leq 30$**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 29 • BACKWARD 3 • Q2/5

**True or false?**

$x^2 \geq 8$ with $x > 0$ means $x \geq 4$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 29 • BACKWARD 3 • Q3/5

1071

QUESTION ID

**S29-B3-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** $x \geq \sqrt{8} = 2\sqrt{2} \approx 2.83$; for example $x = 3$ works.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 29 • BACKWARD 3 • Q3/5

The solution of $(x - 2)(x - 6) \leq 0$ is

**A.** $x = 2$ or $x = 6$

**B.** $2 \leq x \leq 6$

**C.** $x \leq 2$ or $x \geq 6$

**D.** $x \geq 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 29 • BACKWARD 3 • Q4/5

1072

**QUESTION ID**

**S29-B3-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. $2 \leq x \leq 6$**

**GRADING NOTE** Between the roots.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 29 • BACKWARD 3 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve $32/x^2 \leq 4$ for $x > 0$.

**A.** $0 < x \leq 2\sqrt{2}$

**B.** $x \geq 8$

**C.** $x \geq \sqrt{2}$

**D.** $x \geq 2\sqrt{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 29 • BACKWARD 3 • Q5/5

1073

**QUESTION ID**

**S29-B3-Q5**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. $x \geq 2\sqrt{2}$**

**GRADING NOTE** $32 \leq 4x^2$ for $x > 0$ gives $x^2 \geq 8$, so $x \geq \sqrt{8} = 2\sqrt{2}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B3-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 29 • BACKWARD 3 • Q5/5

SET 29 • HSA-CED.A.3

# FORWARD 1

Optimize within nonlinear constraints • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3

A rectangular garden has length L and width W (feet) with L + W = 30. Which width gives the greatest area, and what is that area?

W (ft) = _____

Greatest area (ft²) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 1 • Q1/5

1075

**QUESTION ID**

## S29-F1-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**15; 225**

**GRADING NOTE** Form required: both. A = W(30 − W) has its vertex at W = 15, and 15 · 15 = 225.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 1 • Q1/5

At a ticket price of p dollars, revenue is p(500 − 10p) dollars (0 ≤ p ≤ 50). Which price gives the greatest revenue, and what is that revenue?

p ($) = _____

Greatest revenue ($) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 1 • Q2/5

1076

**QUESTION ID**

**S29-F1-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**25; 6250**

**GRADING NOTE** Form required: both. The zeros are 0 and 50; the vertex is at p = 25, revenue 25 · 250 = 6250.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S29-F1-Q2**

**High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 1 • Q2/5**

**True or false?**

A maximum found from the model must still satisfy every other constraint to be a viable choice.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 1 • Q3/5

1077

QUESTION ID
**S29-F1-Q3**

TYPE
True or false

CORRECT ANSWER
**A. True**

**GRADING NOTE** Check all constraints.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 1 • Q3/5

For the pen with $xy \geq 50$ and $2x + 2y \leq 40$, which is viable?

**A.** (10, 10)

**B.** (15, 10)

**C.** (2, 20)

**D.** (4, 12)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 1 • Q4/5

1078

QUESTION ID

**S29-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**A. (10, 10)**

**GRADING NOTE** (10, 10): area 100, perimeter 40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 1 • Q4/5

A rectangular garden has length L and width W (feet) with L + W = 30, where L and W are positive whole numbers. The area must be at least 216 ft². Give all viable widths as a compound inequality.

Viable W: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 1 • Q5/5

1079

**QUESTION ID**

**S29-F1-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**$12 \le W \le 18$**

**GRADING NOTE** Form required: the compound inequality (W a whole number). $W(30 - W) \ge 216$ gives $W^2 - 30W + 216 \le 0$, $(W - 12)(W - 18) \le 0$.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S29-F1-Q5**

**High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 1 • Q5/5**

SET 29 • HSA-CED.A.3

# FORWARD 2

Locate constraint boundaries as graph intersections • High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

A garden has width W and area $y = W(30 - W)$ ft². How do the graphs of $y = W(30 - W)$ and $y = 200$ show that the widths with area at least 200 ft² are $10 \le W \le 20$?

**A.** The line $y = 200$ meets the parabola only at its vertex, $W = 15$, so 15 ft is the only width.

**B.** The parabola crosses the W-axis at $W = 10$ and $W = 20$, so those are the boundary widths.

**C.** The parabola is below the line $y = 200$ between $W = 10$ and $W = 20$, and above it outside.

**D.** The graphs intersect at $W = 10$ and $W = 20$, and the parabola is on or above the line $y = 200$ between them.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 29 • FORWARD 2 • Q1/5

1081

**QUESTION ID**

**S29-F2-Q1**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. The graphs intersect at W = 10 and W = 20, and the parabola is on or above the line y = 200 between them.**

**GRADING NOTE** Viable W are where the area graph is at or above 200. The others use the zeros (0 and 30), reverse the region, or misplace the intersections.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 29 • FORWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Where do y = p(500 − 10p) and y = 6000 intersect?
Give both values of p.

p = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 29 • FORWARD 2 • Q2/5

1082

**QUESTION ID**
**S29-F2-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**20 or 30**

**GRADING NOTE** Form required: both, in any order. $10p^2 - 500p + 6000 = 0$ gives $(p - 20)(p - 30) = 0$.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S29-F2-Q2**

**High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 29 • FORWARD 2 • Q2/5**

**True or false?**

The x-coordinates where $y = 32/x^2$ meets $y = 4$ solve $32/x^2 = 4$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 29 • FORWARD 2 • Q3/5

1083

QUESTION ID

**S29-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Intersection.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 29 • FORWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

The graphs of $y = 1000(1.05)^t$ and $y = 1500$ meet at $t \approx$

**A.** 15

**B.** 8.3

**C.** 10

**D.** 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 29 • FORWARD 2 • Q4/5

1084

QUESTION ID
**S29-F2-Q4**

TYPE
Multiple choice

CORRECT ANSWER
**B. 8.3**

**GRADING NOTE** log 1.5/log 1.05 ≈ 8.31.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 29 • FORWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A garden of width W ft has area $A = W(30 - W)$ ft² ($0 < W < 30$). On axes with W horizontal and area vertical, the parabola $y = W(30 - W)$ and the line $y = 200$ are drawn. Why do the widths where the parabola is on or above the line give the viable widths for an area of at least 200 ft²?

**A.** At each such W, the height of the parabola is the area $W(30 - W)$, and it is at least 200.

**B.** At each such W, the parabola is below 200, so the area is small enough.

**C.** At each such W, the width W is at least 200.

**D.** At each such W, the area equals exactly 200.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 29 • FORWARD 2 • Q5/5

1085

**QUESTION ID**
**S29-F2-Q5**

**TYPE**
Multiple choice

**CORRECT ANSWER**
**A. At each such W, the height of the parabola is the area W(30 − W), and it is at least 200.**

**GRADING NOTE** Points on the parabola have height equal to the area. The others confuse "at least" with "equal", reverse the region, or read the wrong axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 29 • FORWARD 2 • Q5/5

# Rearrange rational, radical, and exponential formulas

| | | |
|---|---|---|
| **MAIN** | Algebra II main question | **HSA-CED.A.4** |
| **BACKWARD 1** | Rearrange linear formulas | **HSA-CED.A.4** |
| **BACKWARD 2** | Undo squares and square roots | **8.EE.A.2** |
| **BACKWARD 3** | Clear fractions in an equation | **HSA-REI.B.3** |
| **FORWARD 1** | Rearrange formulas where conditions decide the result | **HSA-CED.A.4** |
| **FORWARD 2** | Rearrange exponential formulas with logarithms | **HSF-LE.A.4** |

SET 30 • HSA-CED.A.4

# MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4

The lens formula is $1/f = 1/a + 1/b$, where a, b, and f are nonzero and $a \neq f$. Solve for b.

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • MAIN • Q1/5

1088

**QUESTION ID**
**S30-M-Q1**

**TYPE**
Short answer

**CORRECT ANSWER**
**a·f/(a − f)**

**GRADING NOTE** Any equivalent expression. $1/b = 1/f - 1/a = (a - f)/(af)$, so $b = af/(a - f)$. Taking the reciprocal needs $a - f \neq 0$, which is why $a \neq f$ is required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A pendulum's period is $T = 2\pi\sqrt{(L/g)}$, where $g > 0$ and $L \geq 0$ (so $T \geq 0$). Solve for L.

L = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • MAIN • Q2/5

1089

**QUESTION ID**

**S30-M-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**$g \cdot T^2/(4\pi^2)$**

**GRADING NOTE** Any equivalent expression. $T/(2\pi) = \sqrt{(L/g)}$; squaring is reversible here because both sides are nonnegative, so $L/g = T^2/(4\pi^2)$.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S30-M-Q2**

**High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • MAIN • Q2/5**

An investment grows by $A = P(1 + r)^t$, where $P > 0$, $A > 0$, the growth base $1 + r$ is positive, and $t \neq 0$. Solve for r.

**A.** $r = (A - P)^{1/t}$

**B.** $r = (A/P)^t - 1$

**C.** $r = (A/P)^{1/t} - 1$

**D.** $r = (A/P - 1)^{1/t}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • MAIN • Q3/5

1090

QUESTION ID

**S30-M-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**C. $r = (A/P)^{1/t} - 1$**

**GRADING NOTE** $(1 + r)^t = A/P$ with $1 + r > 0$, so $1 + r = (A/P)^{1/t}$, the unique positive root.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • MAIN • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

The speed of a falling object is $v = \sqrt{(2gh)}$, where $g > 0$ and $h \geq 0$ (so $v \geq 0$). Solve for h.

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • MAIN • Q4/5

1091

QUESTION ID

**S30-M-Q4**

TYPE

Short answer

CORRECT ANSWER

**$v^2/(2g)$**

**GRADING NOTE** Any equivalent expression. Square both sides (both are nonnegative): $v^2 = 2gh$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • MAIN • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

In a circuit, $I = V/(R + r)$, where $R + r \neq 0$ and $I \neq 0$.
Solve for R.

R = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • MAIN • Q5/5

1092

**QUESTION ID**
**S30-M-Q5**

**TYPE**
Short answer

**CORRECT ANSWER**
**V/I − r**

**GRADING NOTE** Any equivalent expression. I(R + r) = V; dividing by I needs I ≠ 0 (if I = 0 then V = 0 and R cannot be isolated); R = V/I − r.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S30-M-Q5**

**High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • MAIN • Q5/5**

SET 30 • HSA-CED.A.4

# BACKWARD 1

Rearrange linear formulas • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve P = 2L + 2W for W.

W = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • BACKWARD 1 • Q1/5

1094

QUESTION ID

**S30-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**(P − 2L)/2**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve d = rt for t.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • BACKWARD 1 • Q2/5

1095

QUESTION ID

**S30-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**d/r**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • BACKWARD 1 • Q2/5

**True or false?**

Solving V = IR for R gives R = VI.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • BACKWARD 1 • Q3/5

1096

QUESTION ID

**S30-B1-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Divide both sides by I: R = V/I.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • BACKWARD 1 • Q3/5

Solving y = mx + b for x gives

**A.** (y − b)/m

**B.** y/m − b

**C.** (y + b)/m

**D.** m(y − b)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • BACKWARD 1 • Q4/5

1097

QUESTION ID

**S30-B1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**A. (y − b)/m**

**GRADING NOTE** Subtract b, divide by m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • BACKWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve A = (1/2)bh for h.

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • BACKWARD 1 • Q5/5

1098

QUESTION ID

**S30-B1-Q5**

TYPE

Short answer

CORRECT ANSWER

**2A/b**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • BACKWARD 1 • Q5/5

SET 30 • HSA-CED.A.4

# BACKWARD 2

Undo squares and square roots • Grade 8 • Expressions & Equations • 8.EE.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve $x^2 = 49$ for $x > 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 30 • BACKWARD 2 • Q1/5

1100

QUESTION ID

**S30-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**7**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 30 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve $\sqrt{y} = 6$.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 30 • BACKWARD 2 • Q2/5

1101

QUESTION ID

**S30-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**36**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 30 • BACKWARD 2 • Q2/5

**True or false?**

If √(L/g) = k, then L/g = √k.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q3

**Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 30 • BACKWARD 2 • Q3/5**

1102

**QUESTION ID**

## S30-B2-Q3

**TYPE**

True or false

**CORRECT ANSWER**

## B. False

**GRADING NOTE** Square both sides: L/g = k².

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q3

**Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 30 • BACKWARD 2 • Q3/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

$(\sqrt{(2gh)})^2$ equals

**A.** $2g\sqrt{h}$

**B.** $2gh$

**C.** $\sqrt{(2gh)}$

**D.** $4g^2h^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 30 • BACKWARD 2 • Q4/5

1103

QUESTION ID

**S30-B2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**B. 2gh**

**GRADING NOTE** Squaring undoes the root.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 30 • BACKWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve T/(2π) = √m for m.

m = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 30 • BACKWARD 2 • Q5/5

1104

QUESTION ID

**S30-B2-Q5**

TYPE

Short answer

CORRECT ANSWER

**$T^2/(4\pi^2)$**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 30 • BACKWARD 2 • Q5/5

SET 30 • HSA-CED.A.4

# BACKWARD 3

Clear fractions in an equation • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve 1/2 + 1/x = 1 for x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 30 • BACKWARD 3 • Q1/5

1106

QUESTION ID

**S30-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**2**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 30 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Write 1/a − 1/f as one fraction with denominator af (a and f nonzero). Give the numerator.

Numerator: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 30 • BACKWARD 3 • Q2/5

1107

QUESTION ID
**S30-B3-Q2**

TYPE
Short answer

CORRECT ANSWER
**f − a**

**GRADING NOTE** Form required: the numerator. 1/a − 1/f = f/(af) − a/(af) = (f − a)/(af).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 30 • BACKWARD 3 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

If $1/b = (a - f)/(af)$, then $b = af/(a - f)$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 30 • BACKWARD 3 • Q3/5

1108

QUESTION ID

**S30-B3-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Take reciprocals.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 30 • BACKWARD 3 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Solving 3 = 12/(R + 2) for R gives

**A.** 36

**B.** 4

**C.** 6

**D.** 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 30 • BACKWARD 3 • Q4/5

1109

**QUESTION ID**

**S30-B3-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. 2**

**GRADING NOTE** R + 2 = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 30 • BACKWARD 3 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve I = V/(R + r) for R + r.

R + r = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 30 • BACKWARD 3 • Q5/5

1110

**QUESTION ID**

**S30-B3-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**V/I**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 30 • BACKWARD 3 • Q5/5

SET 30 • HSA-CED.A.4

# FORWARD 1

Rearrange formulas where conditions decide the result • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve $A = \pi r^2$ for r, where r is a length ($r \geq 0$) and $A \geq 0$.

**A.** $r = \sqrt{A/\pi}$

**B.** $r = \sqrt{(A/\pi)}$

**C.** $r = \pm\sqrt{(A/\pi)}$

**D.** $r = A/(2\pi)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • FORWARD 1 • Q1/5

1112

**QUESTION ID**

**S30-F1-Q1**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. $r = \sqrt{(A/\pi)}$**

**GRADING NOTE** Divide by π, then take the nonnegative root; a length cannot be negative.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S30-F1-Q1**

**High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • FORWARD 1 • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve $y = (x + 1)/(x - 2)$ for $x$, where $x \neq 2$ and $y \neq 1$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • FORWARD 1 • Q2/5

1113

QUESTION ID

**S30-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**$(2y + 1)/(y - 1)$**

**GRADING NOTE** Any equivalent expression. $y(x - 2) = x + 1$ gives $xy - x = 2y + 1$, $x(y - 1) = 2y + 1$; dividing needs $y \neq 1$ ($y = 1$ would give $0 = 3$, no $x$).

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S30-F1-Q2**

**High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • FORWARD 1 • Q2/5**

**True or false?**

Solving $s = \sqrt{(24d)}$ for d gives $d = s^2/24$ for every real s.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • FORWARD 1 • Q3/5

1114

QUESTION ID
**S30-F1-Q3**

TYPE
True or false

CORRECT ANSWER
**B. False**

**GRADING NOTE** Only $s \geq 0$ comes from the original formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • FORWARD 1 • Q3/5

Solving $E = (1/2)mv^2$ for $v$ ($v \geq 0$) gives

**A.** $\sqrt{(2E/m)}$

**B.** $(2E/m)^2$

**C.** $\sqrt{(E/2m)}$

**D.** $2E/m$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • FORWARD 1 • Q4/5

1115

**QUESTION ID**

## S30-F1-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. $\sqrt{(2E/m)}$**

**GRADING NOTE** Multiply by 2/m and take the positive root.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S30-F1-Q4**

**High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • FORWARD 1 • Q4/5**

Solve $P = k/V^2$ for V, where $k > 0$, $P > 0$, and $V > 0$.

**A.** $V = k/(2P)$

**B.** $V = \pm\sqrt{(k/P)}$

**C.** $V = \sqrt{(k/P)}$

**D.** $V = \sqrt{(P/k)}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • FORWARD 1 • Q5/5

1116

QUESTION ID

**S30-F1-Q5**

TYPE

Multiple choice

CORRECT ANSWER

**C. $V = \sqrt{(k/P)}$**

**GRADING NOTE** $V^2 = k/P$, and $V > 0$ selects the positive root.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S30-F1-Q5**

**High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • FORWARD 1 • Q5/5**

SET 30 • HSA-CED.A.4

# FORWARD 2

Rearrange exponential formulas with logarithms • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

$A = Pe^{rt}$, where $P \neq 0$, $A/P > 0$, and $r \neq 0$. Which formula gives t?

**A.** $t = \ln(A/P)/r$

**B.** $t = \ln(A - P)/r$

**C.** $t = r \ln(A/P)$

**D.** $t = (A/P)/(r\ e)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S30-F2-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 30 • FORWARD 2 • Q1/5

1118

**QUESTION ID**

**S30-F2-Q1**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. t = ln(A/P)/r**

**GRADING NOTE** Divide by P ($P \neq 0$): $e^{rt} = A/P$; take ln (allowed because $A/P > 0$): $rt = \ln(A/P)$; divide by r ($r \neq 0$). The others subtract instead of dividing, do not use a logarithm, or multiply by r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S30-F2-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 30 • FORWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

$A = P(1 + r)^t$, where $P > 0$, $A > 0$, and the growth base $1 + r$ is positive and not 1. Which formula gives t?

**A.** $t = (A/P)/(1 + r)$

**B.** $t = \log(A/P)/\log(1 + r)$

**C.** $t = \log(A/P) - \log(1 + r)$

**D.** $t = \log(A - P)/\log(1 + r)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 30 • FORWARD 2 • Q2/5

1119

QUESTION ID

**S30-F2-Q2**

TYPE

Multiple choice

CORRECT ANSWER

**B. $t = \log(A/P)/\log(1 + r)$**

**GRADING NOTE** Divide by P, take logarithms ($A/P > 0$), and divide by $\log(1 + r)$, which is nonzero because $1 + r \neq 1$. The others subtract instead of dividing, skip the logarithm, or subtract the logarithms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 30 • FORWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

Solving $N = N_0 \cdot 2^{t/d}$ for t gives $t = d \cdot \log_2(N/N_0)$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 30 • FORWARD 2 • Q3/5

1120

QUESTION ID

**S30-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Take log base 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 30 • FORWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Solving $y = 5 \cdot 10^x$ for x gives

**A.** y/50

**B.** log(y/5)

**C.** $10^{y/5}$

**D.** log y − 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 30 • FORWARD 2 • Q4/5

1121

**QUESTION ID**

## S30-F2-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

## B. log(y/5)

**GRADING NOTE** $10^x = y/5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 30 • FORWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Use t = log(A/P)/log(1 + r) to find t for A = 2000, P = 1000, r = 0.05. Round to the nearest hundredth. A calculator may be used.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 30 • FORWARD 2 • Q5/5

1122

**QUESTION ID**

## S30-F2-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**14.21**

**GRADING NOTE** Rounding to 0.01. log 2/log 1.05 ≈ 14.2067.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 30 • FORWARD 2 • Q5/5

SET 31 • HSA-REI.A.2 • CREATING EQUATIONS AND REASONING WITH EQUATIONS

# Solve rational equations and identify extraneous solutions

| | | |
|---|---|---|
| **MAIN** | Algebra II main question | **HSA-REI.A.2** |
| **BACKWARD 1** | Solve proportions | **7.RP.A.2** |
| **BACKWARD 2** | Find a common denominator of rational expressions | **HSA-APR.D.7** |
| **BACKWARD 3** | Values that make a denominator zero | **7.NS.A.2.b** |
| **FORWARD 1** | Explain and construct extraneous solutions | **HSA-REI.A.2** |
| **FORWARD 2** | See rational equations as graph intersections | **HSA-REI.D.11** |

SET 31 • HSA-REI.A.2

# MAIN

Algebra II main question • High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve 3/(x − 1) = 6/(x + 2). Give every value found after clearing the denominators, and the actual solutions (enter none if there are none).

Values found after clearing: _____

Actual solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • MAIN • Q1/5

1125

**QUESTION ID**

**S31-M-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**{4}; {4}**

**GRADING NOTE** Form required: both. 3(x + 2) = 6(x − 1) gives x = 4, which does not make a denominator 0, so it is a solution; nothing is extraneous.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve $x/(x - 2) + 1 = 2/(x - 2)$. Give every value found after clearing the denominators, and the actual solutions (enter none if there are none).

Values found after clearing: _____

Actual solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • MAIN • Q2/5

1126

QUESTION ID

**S31-M-Q2**

TYPE

Short answer

CORRECT ANSWER

**{2}; none**

GRADING NOTE Form required: both. $x + (x - 2) = 2$ gives $x = 2$, which makes the denominators 0, so it is extraneous and there is no solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • MAIN • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve 1/x + 1/(x + 2) = 4/3. Give every value found after clearing the denominators, and the actual solutions (enter none if there are none).

Values found after clearing: _____

Actual solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • MAIN • Q3/5

1127

**QUESTION ID**

**S31-M-Q3**

**TYPE**

Short answer

**CORRECT ANSWER**

**{1, −3/2}; {1, −3/2}**

**GRADING NOTE** Form required: both. $3(x + 2) + 3x = 4x(x + 2)$ gives $2x^2 + x - 3 = 0$, so $x = 1$ or $x = -3/2$; neither makes a denominator 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • MAIN • Q3/5

Solve $x^2/(x - 3) = 9/(x - 3)$. Give every value found after clearing the denominators, and the actual solutions (enter none if there are none).

Values found after clearing: _____

Actual solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • MAIN • Q4/5

1128

**QUESTION ID**

## S31-M-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**{3, −3}; {−3}**

**GRADING NOTE** Form required: both. $x^2 = 9$ gives $x = \pm 3$; $x = 3$ makes the denominator 0, so it is extraneous.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • MAIN • Q4/5

Solve $2/(x + 1) - 1/(x - 1) = 2/(x^2 - 1)$. Give every value found after clearing the denominators, and the actual solutions (enter none if there are none).

Values found after clearing: _____

Actual solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • MAIN • Q5/5

1129

**QUESTION ID**

**S31-M-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**{5}; {5}**

**GRADING NOTE** Form required: both. $2(x - 1) - (x + 1) = 2$ gives $x = 5$, which is allowed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • MAIN • Q5/5

SET 31 • HSA-REI.A.2

# BACKWARD 1

Solve proportions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve 3/x = 6/10.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 31 • BACKWARD 1 • Q1/5

1131

QUESTION ID

**S31-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**5**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 31 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve 4/7 = x/21.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 31 • BACKWARD 1 • Q2/5

1132

QUESTION ID

**S31-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**12**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 31 • BACKWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

$a/b = c/d$ implies $ac = bd$ when $b, d \neq 0$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 31 • BACKWARD 1 • Q3/5

1133

QUESTION ID

## S31-B1-Q3

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Cross-multiplying gives ad = bc.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 31 • BACKWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

2/5 = 6/x gives x =

**A.** 12

**B.** 2.4

**C.** 3

**D.** 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 31 • BACKWARD 1 • Q4/5

1134

QUESTION ID

**S31-B1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**D. 15**

GRADING NOTE 2x = 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 31 • BACKWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve $3/(x - 1) = 1$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 31 • BACKWARD 1 • Q5/5

1135

QUESTION ID

**S31-B1-Q5**

TYPE

Short answer

CORRECT ANSWER

**4**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 31 • BACKWARD 1 • Q5/5

SET 31 • HSA-REI.A.2

# BACKWARD 2

Find a common denominator of rational expressions • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.7

**ANSWER KEY** Reduced copy of the question page, with its answer

What is the LCD of 1/x and 1/(x + 2)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 31 • BACKWARD 2 • Q1/5

1137

QUESTION ID
**S31-B2-Q1**

TYPE
Short answer

CORRECT ANSWER
**x(x + 2)**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 31 • BACKWARD 2 • Q1/5

What is the LCD of $\frac{2}{(x + 1)}$, $\frac{1}{(x - 1)}$, and $\frac{2}{(x^2 - 1)}$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 31 • BACKWARD 2 • Q2/5

1138

QUESTION ID
**S31-B2-Q2**

TYPE
Short answer

CORRECT ANSWER
**(x + 1)(x − 1)**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 31 • BACKWARD 2 • Q2/5

**True or false?**

The LCD of $1/(x-2)$ and $2/(x-2)$ is $(x-2)^2$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S31-B2-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 31 • BACKWARD 2 • Q3/5

1139

QUESTION ID
**S31-B2-Q3**

TYPE
True or false

CORRECT ANSWER
**B. False**

**GRADING NOTE** It is $x-2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S31-B2-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 31 • BACKWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Multiplying $1/x + 1/(x + 2) = 4/3$ by $3x(x + 2)$ gives

**A.** $3 + 3 = 4$

**B.** $3(x + 2) + 3x = 4$

**C.** $3(x + 2) + 3x = 4x(x + 2)$

**D.** $x + 2 + x = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q4

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 31 • BACKWARD 2 • Q4/5

1140

QUESTION ID
**S31-B2-Q4**

TYPE
Multiple choice

CORRECT ANSWER
**C.**
**$3(x + 2) + 3x = 4x(x + 2)$**

**GRADING NOTE** Each term times the LCD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q4

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 31 • BACKWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Write 1/x + 1/(x + 2) as one fraction.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 31 • BACKWARD 2 • Q5/5

1141

QUESTION ID
**S31-B2-Q5**

TYPE
Short answer

CORRECT ANSWER
**(2x + 2)/(x(x + 2))**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 31 • BACKWARD 2 • Q5/5

SET 31 • HSA-REI.A.2

# BACKWARD 3

Values that make a denominator zero • Grade 7 • The Number System • 7.NS.A.2.b

**ANSWER KEY** Reduced copy of the question page, with its answer

Which x-values make x/(x − 2) undefined?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 31 • BACKWARD 3 • Q1/5

1143

QUESTION ID

**S31-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**2**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 31 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Which x-values make $2/(x^2 - 1)$ undefined?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 31 • BACKWARD 3 • Q2/5

1144

QUESTION ID

**S31-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**1 or −1**

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 31 • BACKWARD 3 • Q2/5

**True or false?**

A rational equation cannot have a solution that makes any denominator zero.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 31 • BACKWARD 3 • Q3/5

1145

QUESTION ID

**S31-B3-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** The expression is undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 31 • BACKWARD 3 • Q3/5

Which value is excluded from 1/x + 1/(x + 2) = 4/3?

**A.** 2

**B.** 1

**C.** 4/3

**D.** −2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 31 • BACKWARD 3 • Q4/5

1146

**QUESTION ID**

**S31-B3-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. −2**

**GRADING NOTE** x + 2 = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 31 • BACKWARD 3 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

List the excluded values of $x^2/(x - 3) = 9/(x - 3)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 31 • BACKWARD 3 • Q5/5

1147

QUESTION ID

**S31-B3-Q5**

TYPE

Short answer

CORRECT ANSWER

**3**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B3-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 31 • BACKWARD 3 • Q5/5

SET 31 • HSA-REI.A.2

# FORWARD 1

Explain and construct extraneous solutions • High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

Solving $x/(x - 2) + 1 = 2/(x - 2)$, Leo multiplies both sides by $x - 2$ and gets $x = 2$. Why is $x = 2$ extraneous?

**A.** $x = 2$ is extraneous because it is positive, and solutions of rational equations must be negative.

**B.** $x = 2$ makes both sides equal after multiplying by $x - 2$, so it is a solution of the original equation and is not extraneous.

**C.** Leo made an arithmetic error when he multiplied; the correct solution of the equation is $x = -2$.

**D.** Multiplying by $x - 2$ is not reversible when $x - 2 = 0$, so the new equation can be true at $x = 2$ even though the original equation is undefined there.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • FORWARD 1 • Q1/5

1149

**QUESTION ID**

## S31-F1-Q1

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. Multiplying by $x - 2$ is not reversible when $x - 2 = 0$, so the new equation can be true at $x = 2$ even though the original equation is undefined there.**

**GRADING NOTE** At $x = 2$ the original denominators are 0. The others give irrelevant reasons, invent another solution ($x = -2$ gives $-2/(-4) + 1 = 1/2 \neq -1/2$), or ignore the undefined denominators.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • FORWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Which rational equation has x = 4 as an extraneous solution and no actual solutions?

**A.** 1/(x − 4) = 1

**B.** x/(x − 4) = 2

**C.** x/(x − 4) = 4/(x − 4)

**D.** (x − 4)/x = 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • FORWARD 1 • Q2/5

1150

QUESTION ID

**S31-F1-Q2**

TYPE

Multiple choice

CORRECT ANSWER

**C.**

**x/(x − 4) = 4/(x − 4)**

**GRADING NOTE** Clearing x − 4 gives x = 4, which makes the denominators 0. The others have actual solutions: x = 8, x = 4 (allowed there, since x is the denominator), and x = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • FORWARD 1 • Q2/5

**True or false?**

Every rational equation has at least one extraneous solution.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • FORWARD 1 • Q3/5

1151

QUESTION ID

**S31-F1-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** 3/(x − 1) = 6/(x + 2) has none.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • FORWARD 1 • Q3/5

For which a does a/(x − 1) = 2/(x − 1) have infinitely many solutions?

**A.** a = 2

**B.** no value

**C.** a = 0

**D.** a = 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • FORWARD 1 • Q4/5

1152

QUESTION ID

**S31-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**A. a = 2**

**GRADING NOTE** Then every x ≠ 1 works.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • FORWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Which is the solution set of
$(x + 1)/(x - 1) = 2/(x - 1) + 1$?

**A.** $x = 1$ only

**B.** Every real number except 1

**C.** Every real number

**D.** No solution

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • FORWARD 1 • Q5/5

1153

QUESTION ID
**S31-F1-Q5**

TYPE
Multiple choice

CORRECT ANSWER
**B. Every real number except 1**

**GRADING NOTE** For $x \neq 1$, $(x + 1)/(x - 1) - 2/(x - 1) = (x - 1)/(x - 1) = 1$, so the equation is true for every allowed x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • FORWARD 1 • Q5/5

SET 31 • HSA-REI.A.2

# FORWARD 2

See rational equations as graph intersections • High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

Why does $x/(x - 2) + 1 = 2/(x - 2)$ have no solution, in terms of the graphs of $y = x/(x - 2) + 1$ and $y = 2/(x - 2)$?

**A.** For $x \neq 2$ the left side minus the right side is $(2x - 4)/(x - 2) = 2$, so the graphs never meet.

**B.** The graphs meet at $x = 2$, where both expressions have the same value, so $x = 2$ is a solution.

**C.** The two graphs are the same curve, so every $x$ except 2 is a solution of the equation.

**D.** The graphs cross at $x = 0$ and nowhere else, so $x = 0$ is the only solution of the equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • FORWARD 2 • Q1/5

1155

QUESTION ID
**S31-F2-Q1**

TYPE
Multiple choice

CORRECT ANSWER
**A. For $x \neq 2$ the left side minus the right side is $(2x - 4)/(x - 2) = 2$, so the graphs never meet.**

**GRADING NOTE** The difference of the two functions is the constant 2 on their common domain. The others use the excluded $x = 2$ or false claims (at $x = 0$ the sides are 1 and $-1$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • FORWARD 2 • Q1/5

Where do y = 3/(x − 1) and y = 6/(x + 2) intersect?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • FORWARD 2 • Q2/5

1156

**QUESTION ID**

**S31-F2-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**(4, 1)**

**GRADING NOTE** Form required: the point.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S31-F2-Q2**

**High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • FORWARD 2 • Q2/5**

**True or false?**

An extraneous solution corresponds to an intersection point of the two graphs.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • FORWARD 2 • Q3/5

1157

QUESTION ID

## S31-F2-Q3

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Neither graph is defined there.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • FORWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

$y = x^2/(x - 3)$ and $y = 9/(x - 3)$ intersect at

**A.** (3, 3/2)

**B.** (−3, 3/2)

**C.** (−3, −3/2)

**D.** (3, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • FORWARD 2 • Q4/5

1158

**QUESTION ID**

**S31-F2-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. (−3, −3/2)**

**GRADING NOTE** $9/(-6) = -3/2$.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S31-F2-Q4**

**High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • FORWARD 2 • Q4/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Use a graph or table to approximate the solution of $1/x = x - 1$ with $x > 0$ to the nearest hundredth. A calculator may be used.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • FORWARD 2 • Q5/5

1159

**QUESTION ID**

**S31-F2-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**1.62**

**GRADING NOTE** Any value 1.61 to 1.62 ($(1 + \sqrt{5})/2 \approx 1.618$).

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S31-F2-Q5**

**High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • FORWARD 2 • Q5/5**

SET 32 • HSA-REI.A.2 • CREATING EQUATIONS AND REASONING WITH EQUATIONS

# Solve radical equations and identify extraneous solutions

| | | |
|---|---|---|
| **MAIN** | Algebra II main question | **HSA-REI.A.2** |
| **BACKWARD 1** | Square and cube both sides; principal roots | **8.EE.A.2** |
| **BACKWARD 2** | Solve the resulting quadratic | **HSA-REI.B.4.b** |
| **BACKWARD 3** | Check a candidate in the original equation | **6.EE.B.5** |
| **FORWARD 1** | Solve equations with two radicals, rational exponents, and $\sqrt{(x^2)}$ | **HSA-REI.A.2** |
| **FORWARD 2** | Explain extraneous solutions with graphs of radical functions | **HSF-IF.C.7.b** |

SET 32 • HSA-REI.A.2

# MAIN

Algebra II main question • High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

Solve √(x + 7) = x − 5. Give every value found after squaring, and the actual solutions (enter none if there are none).

Values found after squaring: _____

Actual solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • MAIN • Q1/5

1162

**QUESTION ID**

**S32-M-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**{9, 2}; {9}**

**GRADING NOTE** Form required: both. $x + 7 = x^2 - 10x + 25$ gives $x^2 - 11x + 18 = 0$; $x = 2$ gives $\sqrt{9} = 3 \neq -3$, so it is extraneous.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • MAIN • Q1/5

Solve √(2x − 1) = 3. Give every value found after squaring, and the actual solutions (enter none if there are none).

Values found after squaring: _____

Actual solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • MAIN • Q2/5

1163

**QUESTION ID**

**S32-M-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**{5}; {5}**

**GRADING NOTE** Form required: both. 2x − 1 = 9; √9 = 3 checks.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • MAIN • Q2/5

Solve √(x + 3) + 3 = x. Give every value found after squaring, and the actual solutions (enter none if there are none).

Values found after squaring: _____

Actual solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • MAIN • Q3/5

1164

**QUESTION ID**

**S32-M-Q3**

**TYPE**

Short answer

**CORRECT ANSWER**

**{6, 1}; {6}**

**GRADING NOTE** Form required: both. $x + 3 = (x - 3)^2$ gives $x^2 - 7x + 6 = 0$; $x = 1$ gives $\sqrt{4} + 3 = 5 \neq 1$, so it is extraneous.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • MAIN • Q3/5

Solve $\sqrt[3]{(x - 2)} = -3$. Give every value found after cubing, and the actual solutions (enter none if there are none).

Values found after cubing: _____

Actual solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • MAIN • Q4/5

1165

**QUESTION ID**

**S32-M-Q4**

**TYPE**

Short answer

**CORRECT ANSWER**

**{−25}; {−25}**

**GRADING NOTE** Form required: both. $x - 2 = -27$. Cubing is reversible, so no extraneous solutions appear.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • MAIN • Q4/5

Solve √(x − 1) + 5 = 2. Give every value found after squaring, and the actual solutions (enter none if there are none).

Values found after squaring: _____

Actual solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • MAIN • Q5/5

1166

**QUESTION ID**

## S32-M-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**{10}; none**

**GRADING NOTE** Form required: both. √(x − 1) = −3 is impossible; squaring gives x − 1 = 9, x = 10, which gives 8 ≠ 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • MAIN • Q5/5

SET 32 • HSA-REI.A.2

# BACKWARD 1

Square and cube both sides; principal roots • Grade 8 • Expressions & Equations • 8.EE.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve $\sqrt{y} = 3$.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 32 • BACKWARD 1 • Q1/5

1168

QUESTION ID

**S32-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**9**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 32 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve $\sqrt[3]{y} = -3$.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 32 • BACKWARD 1 • Q2/5

1169

QUESTION ID

**S32-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**−27**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 32 • BACKWARD 1 • Q2/5

**True or false?**

$\sqrt{9} = 3$ or $-3$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q3

**Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 32 • BACKWARD 1 • Q3/5**

1170

**QUESTION ID**

## S32-B1-Q3

**TYPE**

True or false

**CORRECT ANSWER**

## B. False

**GRADING NOTE** The principal square root is 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q3

**Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 32 • BACKWARD 1 • Q3/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

$\sqrt{y} = -2$ has

**A.** $y = 2$

**B.** no solution

**C.** $y = 4$

**D.** $y = -4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 32 • BACKWARD 1 • Q4/5

1171

QUESTION ID

**S32-B1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**B. no solution**

**GRADING NOTE** A principal square root is never negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 32 • BACKWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Square both sides of $\sqrt{(x + 7)} = x - 5$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 32 • BACKWARD 1 • Q5/5

1172

QUESTION ID

**S32-B1-Q5**

TYPE

Short answer

CORRECT ANSWER

**$x + 7 = x^2 - 10x + 25$**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 32 • BACKWARD 1 • Q5/5

SET 32 • HSA-REI.A.2

# BACKWARD 2

Solve the resulting quadratic • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve $x^2 - 11x + 18 = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • BACKWARD 2 • Q1/5

1174

QUESTION ID

**S32-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**2 or 9**

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve $x^2 - 7x + 6 = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • BACKWARD 2 • Q2/5

1175

QUESTION ID

**S32-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**1 or 6**

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • BACKWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

$(x - 5)^2 = x^2 + 25.$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • BACKWARD 2 • Q3/5

1176

QUESTION ID

**S32-B2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE $x^2 - 10x + 25.$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • BACKWARD 2 • Q3/5

$x^2 - 6x + 9 = x + 3$ gives

**A.** $x^2 - 7x + 12 = 0$

**B.** $x^2 - 5x + 12 = 0$

**C.** $x^2 - 6x + 6 = 0$

**D.** $x^2 - 7x + 6 = 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • BACKWARD 2 • Q4/5

1177

**QUESTION ID**

## S32-B2-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. $x^2 - 7x + 6 = 0$**

**GRADING NOTE** Subtract x + 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • BACKWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve $x^2 - 4x = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • BACKWARD 2 • Q5/5

1178

QUESTION ID

**S32-B2-Q5**

TYPE

Short answer

CORRECT ANSWER

**0 or 4**

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • BACKWARD 2 • Q5/5

SET 32 • HSA-REI.A.2

# BACKWARD 3

Check a candidate in the original equation • Grade 6 • Expressions & Equations • 6.EE.B.5

**ANSWER KEY** Reduced copy of the question page, with its answer

Check x = 2 in √(x + 7) = x − 5: evaluate each side, then say whether x = 2 is a solution.

Left side: _____

Right side: _____

Solution? (Yes or No) _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 32 • BACKWARD 3 • Q1/5

1180

QUESTION ID

**S32-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**3; −3; No**

GRADING NOTE Form required: all three. √9 = 3 but 2 − 5 = −3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 32 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Check x = 6 in √(x + 3) + 3 = x: evaluate each side, then say whether x = 6 is a solution.

Left side: _____

Right side: _____

Solution? (Yes or No) _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 32 • BACKWARD 3 • Q2/5

1181

QUESTION ID

**S32-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**6; 6; Yes**

GRADING NOTE Form required: all three. √9 + 3 = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 32 • BACKWARD 3 • Q2/5

**True or false?**

$x = 4$ satisfies $\sqrt{(2x - 1)} = 3$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 32 • BACKWARD 3 • Q3/5

1182

QUESTION ID

**S32-B3-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** $\sqrt{(8 - 1)} = \sqrt{7} \neq 3$; the solution is $x = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 32 • BACKWARD 3 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Which value satisfies √(x + 3) + 3 = x?

**A.** 1

**B.** 0

**C.** 6

**D.** −3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 32 • BACKWARD 3 • Q4/5

1183

**QUESTION ID**

**S32-B3-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. 6**

**GRADING NOTE** Check each.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 32 • BACKWARD 3 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Check x = 10 in √(x − 1) + 5 = 2: evaluate each side, then say whether x = 10 is a solution.

Left side: _____

Right side: _____

Solution? (Yes or No) _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 32 • BACKWARD 3 • Q5/5

1184

**QUESTION ID**

**S32-B3-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**8; 2; No**

**GRADING NOTE** Form required: all three. √9 + 5 = 8 ≠ 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 32 • BACKWARD 3 • Q5/5

SET 32 • HSA-REI.A.2

# FORWARD 1

Solve equations with two radicals, rational exponents, and $\sqrt{(x^2)}$ • High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve $\sqrt{(x + 5)} - \sqrt{x} = 1$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • FORWARD 1 • Q1/5

1186

QUESTION ID

**S32-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**4**

**GRADING NOTE** Any equivalent form. $\sqrt{(x + 5)} = 1 + \sqrt{x}$ gives $\sqrt{x} = 2$; check $\sqrt{9} - 2 = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • FORWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve $x^{3/2} = 8$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • FORWARD 1 • Q2/5

1187

QUESTION ID

**S32-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**4**

GRADING NOTE Any equivalent form. $x = 8^{2/3}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • FORWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve $\sqrt{(x^2)} = 5$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • FORWARD 1 • Q3/5

1188

QUESTION ID

**S32-F1-Q3**

TYPE

Short answer

CORRECT ANSWER

**5 or −5**

**GRADING NOTE** Form required: both. $\sqrt{(x^2)} = |x|$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • FORWARD 1 • Q3/5

**True or false?**

$\sqrt{(x^2)} = x$ for every real x.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • FORWARD 1 • Q4/5

1189

QUESTION ID

**S32-F1-Q4**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE $\sqrt{(x^2)} = |x|$; for example $x = -3$ gives 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • FORWARD 1 • Q4/5

The solution set of $\sqrt{(x^2)} = x$ is

**A.** $x \geq 0$

**B.** $x \leq 0$

**C.** $x = 0$ only

**D.** all real x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • FORWARD 1 • Q5/5

1190

QUESTION ID

**S32-F1-Q5**

TYPE

Multiple choice

CORRECT ANSWER

**A. x ≥ 0**

**GRADING NOTE** $|x| = x$ exactly when $x \geq 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • FORWARD 1 • Q5/5

SET 32 • HSA-REI.A.2

# FORWARD 2

Explain extraneous solutions with graphs of radical functions • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b

**ANSWER KEY** Reduced copy of the question page, with its answer

The graphs of $y = \sqrt{(x + 7)}$ and $y = x - 5$ meet at one point. Give it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 32 • FORWARD 2 • Q1/5

1192

QUESTION ID
**S32-F2-Q1**

TYPE
Short answer

CORRECT ANSWER
**(9, 4)**

GRADING NOTE Form required: the point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 32 • FORWARD 2 • Q1/5

Squaring $\sqrt{(x + 7)} = x - 5$ gives $x + 7 = (x - 5)^2$, with solutions $x = 9$ and $x = 2$. Squaring cannot tell $\sqrt{(x + 7)}$ from $-\sqrt{(x + 7)}$. At $x = 2$, the line $y = x - 5$ meets which graph?

**A.** $y = (x - 5)^2$

**B.** $y = x + 7$

**C.** $y = \sqrt{(x + 7)}$

**D.** $y = -\sqrt{(x + 7)}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 32 • FORWARD 2 • Q2/5

1193

QUESTION ID

**S32-F2-Q2**

TYPE

Multiple choice

CORRECT ANSWER

**D. $y = -\sqrt{(x + 7)}$**

**GRADING NOTE** At $x = 2$, $x - 5 = -3 = -\sqrt{9}$, so the line meets $y = -\sqrt{(x + 7)}$; the original equation uses $+\sqrt{}$, so $x = 2$ is extraneous. At $x = 9$ the line meets $y = \sqrt{(x + 7)}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 32 • FORWARD 2 • Q2/5

**True or false?**

The graph of $y = \sqrt{(x + 7)}$ starts at (7, 0) and increases.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 32 • FORWARD 2 • Q3/5

1194

QUESTION ID
**S32-F2-Q3**

TYPE
True or false

CORRECT ANSWER
**B. False**

**GRADING NOTE** $x + 7 = 0$ at $x = -7$, so it starts at $(-7, 0)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 32 • FORWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

The graph of $y = \sqrt{(x - 1)} + 5$ lies

**A.** below $y = 2$

**B.** below the x-axis

**C.** at or below $y = 5$

**D.** at or above $y = 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 32 • FORWARD 2 • Q4/5

1195

**QUESTION ID**

**S32-F2-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. at or above y = 5**

**GRADING NOTE** $\sqrt{} \geq 0$.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S32-F2-Q4**

**High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 32 • FORWARD 2 • Q4/5**

Why does $\sqrt{(x - 1)} + 5 = 2$ have no solution, in terms of the graphs of $y = \sqrt{(x - 1)} + 5$ and $y = 2$?

**A.** $y = \sqrt{(x - 1)} + 5$ is always below the line $y = 2$, so the graphs never meet.

**B.** The graph of $y = \sqrt{(x - 1)} + 5$ is not defined for any x, so there is nothing to meet $y = 2$.

**C.** The graphs meet at $x = 10$, because squaring $\sqrt{(x - 1)} = -3$ gives $x = 10$.

**D.** $y = \sqrt{(x - 1)} + 5$ is always at least 5 (a square root is never negative), so it never meets the line $y = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 32 • FORWARD 2 • Q5/5

1196

**QUESTION ID**

## S32-F2-Q5

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. $y = \sqrt{(x - 1)} + 5$ is always at least 5 (a square root is never negative), so it never meets the line $y = 2$.**

**GRADING NOTE** The radical graph starts at (1, 5) and rises. The others use the extraneous $x = 10$ ($\sqrt{9} + 5 = 8 \neq 2$), reverse the position, or misstate the domain ($x \geq 1$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 32 • FORWARD 2 • Q5/5

# Explain why intersections of nonlinear graphs solve f(x) = g(x)

| | | |
|---|---|---|
| **MAIN** | Algebra II main question | **HSA-REI.D.11** |
| **BACKWARD 1** | A graph is the set of its solutions | **HSA-REI.D.10** |
| **BACKWARD 2** | Solve a linear system by graphing | **8.EE.C.8.a** |
| **BACKWARD 3** | Evaluate nonlinear functions at an input | **HSF-IF.A.2** |
| **FORWARD 1** | Relate intersections to zeros of f - g | **HSA-REI.D.11** |
| **FORWARD 2** | Find all solutions of a trigonometric equation on an interval from a… | **HSF-TF.B.7** |

SET 33 • HSA-REI.D.11

# MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

**ANSWER KEY** Reduced copy of the question page, with its answer

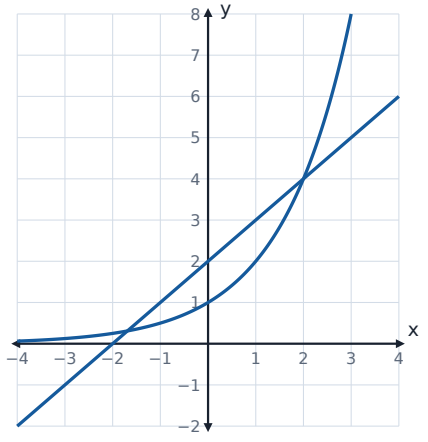
The graphs of $y = 2^x$ and $y = x + 2$ are shown. They meet at $x = 2$ and at a negative $x$. Why does $x = 2$ solve $2^x = x + 2$?

**A.** Because $2^x$ and $x + 2$ are equal for every $x \geq 0$, including $x = 2$.

**B.** At an intersection both functions have the same output for the same input; at $x = 2$, $2^2 = 4 = 2 + 2$.

**C.** Because $x = 2$ is where the graph of $y = 2^x$ crosses the x-axis.

**D.** Because the y-coordinate of the intersection point is 2, not its x-coordinate.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • MAIN • Q1/5

1199

QUESTION ID
**S33-M-Q1**

TYPE
Multiple choice

CORRECT ANSWER
**B. At an intersection both functions have the same output for the same input; at $x = 2$, $2^2 = 4 = 2 + 2$.**

**GRADING NOTE** Reading the x-coordinate of an intersection solves $f(x) = g(x)$. The others read the wrong coordinate ($y = 4$ there), overgeneralize (at $x = 0$: $1 \neq 2$), or state a false fact ($2^x$ is never 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

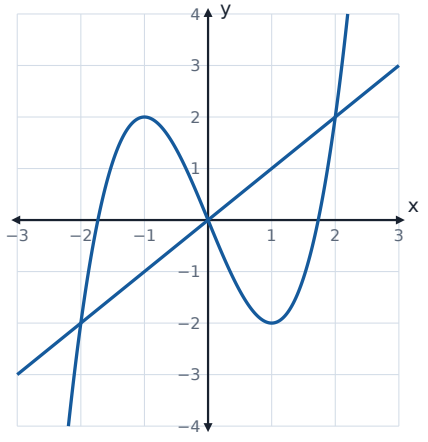
The graphs of $y = x^3 - 3x$ and $y = x$ are shown. Which gives every solution of $x^3 - 3x = x$ with a correct reason?

**A.** $x = 0$ only: that is where both graphs pass through the origin.

**B.** $x = -\sqrt{3}, 0, \sqrt{3}$: these are where $y = x^3 - 3x$ crosses the x-axis.

**C.** $x = -2, 0, 2$: the graphs meet at these x-values, where the two outputs are equal.

**D.** $y = -2, 0, 2$: these are the heights where the graphs meet.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • MAIN • Q2/5

1200

**QUESTION ID**

## S33-M-Q2

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. $x = -2, 0, 2$: the graphs meet at these x-values, where the two outputs are equal.**

**GRADING NOTE** The solutions are the x-coordinates of all intersections. The others give heights, use the zeros of one function, or miss two intersections.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • MAIN • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

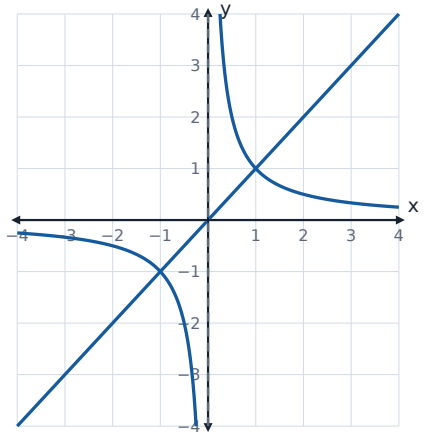
The graphs of y = 1/x and y = x are shown. Which gives every solution of 1/x = x with a correct reason?

**A.** x = 0: the graphs are closest near the origin, where 1/x and x both approach 0.

**B.** x = 1 only: the graphs meet only in the first quadrant, at (1, 1).

**C.** x = −1 and x = 1: the graphs meet at (−1, −1) and (1, 1), where 1/x and x are equal.

**D.** There is no solution, because y = 1/x has a vertical asymptote at x = 0.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • MAIN • Q3/5

1201

QUESTION ID

**S33-M-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**C. x = −1 and x = 1: the graphs meet at (−1, −1) and (1, 1), where 1/x and x are equal.**

**GRADING NOTE** Both intersections count. The others miss one, use the excluded x = 0, or confuse an asymptote with having no intersections.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • MAIN • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

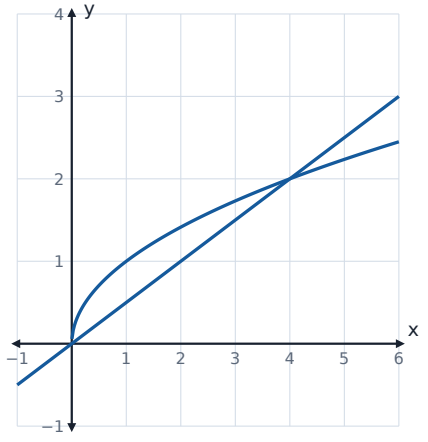
The graphs of $y = \sqrt{x}$ and $y = x/2$ are shown. Which gives every solution of $\sqrt{x} = x/2$ with a correct check?

**A.** $x = 0$ and $x = 4$: $\sqrt{0} = 0 = 0/2$ and $\sqrt{4} = 2 = 4/2$.

**B.** $x = 2$ and $x = 4$: these are the heights and the widths of the intersection.

**C.** $x = 4$ only: $\sqrt{4} = 2 = 4/2$; the origin does not count.

**D.** $x = 0$ and $x = 2$: $\sqrt{2} = 1 = 2/2$.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • MAIN • Q4/5

1202

QUESTION ID
**S33-M-Q4**

TYPE
Multiple choice

CORRECT ANSWER
**A. $x = 0$ and $x = 4$: $\sqrt{0} = 0 = 0/2$ and $\sqrt{4} = 2 = 4/2$.**

**GRADING NOTE** Both intersections are solutions, and each checks. The others drop $x = 0$, mix coordinates, or use a false check ($\sqrt{2} \approx 1.41$).

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S33-M-Q4**

**High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • MAIN • Q4/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

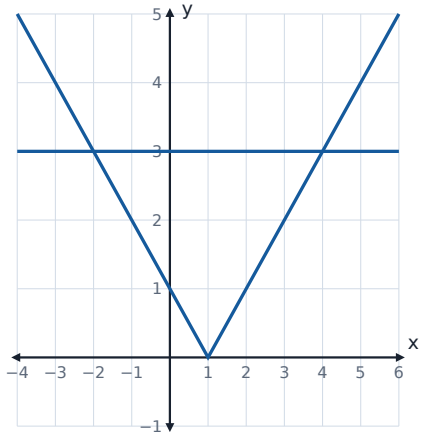
The graphs of $y = |x - 1|$ and $y = 3$ are shown. Which gives every solution of $|x - 1| = 3$ with a correct reason?

**A.** $x = -4$ and $x = 2$: these x-values are 3 units from 1 on the number line.

**B.** $x = -2$ and $x = 4$: the V-shaped graph is at height 3 at these x-values.

**C.** $x = 3$: the horizontal line is at height 3, so $x = 3$ is the solution.

**D.** $x = 4$ only: $|x - 1|$ is positive only for $x > 1$, so only one solution.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • MAIN • Q5/5

1203

**QUESTION ID**

## S33-M-Q5

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. $x = -2$ and $x = 4$: the V-shaped graph is at height 3 at these x-values.**

**GRADING NOTE** Both intersections count. The others miss one, give a height, or compute distances from −1 instead of 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • MAIN • Q5/5

SET 33 • HSA-REI.D.11

# BACKWARD 1

A graph is the set of its solutions • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10

**ANSWER KEY** Reduced copy of the question page, with its answer

Is the point (2, 4) on the graph of $y = 2^x$? (Yes or No)

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 33 • BACKWARD 1 • Q1/5

1205

**QUESTION ID**

## S33-B1-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**Yes**

**GRADING NOTE** $2^2 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 33 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Is the point (3, 6) on the graph of y = x + 2? (Yes or No)

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 33 • BACKWARD 1 • Q2/5

1206

**QUESTION ID**

**S33-B1-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**No**

**GRADING NOTE** $3 + 2 = 5 \neq 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 33 • BACKWARD 1 • Q2/5

**True or false?**

Every point on the graph of y = f(x) has the form (x, f(x)).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 33 • BACKWARD 1 • Q3/5

1207

QUESTION ID
**S33-B1-Q3**

TYPE
True or false

CORRECT ANSWER
**A. True**

**GRADING NOTE** Definition of the graph.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 33 • BACKWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Which point is on $y = x^3 - 3x$?

**A.** (0, 3)

**B.** (2, 6)

**C.** (2, 2)

**D.** (1, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 33 • BACKWARD 1 • Q4/5

1208

**QUESTION ID**

## S33-B1-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. (2, 2)**

**GRADING NOTE** $8 - 6 = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 33 • BACKWARD 1 • Q4/5

Is the point (−1, −1) on the graph of y = 1/x? (Yes or No)

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 33 • BACKWARD 1 • Q5/5

1209

**QUESTION ID**

**S33-B1-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**Yes**

**GRADING NOTE** 1/(−1) = −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 33 • BACKWARD 1 • Q5/5

SET 33 • HSA-REI.D.11

# BACKWARD 2

Solve a linear system by graphing • Grade 8 • Expressions & Equations • 8.EE.C.8.a

**ANSWER KEY** Reduced copy of the question page, with its answer

The lines $y = x + 1$ and $y = -x + 5$ meet at $(2, 3)$. Why does $x = 2$ solve $x + 1 = -x + 5$?

**A.** Because the two lines have opposite slopes, 1 and −1.

**B.** Because the y-coordinate of the intersection point (2, 3) is 2.

**C.** At $x = 2$ both expressions equal 3, so both sides are equal.

**D.** Because $x = 2$ is where both lines cross the x-axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 33 • BACKWARD 2 • Q1/5

1211

QUESTION ID

**S33-B2-Q1**

TYPE

Multiple choice

CORRECT ANSWER

**C. At x = 2 both expressions equal 3, so both sides are equal.**

GRADING NOTE An intersection point is on both graphs, so both outputs are equal there. The others misread the coordinates or give an irrelevant fact.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 33 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Where do y = 2x and y = 6 meet?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 33 • BACKWARD 2 • Q2/5

1212

QUESTION ID

**S33-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**(3, 6)**

GRADING NOTE Form required: the point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 33 • BACKWARD 2 • Q2/5

**True or false?**

Two lines that do not intersect give an equation with no solution.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 33 • BACKWARD 2 • Q3/5

1213

QUESTION ID

**S33-B2-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Parallel lines.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 33 • BACKWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

$y = 3$ and $y = x - 1$ meet at $x =$

**A.** 4

**B.** 3

**C.** $-1$

**D.** 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 33 • BACKWARD 2 • Q4/5

1214

QUESTION ID

**S33-B2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**A. 4**

GRADING NOTE $x - 1 = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 33 • BACKWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve x + 1 = −x + 5 algebraically.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 33 • BACKWARD 2 • Q5/5

1215

QUESTION ID

**S33-B2-Q5**

TYPE

Short answer

CORRECT ANSWER

**2**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 33 • BACKWARD 2 • Q5/5

SET 33 • HSA-REI.D.11

# BACKWARD 3

Evaluate nonlinear functions at an input • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

Find $2^x$ and $x + 2$ when $x = 2$.

$2^x =$ _____

$x + 2 =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 33 • BACKWARD 3 • Q1/5

1217

QUESTION ID

**S33-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**4; 4**

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 33 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Find $x^3 - 3x$ when $x = -2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 33 • BACKWARD 3 • Q2/5

1218

QUESTION ID

**S33-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**−2**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 33 • BACKWARD 3 • Q2/5

**True or false?**

$\sqrt{9} = 9/2$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 33 • BACKWARD 3 • Q3/5

1219

QUESTION ID

**S33-B3-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** $\sqrt{9} = 3$, but $9/2 = 4.5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 33 • BACKWARD 3 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

$|x - 1|$ when $x = -2$ is

**A.** 3

**B.** 1

**C.** $-1$

**D.** $-3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 33 • BACKWARD 3 • Q4/5

1220

**QUESTION ID**

**S33-B3-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. 3**

**GRADING NOTE** $|-3|$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 33 • BACKWARD 3 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Find 1/x when x = −1.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 33 • BACKWARD 3 • Q5/5

1221

QUESTION ID

**S33-B3-Q5**

TYPE

Short answer

CORRECT ANSWER

**−1**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 33 • BACKWARD 3 • Q5/5

SET 33 • HSA-REI.D.11

# FORWARD 1

Relate intersections to zeros of f − g • High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

Why are the solutions of $x^3 - 3x = x$ the zeros of $h(x) = x^3 - 4x$?

**A.** Because $h(x) = f(x) \cdot g(x)$ and a product is 0 when a factor is 0.

**B.** Because adding x to both sides gives $x^3 - 4x = 0$.

**C.** Because $x^3 - 4x$ has three terms.

**D.** $f(x) = g(x)$ exactly when $f(x) - g(x) = 0$, and $(x^3 - 3x) - x = x^3 - 4x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • FORWARD 1 • Q1/5

1223

**QUESTION ID**

**S33-F1-Q1**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. $f(x) = g(x)$ exactly when $f(x) - g(x) = 0$, and $(x^3 - 3x) - x = x^3 - 4x$.**

**GRADING NOTE** Subtracting g(x) from both sides. The others give a non-reason, an algebra error (adding x gives $x^3 - 2x = 2x$), or the wrong combination.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • FORWARD 1 • Q1/5

Find the zeros of $h(x) = x^3 - 4x$ algebraically.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • FORWARD 1 • Q2/5

1224

**QUESTION ID**

**S33-F1-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**−2, 0, 2**

**GRADING NOTE** Form required: all three. x(x − 2)(x + 2).

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S33-F1-Q2**

**High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • FORWARD 1 • Q2/5**

**True or false?**

If y = f(x) and y = g(x) meet at x = a, then the graph of y = f(x) − g(x) crosses or touches the x-axis at a.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • FORWARD 1 • Q3/5

1225

QUESTION ID

## S33-F1-Q3

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** f(a) − g(a) = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • FORWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

The solutions of $\sqrt{x} = x/2$ are the zeros of

**A.** $\sqrt{x} + x/2$

**B.** $\sqrt{x} - x/2$

**C.** $\sqrt{x} \cdot x/2$

**D.** $x/2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • FORWARD 1 • Q4/5

1226

QUESTION ID
**S33-F1-Q4**

TYPE
Multiple choice

CORRECT ANSWER
**B. $\sqrt{x} - x/2$**

**GRADING NOTE** Subtract.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S33-F1-Q4**

**High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • FORWARD 1 • Q4/5**

$h(x) = 2^x - x - 2$, with $h(-2) = 0.25$, $h(0) = -1$, $h(3) = 3$. h decreases and then increases (it has one turning point). Why does $2^x = x + 2$ have exactly two solutions?

**A.** h is continuous and changes sign twice (on −2 to 0 and 0 to 3), and with one turning point it has at most two zeros.

**B.** Every equation that sets an exponential function equal to a line has exactly two solutions, so $2^x = x + 2$ has two.

**C.** h(−2) and h(3) are both positive, so h cannot change sign anywhere between them, and therefore it has no zeros at all.

**D.** h has degree 2 because of the $2^x$ term, so like every quadratic it has exactly two zeros.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • FORWARD 1 • Q5/5

1227

**QUESTION ID**
**S33-F1-Q5**

**TYPE**
Multiple choice

**CORRECT ANSWER**
**A. h is continuous and changes sign twice (on −2 to 0 and 0 to 3), and with one turning point it has at most two zeros.**

**GRADING NOTE** Sign changes give at least two zeros; one turning point allows at most two. The others ignore the sign changes, overgeneralize ($2^x = 1$ has one solution), or treat h as a polynomial.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S33-F1-Q5**

**High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • FORWARD 1 • Q5/5**

SET 33 • HSA-REI.D.11

# FORWARD 2

Find all solutions of a trigonometric equation on an interval from a graph • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7

**ANSWER KEY** Reduced copy of the question page, with its answer

On $0 \le x < 2\pi$, where does $y = \sin x$ meet $y = 1/2$? Give exact values.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 33 • FORWARD 2 • Q1/5

1229

**QUESTION ID**

## S33-F2-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

## π/6 and 5π/6

**GRADING NOTE** Form required: both.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S33-F2-Q1**

**High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 33 • FORWARD 2 • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

On $0 \le x < 2\pi$, where does $y = \cos x$ meet $y = 0$?

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 33 • FORWARD 2 • Q2/5

1230

**QUESTION ID**

## S33-F2-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

## π/2 and 3π/2

**GRADING NOTE** Form required: both.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S33-F2-Q2**

**High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 33 • FORWARD 2 • Q2/5**

**True or false?**

sin x = 1/2 has exactly two real solutions.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 33 • FORWARD 2 • Q3/5

1231

QUESTION ID

## S33-F2-Q3

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** It has two solutions in each period, so infinitely many real solutions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 33 • FORWARD 2 • Q3/5

How many solutions does $\sin x = 0.3$ have on $0 \le x < 2\pi$?

**A.** 1

**B.** 0

**C.** 2

**D.** 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 33 • FORWARD 2 • Q4/5

1232

**QUESTION ID**

**S33-F2-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. 2**

**GRADING NOTE** The horizontal line crosses twice in one period.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S33-F2-Q4**

**High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 33 • FORWARD 2 • Q4/5**

Why does sin x = 2 have no solution?

**A.** The graph of y = sin x meets y = 2 only at x = π/2.

**B.** sin x = 2 has solutions only when x is in degrees.

**C.** The graph of y = sin x is not periodic, so it never returns to the height 2.

**D.** The graph of y = sin x stays between −1 and 1, so it never meets y = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 33 • FORWARD 2 • Q5/5

1233

**QUESTION ID**

## S33-F2-Q5

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. The graph of y = sin x stays between −1 and 1, so it never meets y = 2.**

**GRADING NOTE** The range of sine is [−1, 1]. The others misread sin(π/2) = 1, confuse units, or state a false fact.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 33 • FORWARD 2 • Q5/5

# Approximate solutions of f(x) = g(x)

| | | |
|---|---|---|
| **MAIN** | Algebra II main question | **HSA-REI.D.11** |
| **BACKWARD 1** | Read solutions of linear-quadratic systems from graphs | **HSA-REI.D.11** |
| **BACKWARD 2** | Evaluate exponential and logarithmic expressions with a calculat… | **HSF-IF.A.2** |
| **BACKWARD 3** | Round to a place value | **5.NBT.A.4** |
| **FORWARD 1** | Guarantee the accuracy of an approximation | **HSA-REI.D.11** |
| **FORWARD 2** | Compare an approximation with an exact logarithmic solution | **HSF-LE.A.4** |

SET 34 • HSA-REI.D.11

# MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

**ANSWER KEY** Reduced copy of the question page, with its answer

Use a table or successive approximation to solve $2^x = 5 - x$ to the nearest hundredth. A calculator may be used.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • MAIN • Q1/5

1236

**QUESTION ID**
**S34-M-Q1**

**TYPE**
Short answer

**CORRECT ANSWER**
**1.72**

**GRADING NOTE** Accept 1.71 to 1.72 (exact root 1.7156...).

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S34-M-Q1**

**High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • MAIN • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve $x^3 = 7 - x$ to the nearest hundredth with a table or successive approximation. A calculator may be used.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • MAIN • Q2/5

1237

**QUESTION ID**
**S34-M-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**1.74**

**GRADING NOTE** Accept 1.73 to 1.74 (root 1.7392...).

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S34-M-Q2**

**High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • MAIN • Q2/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve log x = 2 − x (log base 10) to the nearest hundredth. A calculator may be used.

x ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • MAIN • Q3/5

1238

**QUESTION ID**
**S34-M-Q3**

**TYPE**
Short answer

**CORRECT ANSWER**
**1.76**

**GRADING NOTE** Accept 1.75 to 1.76 (root 1.7556...).

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S34-M-Q3**

**High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • MAIN • Q3/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

The table shows $f(x) = 3(0.8)^x$ and $g(x) = x$. Between which two consecutive table inputs is the solution of $f(x) = g(x)$? Then estimate the solution to the nearest tenth.

| x | 0 | 0.5 | 1 | 1.5 | 2 | 2.5 |
|---|---|---|---|---|---|---|
| f(x) | 3 | 2.683 | 2.4 | 2.147 | 1.92 | 1.717 |
| g(x) | 0 | 0.5 | 1 | 1.5 | 2 | 2.5 |

Between x = _____

and x = _____

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • MAIN • Q4/5

1239

QUESTION ID

**S34-M-Q4**

TYPE

Short answer

CORRECT ANSWER

**1.5; 2; 1.9**

GRADING NOTE Form required: all three. f − g changes sign between 1.5 (0.647) and 2 (−0.08); the root is about 1.944, so 1.9 to the nearest tenth.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • MAIN • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Use a graph or table to find the negative solution of $2^x = x + 2$ to the nearest tenth. A calculator may be used.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • MAIN • Q5/5

1240

**QUESTION ID**
**S34-M-Q5**

**TYPE**
Short answer

**CORRECT ANSWER**
**−1.7**

**GRADING NOTE** Any equivalent form (root −1.690...).

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S34-M-Q5**

**High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • MAIN • Q5/5**

SET 34 • HSA-REI.D.11

# BACKWARD 1

Read solutions of linear-quadratic systems from graphs • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

**ANSWER KEY** Reduced copy of the question page, with its answer

Where do $y = x^2$ and $y = 4$ meet? Give both x-values.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • BACKWARD 1 • Q1/5

1242

QUESTION ID

**S34-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**2 or −2**

GRADING NOTE Form required: both, in any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Where do $y = x^2$ and $y = x + 2$ meet? Give both x-values.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • BACKWARD 1 • Q2/5

1243

**QUESTION ID**

## S34-B1-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**−1 or 2**

**GRADING NOTE** Form required: both, in any order. $x^2 - x - 2 = (x - 2)(x + 1) = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • BACKWARD 1 • Q2/5

**True or false?**

If f − g is continuous on an interval and changes sign between two of its table values, then f(x) = g(x) has a solution between those two inputs.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • BACKWARD 1 • Q3/5

1244

**QUESTION ID**

## S34-B1-Q3

**TYPE**

True or false

**CORRECT ANSWER**

## A. True

**GRADING NOTE** Intermediate Value Theorem. Without continuity a sign change does not guarantee a solution (a jump can skip 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • BACKWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

f and g are continuous on [1, 2], with f(1) − g(1) = −2 and f(2) − g(2) = 1. A solution of f(x) = g(x) lies

**A.** below 1

**B.** at 1.5 exactly

**C.** above 2

**D.** between 1 and 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • BACKWARD 1 • Q4/5

1245

QUESTION ID

**S34-B1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**D. between 1 and 2**

**GRADING NOTE** f − g is continuous and changes sign on [1, 2], so it has a zero there. Without continuity (for example a jump from −2 to 1) there might be none.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • BACKWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

$f(x) = x^2$ and $g(x) = 3$. Between which two consecutive integers is the positive solution of $f(x) = g(x)$?

Smaller integer: _____

Larger integer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • BACKWARD 1 • Q5/5

1246

QUESTION ID

**S34-B1-Q5**

TYPE

Short answer

CORRECT ANSWER

**1; 2**

**GRADING NOTE** Form required: both. $1^2 = 1 < 3 < 4 = 2^2$ (the solution is $\sqrt{3} \approx 1.73$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • BACKWARD 1 • Q5/5

SET 34 • HSA-REI.D.11

# BACKWARD 2

Evaluate exponential and logarithmic expressions with a calculator • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

Find $2^{1.7}$ to the nearest hundredth. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 34 • BACKWARD 2 • Q1/5

1248

**QUESTION ID**

**S34-B2-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**3.25**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 34 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Find log 1.75 (base 10) to the nearest thousandth. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 34 • BACKWARD 2 • Q2/5

1249

**QUESTION ID**

## S34-B2-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**0.243**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 34 • BACKWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

$1.7^3 = 5.1$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 34 • BACKWARD 2 • Q3/5

1250

QUESTION ID

**S34-B2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** $1.7 \cdot 1.7 \cdot 1.7 = 4.913$ (5.1 is $3 \cdot 1.7$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 34 • BACKWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

$3(0.8)^2$ equals

**A.** 0.192

**B.** 4.8

**C.** 1.92

**D.** 2.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 34 • BACKWARD 2 • Q4/5

1251

**QUESTION ID**

**S34-B2-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. 1.92**

**GRADING NOTE** $3 \cdot 0.64$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 34 • BACKWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Find $2^{-1.7}$ to the nearest hundredth. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 34 • BACKWARD 2 • Q5/5

1252

**QUESTION ID**

**S34-B2-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**0.31**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 34 • BACKWARD 2 • Q5/5

SET 34 • HSA-REI.D.11

# BACKWARD 3

Round to a place value • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4

**ANSWER KEY** Reduced copy of the question page, with its answer

Round 1.7156 to the nearest hundredth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 34 • BACKWARD 3 • Q1/5

1254

QUESTION ID

**S34-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**1.72**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 34 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Round 1.944 to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 34 • BACKWARD 3 • Q2/5

1255

**QUESTION ID**

**S34-B3-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**1.9**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 34 • BACKWARD 3 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

1.755 rounded to the nearest hundredth is 1.76.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 34 • BACKWARD 3 • Q3/5

1256

QUESTION ID

**S34-B3-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Round half up.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 34 • BACKWARD 3 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

$-1.690$ to the nearest tenth is

**A.** $-1.6$

**B.** $-1.69$

**C.** $-2$

**D.** $-1.7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 34 • BACKWARD 3 • Q4/5

1257

QUESTION ID

**S34-B3-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**D. $-1.7$**

GRADING NOTE Nearest tenth.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 34 • BACKWARD 3 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Round 1.7392 to the nearest hundredth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 34 • BACKWARD 3 • Q5/5

1258

**QUESTION ID**

**S34-B3-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**1.74**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B3-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 34 • BACKWARD 3 • Q5/5

SET 34 • HSA-REI.D.11

# FORWARD 1

Guarantee the accuracy of an approximation • High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

**ANSWER KEY** Reduced copy of the question page, with its answer

$h(x) = 2^x + x - 5$. Find h(1.71) and h(1.72), each to 3 decimal places. A calculator may be used.

h(1.71) = _____

h(1.72) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • FORWARD 1 • Q1/5

1260

**QUESTION ID**

**S34-F1-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**−0.018; 0.014**

**GRADING NOTE** Rounding to 0.001. h(1.71) ≈ −0.0184 and h(1.72) ≈ 0.0144; h is continuous and changes sign, so h has a zero between 1.71 and 1.72.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S34-F1-Q1**

**High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • FORWARD 1 • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

$h(x) = x^3 + x - 7$ is continuous. One step of bisection on [1, 2] evaluates h(1.5). Which half contains the solution of $h(x) = 0$, and why?

**A.** [1, 1.5], because h(1.5) is closer to 0 than h(1).

**B.** [1.5, 2], because $h(1.5) = -2.125 < 0$ and $h(2) = 3 > 0$.

**C.** [1.5, 2], because 1.5 is the midpoint.

**D.** [1, 1.5], because $h(1) = -5 < 0$ and $h(1.5) = -2.125 < 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • FORWARD 1 • Q2/5

1261

QUESTION ID
**S34-F1-Q2**

TYPE
Multiple choice

CORRECT ANSWER
**B. [1.5, 2], because $h(1.5) = -2.125 < 0$ and $h(2) = 3 > 0$.**

**GRADING NOTE** Keep the half where h changes sign. The others keep a half without a sign change or give a non-reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • FORWARD 1 • Q2/5

**True or false?**

After 4 bisection steps starting from an interval of length 1, the interval has length 1/8.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • FORWARD 1 • Q3/5

1262

QUESTION ID

## S34-F1-Q3

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Each step halves the length: $(1/2)^4 = 1/16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • FORWARD 1 • Q3/5

How many bisection steps from [1, 2] guarantee an interval shorter than 0.01?

**A.** 5

**B.** 100

**C.** 10

**D.** 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • FORWARD 1 • Q4/5

1263

QUESTION ID

**S34-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**D. 7**

**GRADING NOTE** $2^7 = 128 > 100$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • FORWARD 1 • Q4/5

$h(x) = 2^x + x - 5$ is continuous and increasing, so $h(x) = 0$ has exactly one solution. $h(1.715) \approx -0.002$ and $h(1.725) \approx 0.031$. What can be concluded?

**A.** The solution is strictly between 1.715 and 1.725, so it rounds to 1.72 to two decimal places.

**B.** Nothing can be concluded about the solution, because neither h(1.715) nor h(1.725) is exactly 0.

**C.** The solution is exactly 1.72, because 1.72 is halfway between 1.715 and 1.725.

**D.** The solution could round to 1.73, because h(1.725) is positive and 1.725 rounds up to 1.73.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • FORWARD 1 • Q5/5

1264

QUESTION ID
**S34-F1-Q5**

TYPE
Multiple choice

CORRECT ANSWER
**A. The solution is strictly between 1.715 and 1.725, so it rounds to 1.72 to two decimal places.**

**GRADING NOTE** The sign change brackets the root inside the rounding interval [1.715, 1.725) (strictly inside, so no tie). The others claim exactness, misread the bracket, or ignore the sign change.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • FORWARD 1 • Q5/5

SET 34 • HSA-REI.D.11

# FORWARD 2

Compare an approximation with an exact logarithmic solution • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Solve $2^x = 5$. Give x to the nearest hundredth. A calculator may be used.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 34 • FORWARD 2 • Q1/5

1266

**QUESTION ID**

**S34-F2-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**2.32**

**GRADING NOTE** Rounding to 0.01. $x = \log_2 5 = \log 5/\log 2 \approx 2.3219$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 34 • FORWARD 2 • Q1/5

Solve $3(0.8)^x = 1$. Give x to the nearest hundredth. A calculator may be used.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 34 • FORWARD 2 • Q2/5

1267

**QUESTION ID**
**S34-F2-Q2**

**TYPE**
Short answer

**CORRECT ANSWER**
**4.92**

**GRADING NOTE** Rounding to 0.01. $(0.8)^x = 1/3$, so $x = \log(1/3)/\log 0.8 \approx 4.9233$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 34 • FORWARD 2 • Q2/5

**True or false?**

$2^x = 5 - x$ can be solved exactly with one logarithm.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 34 • FORWARD 2 • Q3/5

1268

QUESTION ID

## S34-F2-Q3

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** x appears both in an exponent and outside it; approximation is needed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 34 • FORWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Which equation can be solved exactly with logarithms?

**A.** $\log x = 2 - x$

**B.** $2^x = x + 2$

**C.** $5(1.2)^x = 30$

**D.** $3(0.8)^x = x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 34 • FORWARD 2 • Q4/5

1269

QUESTION ID

**S34-F2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**C. $5(1.2)^x = 30$**

**GRADING NOTE** Only one exponential term and no other x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 34 • FORWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

$2^x = x + 2$ has the exact solution $x = 2$ ($2^2 = 4 = 2 + 2$) and a second, negative solution. Why is the negative solution found by approximation, while $2^x = 4$ is solved exactly?

**A.** Negative numbers cannot be exact solutions of an exponential equation, so any negative solution of $2^x = x + 2$ must be approximated, while $x = 2$ is exact.

**B.** The negative solution is approximated because $2^x$ is negative for $x < 0$, so logarithms cannot be taken of it; $2^x = 4$ has a positive power, so $x = \log_2 4 = 2$.

**C.** $2^x = 4$ isolates the power, so $x = \log_2 4 = 2$; in $2^x = x + 2$ the variable is both in an exponent and outside it, and no algebra step isolates it, so the negative solution (about −1.69) is approximated.

**D.** Approximation is needed only because $x = 2$ is already a solution: once one exact solution is known, the other solution of the equation cannot also be found exactly.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 34 • FORWARD 2 • Q5/5

1270

**QUESTION ID**

## S34-F2-Q5

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. $2^x = 4$ isolates the power, so $x = \log_2 4 = 2$; in $2^x = x + 2$ the variable is both in an exponent and outside it, and no algebra step isolates it, so the negative solution (about −1.69) is approximated.**

**GRADING NOTE** Equations that mix exponential and linear terms usually have no algebraic solution, even when one root (here $x = 2$) can be found by inspection. The others make false claims ($2^x > 0$ for all $x$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 34 • FORWARD 2 • Q5/5

SET 35 • HSF-TF.A.1 • TRIGONOMETRIC FUNCTIONS

# Use radian measure as arc length on the unit circle

| | | |
|---|---|---|
| **MAIN** | Algebra II main question | **HSF-TF.A.1** |
| **BACKWARD 1** | Circumference of a circle | **7.G.B.4** |
| **BACKWARD 2** | Arc length is proportional to the angle | **HSG-C.B.5** |
| **BACKWARD 3** | Degrees as fractions of a turn | **4.MD.C.5** |
| **FORWARD 1** | Measure negative angles and angles beyond one turn | **HSF-TF.A.1** |
| **FORWARD 2** | Read sine and cosine of radian angles from the unit circle | **HSF-TF.A.2** |

SET 35 • HSF-TF.A.1

# MAIN

Algebra II main question • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1

On the unit circle, a central angle subtends an arc of length 2.5. What is the radian measure of the angle?

Radians: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • MAIN • Q1/5

1273

**QUESTION ID**

## S35-M-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**2.5**

**GRADING NOTE** Form required: the value. On the unit circle (radius 1) the radian measure equals the length of the subtended arc: $\theta = s/r = 2.5/1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Convert 150° to radians. Give an exact answer.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • MAIN • Q2/5

1274

**QUESTION ID**

## S35-M-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**5π/6**

**GRADING NOTE** Form required: exact. 150 · π/180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • MAIN • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Convert 7π/4 radians to degrees.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • MAIN • Q3/5

1275

QUESTION ID

**S35-M-Q3**

TYPE

Short answer

CORRECT ANSWER

**315°**

**GRADING NOTE** Any equivalent form. (7/4)·180.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • MAIN • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A central angle of 2π/3 radians is drawn in a circle of radius 6 cm. Find the exact length of the arc it subtends, in cm, in terms of π.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • MAIN • Q4/5

1276

**QUESTION ID**

**S35-M-Q4**

**TYPE**

Short answer

**CORRECT ANSWER**

**4π**

**GRADING NOTE** s = rθ = 6 · 2π/3 = 4π cm (about 12.57 cm).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • MAIN • Q4/5

Why does one full turn measure 2π radians?

**A.** A full turn subtends the whole unit circle, whose circumference is 2π · 1 = 2π, and radian measure equals arc length on the unit circle.

**B.** Because π radians is a quarter turn of the unit circle, so a full turn of four quarters measures 2π radians.

**C.** Because the area of the unit circle is 2π · 1 = 2π, and radian measure equals the area swept out by the angle.

**D.** Because a full turn is 360°, and 360 = 2π when degrees are written as numbers of radians.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • MAIN • Q5/5

1277

QUESTION ID

## S35-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

**A. A full turn subtends the whole unit circle, whose circumference is 2π · 1 = 2π, and radian measure equals arc length on the unit circle.**

**GRADING NOTE** Radian measure is arc length on the unit circle. The others confuse degrees and radians, use the area (π), or misstate π as a quarter turn (it is a half turn).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • MAIN • Q5/5

SET 35 • HSF-TF.A.1

# BACKWARD 1

Circumference of a circle • Grade 7 • Geometry • 7.G.B.4

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the circumference of a circle of radius 1.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 1 • Q1/5

1279

**QUESTION ID**

**S35-B1-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**2π**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the circumference of a circle of radius 6 cm.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 1 • Q2/5

1280

QUESTION ID

**S35-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**12π cm**

GRADING NOTE Any equivalent form with units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 1 • Q2/5

**True or false?**

The circumference of a circle is $\pi r^2$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 1 • Q3/5

1281

QUESTION ID

**S35-B1-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE $\pi r^2$ is the area; the circumference is $\pi d = 2\pi r$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 1 • Q3/5

Half of the unit circle has length

**A.** 2π

**B.** 1

**C.** π

**D.** π/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 1 • Q4/5

1282

QUESTION ID

**S35-B1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**C. π**

GRADING NOTE 2π/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

What length is one quarter of the unit circle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 1 • Q5/5

1283

QUESTION ID

**S35-B1-Q5**

TYPE

Short answer

CORRECT ANSWER

**π/2**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 1 • Q5/5

SET 35 • HSF-TF.A.1

# BACKWARD 2

Arc length is proportional to the angle • High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5

**ANSWER KEY** Reduced copy of the question page, with its answer

A 60° angle subtends what fraction of a full circle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q1

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 35 • BACKWARD 2 • Q1/5

1285

QUESTION ID

**S35-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**1/6**

**GRADING NOTE** Any equivalent form.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S35-B2-Q1**

**High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 35 • BACKWARD 2 • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the arc length of a 90° angle in a circle of radius 4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q2

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 35 • BACKWARD 2 • Q2/5

1286

**QUESTION ID**

**S35-B2-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**2π**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q2

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 35 • BACKWARD 2 • Q2/5

**True or false?**

Doubling the radius doubles the arc length for the same angle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q3

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 35 • BACKWARD 2 • Q3/5

1287

QUESTION ID

**S35-B2-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Arc length is proportional to the radius.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q3

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 35 • BACKWARD 2 • Q3/5

The arc length for angle θ (radians) in a circle of radius r is

**A.** θ/r

**B.** $r^2\theta$

**C.** $2\pi\theta$

**D.** rθ

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q4

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 35 • BACKWARD 2 • Q4/5

1288

QUESTION ID

**S35-B2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**D. rθ**

**GRADING NOTE** Definition of radian measure scaled by r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q4

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 35 • BACKWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

What fraction of the circle is an angle of π/3 radians?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q5

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 35 • BACKWARD 2 • Q5/5

1289

QUESTION ID

**S35-B2-Q5**

TYPE

Short answer

CORRECT ANSWER

**1/6**

**GRADING NOTE** Any equivalent form. (π/3)/(2π).

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S35-B2-Q5**

**High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 35 • BACKWARD 2 • Q5/5**

SET 35 • HSF-TF.A.1

# BACKWARD 3

Degrees as fractions of a turn • Grade 4 • Measurement & Data • 4.MD.C.5

**ANSWER KEY** Reduced copy of the question page, with its answer

How many degrees are in a full turn?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q1

Grade 4 • Measurement & Data • 4.MD.C.5 • Set 35 • BACKWARD 3 • Q1/5

1291

QUESTION ID

**S35-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**360°**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q1

Grade 4 • Measurement & Data • 4.MD.C.5 • Set 35 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

What fraction of a turn is 150°?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q2

Grade 4 • Measurement & Data • 4.MD.C.5 • Set 35 • BACKWARD 3 • Q2/5

1292

QUESTION ID

**S35-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**5/12**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q2

Grade 4 • Measurement & Data • 4.MD.C.5 • Set 35 • BACKWARD 3 • Q2/5

**True or false?**

A 315° angle is 3/4 of a turn.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q3

Grade 4 • Measurement & Data • 4.MD.C.5 • Set 35 • BACKWARD 3 • Q3/5

1293

QUESTION ID

**S35-B3-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** 315/360 = 7/8; 3/4 of a turn is 270°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q3

Grade 4 • Measurement & Data • 4.MD.C.5 • Set 35 • BACKWARD 3 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A quarter turn is

**A.** 90°

**B.** 180°

**C.** 45°

**D.** 360°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q4

Grade 4 • Measurement & Data • 4.MD.C.5 • Set 35 • BACKWARD 3 • Q4/5

1294

QUESTION ID

**S35-B3-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**A. 90°**

GRADING NOTE 360/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q4

Grade 4 • Measurement & Data • 4.MD.C.5 • Set 35 • BACKWARD 3 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

How many degrees are in 2/3 of a turn?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q5

Grade 4 • Measurement & Data • 4.MD.C.5 • Set 35 • BACKWARD 3 • Q5/5

1295

QUESTION ID

**S35-B3-Q5**

TYPE

Short answer

CORRECT ANSWER

**240°**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B3-Q5

Grade 4 • Measurement & Data • 4.MD.C.5 • Set 35 • BACKWARD 3 • Q5/5

SET 35 • HSF-TF.A.1

# FORWARD 1

Measure negative angles and angles beyond one turn • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

Give a positive angle less than $2\pi$ that is coterminal with $-\pi/3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • FORWARD 1 • Q1/5

1297

**QUESTION ID**

**S35-F1-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**$5\pi/3$**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • FORWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Give the angle between 0 and 2π that is coterminal with 13π/6.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • FORWARD 1 • Q2/5

1298

QUESTION ID

**S35-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**π/6**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • FORWARD 1 • Q2/5

**True or false?**

An arc of length 10 on the unit circle goes more than once around the circle.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • FORWARD 1 • Q3/5

1299

QUESTION ID

**S35-F1-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE $10 > 2\pi \approx 6.28$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • FORWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

An angle of $-\pi/2$ radians is traced

**A.** a quarter turn clockwise

**B.** one full turn

**C.** half a turn

**D.** a quarter turn counterclockwise

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • FORWARD 1 • Q4/5

1300

**QUESTION ID**

## S35-F1-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. a quarter turn clockwise**

**GRADING NOTE** Negative means clockwise.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • FORWARD 1 • Q4/5

1300

A point travels 10 units counterclockwise around the unit circle from (1, 0). How far past one full turn is it (along the circle)? Round to the nearest hundredth. A calculator may be used.

Distance past one turn: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • FORWARD 1 • Q5/5

1301

**QUESTION ID**

**S35-F1-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**3.72**

**GRADING NOTE** Rounding to 0.01. One turn is $2\pi \approx 6.2832$, and $10 - 2\pi \approx 3.7168$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • FORWARD 1 • Q5/5

SET 35 • HSF-TF.A.1

# FORWARD 2

Read sine and cosine of radian angles from the unit circle • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

The point of the unit circle at angle π is (cos π, sin π). Give that point.

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 35 • FORWARD 2 • Q1/5

1303

**QUESTION ID**

## S35-F2-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**(−1, 0)**

**GRADING NOTE** Form required: the ordered pair, so cos π = −1 and sin π = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 35 • FORWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Find cos(3π/2) and sin(3π/2).

cos(3π/2) = _____

sin(3π/2) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 35 • FORWARD 2 • Q2/5

1304

**QUESTION ID**

**S35-F2-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**0; −1**

**GRADING NOTE** Form required: both. The point at 3π/2 is (0, −1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 35 • FORWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

$\cos(2\pi) = 1$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 35 • FORWARD 2 • Q3/5

1305

QUESTION ID

**S35-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Back at (1, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 35 • FORWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

sin(π/2) equals

**A.** −1

**B.** 0

**C.** π/2

**D.** 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 35 • FORWARD 2 • Q4/5

1306

**QUESTION ID**

## S35-F2-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. 1**

**GRADING NOTE** Point (0, 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 35 • FORWARD 2 • Q4/5

Why do cos θ and sin θ make sense for θ = 13π/6?

**A.** 13π/6 ends at the same point as 13π/6 − π = 7π/6, so the values are those at 7π/6.

**B.** 13π/6 = 2π + π/6 ends at the same point as π/6, so cos and sin take the values at that point.

**C.** 13π/6 is greater than 2π, so cos and sin are both equal to 0 at that angle.

**D.** cos and sin are defined only for 0 ≤ θ ≤ 2π, so 13π/6 is first divided by 13.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 35 • FORWARD 2 • Q5/5

1307

QUESTION ID

**S35-F2-Q5**

TYPE

Multiple choice

CORRECT ANSWER

**B. 13π/6 = 2π + π/6 ends at the same point as π/6, so cos and sin take the values at that point.**

**GRADING NOTE** Angles that differ by 2π end at the same point on the unit circle. The others subtract π (a half turn, the opposite point), claim both values are 0, or invent a rule that angles are divided down.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 35 • FORWARD 2 • Q5/5

SET 36 • HSF-TF.A.2 • TRIGONOMETRIC FUNCTIONS

# Find exact trigonometric values beyond the first quadrant

| | | |
|---|---|---|
| **MAIN** | Algebra II main question | **HSF-TF.A.2** |
| **BACKWARD 1** | Special right triangle ratios | **HSG-SRT.C.6** |
| **BACKWARD 2** | Reference angles for radian measures | **HSF-TF.A.1** |
| **BACKWARD 3** | Signs of coordinates by quadrant | **6.NS.C.6.b** |
| **FORWARD 1** | Values for angles beyond 2π and solving for angles | **HSF-TF.A.2** |
| **FORWARD 2** | Verify the Pythagorean identity at exact values | **HSF-TF.C.8** |

SET 36 • HSF-TF.A.2

# MAIN

Algebra II main question • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

Use the unit circle to find sin(5π/6) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • MAIN • Q1/5

1310

**QUESTION ID**

**S36-M-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**1/2**

**GRADING NOTE** Form required: exact. Reference angle π/6, quadrant II.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Use the unit circle to find cos(4π/3) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • MAIN • Q2/5

1311

**QUESTION ID**

## S36-M-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**−1/2**

**GRADING NOTE** Form required: exact. Reference angle π/3, quadrant III.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • MAIN • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Use the unit circle to find tan(7π/4) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • MAIN • Q3/5

1312

**QUESTION ID**

## S36-M-Q3

**TYPE**

Short answer

**CORRECT ANSWER**

**−1**

**GRADING NOTE** Form required: exact. Point (√2/2, −√2/2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • MAIN • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Use the unit circle to find $\sin(-\pi/2)$ exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • MAIN • Q4/5

1313

QUESTION ID

**S36-M-Q4**

TYPE

Short answer

CORRECT ANSWER

**−1**

**GRADING NOTE** Form required: exact. Point (0, −1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • MAIN • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Use the unit circle to find cos(11π/6) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • MAIN • Q5/5

1314

**QUESTION ID**

**S36-M-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**√3/2**

**GRADING NOTE** Form required: exact. Reference angle π/6, quadrant IV.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • MAIN • Q5/5

SET 36 • HSF-TF.A.2

# BACKWARD 1

Special right triangle ratios • High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

**ANSWER KEY** Reduced copy of the question page, with its answer

In a 30°−60°−90° triangle with hypotenuse 1, give the side lengths.

Side opposite 30°: _____

Side opposite 60°: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q1

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 36 • BACKWARD 1 • Q1/5

1316

**QUESTION ID**

## S36-B1-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**1/2; √3/2**

**GRADING NOTE** Form required: both, exact.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S36-B1-Q1**

**High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 36 • BACKWARD 1 • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Find sin 45° exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q2

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 36 • BACKWARD 1 • Q2/5

1317

QUESTION ID
**S36-B1-Q2**

TYPE
Short answer

CORRECT ANSWER
**√2/2**

**GRADING NOTE** Any equivalent form.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S36-B1-Q2**

**High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 36 • BACKWARD 1 • Q2/5**

**True or false?**

cos 60° = √3/2.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q3

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 36 • BACKWARD 1 • Q3/5

1318

QUESTION ID

## S36-B1-Q3

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** cos 60° = 1/2; √3/2 is sin 60°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q3

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 36 • BACKWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

sin 60° equals

**A.** √2/2

**B.** √3/2

**C.** √3

**D.** 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q4

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 36 • BACKWARD 1 • Q4/5

1319

QUESTION ID
**S36-B1-Q4**

TYPE
Multiple choice

CORRECT ANSWER
**B. √3/2**

**GRADING NOTE** Opposite/hypotenuse.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S36-B1-Q4**

**High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 36 • BACKWARD 1 • Q4/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

Find tan 30° exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q5

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 36 • BACKWARD 1 • Q5/5

1320

QUESTION ID
**S36-B1-Q5**

TYPE
Short answer

CORRECT ANSWER
**√3/3**

**GRADING NOTE** Any equivalent form (1/√3).

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S36-B1-Q5**

**High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 36 • BACKWARD 1 • Q5/5**

SET 36 • HSF-TF.A.2

# BACKWARD 2

Reference angles for radian measures • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the reference angle of 5π/6.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 36 • BACKWARD 2 • Q1/5

1322

QUESTION ID

**S36-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**π/6**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 36 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the reference angle of 4π/3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 36 • BACKWARD 2 • Q2/5

1323

QUESTION ID

**S36-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**π/3**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 36 • BACKWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

The reference angle of 7π/4 is 3π/4.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 36 • BACKWARD 2 • Q3/5

1324

QUESTION ID

**S36-B2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE $2\pi - 7\pi/4 = \pi/4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 36 • BACKWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Which quadrant contains 11π/6?

**A.** I

**B.** IV

**C.** III

**D.** II

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 36 • BACKWARD 2 • Q4/5

1325

QUESTION ID

**S36-B2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**B. IV**

**GRADING NOTE** Between 3π/2 and 2π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 36 • BACKWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Convert 4π/3 to degrees.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 36 • BACKWARD 2 • Q5/5

1326

**QUESTION ID**
**S36-B2-Q5**

**TYPE**
Short answer

**CORRECT ANSWER**
**240°**

**GRADING NOTE** Any equivalent form.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S36-B2-Q5**

**High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 36 • BACKWARD 2 • Q5/5**

SET 36 • HSF-TF.A.2

# BACKWARD 3

Signs of coordinates by quadrant • Grade 6 • The Number System • 6.NS.C.6.b

In which quadrant are both coordinates negative?
Answer I, II, III, or IV.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 36 • BACKWARD 3 • Q1/5

1328

QUESTION ID
**S36-B3-Q1**

TYPE
Short answer

CORRECT ANSWER
**III**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 36 • BACKWARD 3 • Q1/5

What are the signs of the coordinates of a point in quadrant II?

**A.** both positive

**B.** both negative

**C.** x positive, y negative

**D.** x negative, y positive

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 36 • BACKWARD 3 • Q2/5

1329

**QUESTION ID**

**S36-B3-Q2**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. x negative, y positive**

**GRADING NOTE** Quadrant II is to the left of the y-axis and above the x-axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 36 • BACKWARD 3 • Q2/5

**True or false?**

A point in quadrant IV has a positive y-coordinate.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 36 • BACKWARD 3 • Q3/5

1330

QUESTION ID

**S36-B3-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Quadrant IV points have $x > 0$ and $y < 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 36 • BACKWARD 3 • Q3/5

The point $(-\sqrt{3}/2, 1/2)$ is in quadrant

**A.** I

**B.** II

**C.** III

**D.** IV

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 36 • BACKWARD 3 • Q4/5

1331

**QUESTION ID**

**S36-B3-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. II**

**GRADING NOTE** (−, +).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 36 • BACKWARD 3 • Q4/5

What are the signs of the coordinates of a point in quadrant IV?

**A.** both negative

**B.** x negative, y positive

**C.** x positive, y negative

**D.** both positive

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 36 • BACKWARD 3 • Q5/5

1332

**QUESTION ID**

**S36-B3-Q5**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. x positive, y negative**

**GRADING NOTE** Quadrant IV is to the right of the y-axis and below the x-axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B3-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 36 • BACKWARD 3 • Q5/5

SET 36 • HSF-TF.A.2

# FORWARD 1

Values for angles beyond 2π and solving for angles • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

Find sin(13π/6) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • FORWARD 1 • Q1/5

1334

QUESTION ID

**S36-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**1/2**

GRADING NOTE Form required: exact.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • FORWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Find cos(−3π/4) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • FORWARD 1 • Q2/5

1335

QUESTION ID

**S36-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**−√2/2**

GRADING NOTE Form required: exact.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • FORWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Find every θ with $0 \le \theta < 2\pi$ and $\cos\theta = -1/2$.

θ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • FORWARD 1 • Q3/5

1336

QUESTION ID

**S36-F1-Q3**

TYPE

Short answer

CORRECT ANSWER

**2π/3 and 4π/3**

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • FORWARD 1 • Q3/5

**True or false?**

tan(π/2) is undefined.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • FORWARD 1 • Q4/5

1337

QUESTION ID

## S36-F1-Q4

TYPE

True or false

CORRECT ANSWER

## A. True

GRADING NOTE cos(π/2) = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • FORWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Which angle has sin θ = −√3/2 and cos θ = 1/2?

**A.** 5π/3

**B.** 4π/3

**C.** 2π/3

**D.** 7π/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • FORWARD 1 • Q5/5

1338

**QUESTION ID**

**S36-F1-Q5**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. 5π/3**

**GRADING NOTE** Quadrant IV, reference π/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • FORWARD 1 • Q5/5

SET 36 • HSF-TF.A.2

# FORWARD 2

Verify the Pythagorean identity at exact values • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8

**ANSWER KEY** Reduced copy of the question page, with its answer

For $\theta = 5\pi/6$, find $\sin^2\theta$ and $\cos^2\theta$, then their sum.

$\sin^2\theta =$ _____

$\cos^2\theta =$ _____

Sum = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 36 • FORWARD 2 • Q1/5

1340

**QUESTION ID**

**S36-F2-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**1/4; 3/4; 1**

**GRADING NOTE** Form required: all three, exact. $\sin(5\pi/6) = 1/2$ and $\cos(5\pi/6) = -\sqrt{3}/2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 36 • FORWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

For $\theta = 7\pi/4$, find $\sin^2\theta$ and $\cos^2\theta$, then their sum.

$\sin^2\theta =$ _____

$\cos^2\theta =$ _____

Sum = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 36 • FORWARD 2 • Q2/5

1341

**QUESTION ID**

**S36-F2-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**1/2; 1/2; 1**

**GRADING NOTE** Form required: all three, exact. $\sin(7\pi/4) = -\sqrt{2}/2$ and $\cos(7\pi/4) = \sqrt{2}/2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 36 • FORWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

$\sin^2\theta + \cos^2\theta = 1$ is true only for first-quadrant angles.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 36 • FORWARD 2 • Q3/5

1342

QUESTION ID

**S36-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Every point (cos θ, sin θ) is on the unit circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 36 • FORWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

If cos θ = −1/2 and θ is in quadrant III, then sin θ =

**A.** √3/2

**B.** −√3/2

**C.** 1/2

**D.** −1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 36 • FORWARD 2 • Q4/5

1343

**QUESTION ID**

**S36-F2-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. −√3/2**

**GRADING NOTE** $\sin^2 = 3/4$, negative in III.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 36 • FORWARD 2 • Q4/5

Why do cos θ and sin θ, read from the unit circle, always satisfy $\sin^2\theta + \cos^2\theta = 1$?

**A.** Because sin θ + cos θ = 1 for every θ.

**B.** Because sin θ and cos θ are always equal.

**C.** Because the values were rounded so that they work.

**D.** The point (cos θ, sin θ) lies on the unit circle $x^2 + y^2 = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 36 • FORWARD 2 • Q5/5

1344

**QUESTION ID**

**S36-F2-Q5**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. The point (cos θ, sin θ) lies on the unit circle $x^2 + y^2 = 1$.**

**GRADING NOTE** The coordinates satisfy the circle equation. The others are false (sin θ + cos θ = √2 at π/4; the values differ at π/6) or a non-reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 36 • FORWARD 2 • Q5/5

SET 37 • HSF-TF.A.2 • TRIGONOMETRIC FUNCTIONS

# Explain the extension of sine and cosine to all real numbers

| | | |
|---|---|---|
| **MAIN** | Algebra II main question | **HSF-TF.A.2** |
| **BACKWARD 1** | Right-triangle definitions of sine and cosine | **HSG-SRT.C.6** |
| **BACKWARD 2** | Points on the unit circle | **HSG-GPE.A.1** |
| **BACKWARD 3** | Radian measure as directed arc length | **HSF-TF.A.1** |
| **FORWARD 1** | Explain where tangent is defined and how values repeat | **HSF-TF.A.2** |
| **FORWARD 2** | Explain symmetry and periodicity | **HSF-TF.A.4** |

SET 37 • HSF-TF.A.2

# MAIN

Algebra II main question • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

Which statement correctly explains how the unit circle defines cos t and sin t for every real number t?

**A.** Start at (1, 0) and move t units straight up; cos t and sin t are the coordinates of the point reached.

**B.** Start at (1, 0) and travel |t| units along the unit circle, counterclockwise if t > 0 and clockwise if t < 0; cos t and sin t are the x- and y-coordinates of the endpoint.

**C.** Travel t units counterclockwise from (1, 0); for t < 0 the values are undefined.

**D.** Draw a right triangle with an acute angle of t radians; cos t and sin t are the ratios adjacent/hypotenuse and opposite/hypotenuse, so the definition works only for t between 0 and π/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • MAIN • Q1/5

1347

QUESTION ID

**S37-M-Q1**

TYPE

Multiple choice

CORRECT ANSWER

**B. Start at (1, 0) and travel |t| units along the unit circle, counterclockwise if t > 0 and clockwise if t < 0; cos t and sin t are the x- and y-coordinates of the endpoint.**

**GRADING NOTE** Arc length |t| from (1, 0), direction by the sign of t, coordinates of the endpoint. The others use the right-triangle definition (only acute angles), move along a line instead of the circle (the point leaves the circle), or leave negative t undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Which explanation shows why sin(t + 2π) = sin t for every real number t?

**A.** For t = π/2 both sides equal 1, so they are always equal.

**B.** Sine distributes over addition, so sin(t + 2π) = sin t + sin 2π, and sin 2π = 0 because the point at 2π is (1, 0); therefore sin(t + 2π) = sin t.

**C.** Adding 2π to t reflects the endpoint on the unit circle over the x-axis, and a reflection over the x-axis keeps the y-coordinate, so sin(t + 2π) = sin t.

**D.** The unit circle has circumference 2π, so adding 2π adds one full trip around it: t and t + 2π end at the same point, which has the same y-coordinate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • MAIN • Q2/5

1348

QUESTION ID
**S37-M-Q2**

TYPE
Multiple choice

CORRECT ANSWER
**D. The unit circle has circumference 2π, so adding 2π adds one full trip around it: t and t + 2π end at the same point, which has the same y-coordinate.**

**GRADING NOTE** One full turn returns to the same point. The others distribute sine over a sum (false in general), check one value, or describe the wrong motion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • MAIN • Q2/5

Why is cos t < 0 for π/2 < t < π?

**A.** For these t the endpoint on the unit circle is in quadrant II, where x-coordinates are negative, and cos t is the x-coordinate.

**B.** For these t the endpoint is in quadrant II, where y-coordinates are negative, and cos t is the y-coordinate.

**C.** cos(2π/3) = −1/2, so cos t is negative on this interval.

**D.** On the unit circle cos t decreases steadily as t increases from 0, so once t passes π/2, where cos t = 0, the value of cos t is negative for every larger t.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • MAIN • Q3/5

1349

QUESTION ID

**S37-M-Q3**

TYPE

Multiple choice

CORRECT ANSWER

**A. For these t the endpoint on the unit circle is in quadrant II, where x-coordinates are negative, and cos t is the x-coordinate.**

**GRADING NOTE** Quadrant II and the x-coordinate. The others use the wrong coordinate and sign, a false claim (cos t increases again after π and is positive after 3π/2), or one example.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q3

**High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • MAIN • Q3/5**

Why is $-1 \le \sin t \le 1$ for every real number t?

**A.** $\sin(\pi/2) = 1$ and $\sin(3\pi/2) = -1$, so every value lies between them.

**B.** In a right triangle the hypotenuse is longer than either leg, so opposite/hypotenuse is less than 1, which means $\sin t < 1$ for every real number t.

**C.** sin t is the y-coordinate of a point on the unit circle $x^2 + y^2 = 1$, so $y^2 \le 1$, which means $-1 \le y \le 1$.

**D.** sin t is an arc length on the unit circle, and those arcs are at most 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • MAIN • Q4/5

1350

QUESTION ID

## S37-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

**C. sin t is the y-coordinate of a point on the unit circle $x^2 + y^2 = 1$, so $y^2 \le 1$, which means $-1 \le y \le 1$.**

**GRADING NOTE** Coordinates of points on the unit circle satisfy $x^2 + y^2 = 1$. The others only cover acute angles (and wrongly exclude sin t = 1), misstate the definition, or reason from two values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • MAIN • Q4/5

How does the unit-circle definition give sin(−π/6)?

**A.** Travel π/6 clockwise from (1, 0) to (1/2, −√3/2), so sin(−π/6) = −√3/2.

**B.** Negative inputs are not on the unit circle, so sin(−π/6) is undefined.

**C.** Travel π/6 clockwise from (1, 0) to the point (√3/2, −1/2), so sin(−π/6) = −1/2.

**D.** Travel π/6 counterclockwise from (1, 0) to the point (√3/2, 1/2), so sin(−π/6) = 1/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • MAIN • Q5/5

1351

QUESTION ID

**S37-M-Q5**

TYPE

Multiple choice

CORRECT ANSWER

**C. Travel π/6 clockwise from (1, 0) to the point (√3/2, −1/2), so sin(−π/6) = −1/2.**

**GRADING NOTE** A negative t means clockwise travel; the endpoint is (√3/2, −1/2) and the sine is its y-coordinate. The others go the wrong way, use the point for π/3, or leave negative t undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • MAIN • Q5/5

SET 37 • HSF-TF.A.2

# BACKWARD 1

Right-triangle definitions of sine and cosine • High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

**ANSWER KEY** Reduced copy of the question page, with its answer

In a right triangle, sin A (A an acute angle) is defined as

**A.** opposite/adjacent

**B.** adjacent/hypotenuse

**C.** opposite/hypotenuse

**D.** hypotenuse/opposite

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q1

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 37 • BACKWARD 1 • Q1/5

1353

**QUESTION ID**

**S37-B1-Q1**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. opposite/hypotenuse**

**GRADING NOTE** The side opposite A divided by the hypotenuse.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S37-B1-Q1**

**High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 37 • BACKWARD 1 • Q1/5**

In a right triangle with hypotenuse 1, what are the lengths of the leg adjacent to an acute angle θ and the leg opposite θ?

**A.** adjacent cos θ, opposite sin θ

**B.** adjacent 1, opposite tan θ

**C.** adjacent sin θ, opposite cos θ

**D.** adjacent tan θ, opposite 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q2

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 37 • BACKWARD 1 • Q2/5

1354

**QUESTION ID**

## S37-B1-Q2

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. adjacent cos θ, opposite sin θ**

**GRADING NOTE** cos θ = adjacent/1 and sin θ = opposite/1.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S37-B1-Q2**

**High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 37 • BACKWARD 1 • Q2/5**

**True or false?**

The right-triangle definition works for θ = 120°.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q3

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 37 • BACKWARD 1 • Q3/5

1355

QUESTION ID

## S37-B1-Q3

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** A right triangle has no 120° angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q3

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 37 • BACKWARD 1 • Q3/5

In a right triangle with hypotenuse 1, the leg opposite θ is

**A.** sin θ

**B.** tan θ

**C.** 1

**D.** cos θ

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q4

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 37 • BACKWARD 1 • Q4/5

1356

**QUESTION ID**
**S37-B1-Q4**

**TYPE**
Multiple choice

**CORRECT ANSWER**
**A. sin θ**

**GRADING NOTE** opposite/1.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S37-B1-Q4**

**High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 37 • BACKWARD 1 • Q4/5**

Why does the right-triangle definition of sine fail for negative angles?

**A.** Because sine is always negative for negative angles, and side ratios in a triangle are positive.

**B.** Because the hypotenuse of a triangle with a negative angle would have negative length.

**C.** The acute angles of a right triangle are between 0° and 90°, so there is no right triangle with a negative angle.

**D.** It does not fail: a negative angle just flips the right triangle over, and the same side ratio is used.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q5

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 37 • BACKWARD 1 • Q5/5

1357

**QUESTION ID**

## S37-B1-Q5

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. The acute angles of a right triangle are between 0° and 90°, so there is no right triangle with a negative angle.**

**GRADING NOTE** The triangle definition only covers acute angles, which is why the unit circle definition is used. The others are false or confuse a reflection with a negative angle in a triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q5

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 37 • BACKWARD 1 • Q5/5

SET 37 • HSF-TF.A.2

# BACKWARD 2

Points on the unit circle • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

Write the equation of the circle with center (0, 0) and radius 1.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 37 • BACKWARD 2 • Q1/5

1359

QUESTION ID
**S37-B2-Q1**

TYPE
Short answer

CORRECT ANSWER
$x^2 + y^2 = 1$

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 37 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Is $(-1/2, \sqrt{3}/2)$ on the unit circle $x^2 + y^2 = 1$? Find $x^2 + y^2$, then answer Yes or No.

$x^2 + y^2 =$ _____

On the circle? (Yes or No) _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q2

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 37 • BACKWARD 2 • Q2/5

1360

QUESTION ID

**S37-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**1; Yes**

GRADING NOTE Form required: both. 1/4 + 3/4 = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q2

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 37 • BACKWARD 2 • Q2/5

**True or false?**

Every point on the unit circle has $-1 \le x \le 1$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S37-B2-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 37 • BACKWARD 2 • Q3/5

1361

QUESTION ID

**S37-B2-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE $x^2 \le 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S37-B2-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 37 • BACKWARD 2 • Q3/5

Which point is on the unit circle?

**A.** (0.9, 0.9)

**B.** (0.5, 0.5)

**C.** (1, 1)

**D.** (0.6, −0.8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 37 • BACKWARD 2 • Q4/5

1362

**QUESTION ID**
## S37-B2-Q4

**TYPE**
Multiple choice

**CORRECT ANSWER**
## D. (0.6, −0.8)

**GRADING NOTE** 0.36 + 0.64 = 1.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S37-B2-Q4**
**High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations**
**HSG-GPE.A.1 • Set 37 • BACKWARD 2 • Q4/5**

A point on the unit circle has x = 0. Give both possible points.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 37 • BACKWARD 2 • Q5/5

1363

QUESTION ID
**S37-B2-Q5**

TYPE
Short answer

CORRECT ANSWER
**(0, 1) and (0, −1)**

**GRADING NOTE** Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 37 • BACKWARD 2 • Q5/5

SET 37 • HSF-TF.A.2

# BACKWARD 3

Radian measure as directed arc length • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1

**ANSWER KEY** Reduced copy of the question page, with its answer

What arc length on the unit circle does t = π/6 cut off?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 37 • BACKWARD 3 • Q1/5

1365

**QUESTION ID**

**S37-B3-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**π/6**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 37 • BACKWARD 3 • Q1/5

On the unit circle, which direction does t = −π/6 travel from (1, 0)? Answer Clockwise or Counterclockwise.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 37 • BACKWARD 3 • Q2/5

1366

**QUESTION ID**

**S37-B3-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**Clockwise**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 37 • BACKWARD 3 • Q2/5

**True or false?**

t = π returns to the starting point (1, 0).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 37 • BACKWARD 3 • Q3/5

1367

QUESTION ID

**S37-B3-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** t = π is half a turn and ends at (−1, 0); t = 2π returns to (1, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 37 • BACKWARD 3 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

An arc of length π from (1, 0) ends at

**A.** (0, 1)

**B.** (0, −1)

**C.** (1, 0)

**D.** (−1, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 37 • BACKWARD 3 • Q4/5

1368

**QUESTION ID**

**S37-B3-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. (−1, 0)**

**GRADING NOTE** Half a turn.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 37 • BACKWARD 3 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Find the value of t with $-2\pi < t < 0$ for which travel of t units from (1, 0) on the unit circle ends at (0, −1). Give it exactly, in terms of π.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 37 • BACKWARD 3 • Q5/5

1369

QUESTION ID

**S37-B3-Q5**

TYPE

Short answer

CORRECT ANSWER

**−π/2**

**GRADING NOTE** Form required: the exact value. A quarter turn clockwise; −π/2 is the only such t in this interval.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B3-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 37 • BACKWARD 3 • Q5/5

SET 37 • HSF-TF.A.2

# FORWARD 1

Explain where tangent is defined and how values repeat • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

Why is tan t undefined at t = π/2 + kπ (k an integer)?

**A.** At these t the endpoint on the unit circle is (0, 1) or (0, −1), so cos t = 0 and tan t = sin t/cos t would divide by 0.

**B.** tan t is undefined whenever sin t = 1.

**C.** At these t the endpoint on the unit circle is (1, 0) or (−1, 0), so sin t = 0 and tan t = sin t/cos t would have to divide by 0.

**D.** tan t = cos t/sin t, and sin t = 0 at these t.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • FORWARD 1 • Q1/5

1371

QUESTION ID
**S37-F1-Q1**

TYPE
Multiple choice

CORRECT ANSWER
**A. At these t the endpoint on the unit circle is (0, 1) or (0, −1), so cos t = 0 and tan t = sin t/cos t would divide by 0.**

**GRADING NOTE** The endpoints are (0, ±1), where the x-coordinate cos t is 0. The others use the wrong points, invert the ratio, or give a rule that misses t = 3π/2 (where sin t = −1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • FORWARD 1 • Q1/5

What are the domain and range of y = sin t?

**A.** Domain: $0 \le t \le 2\pi$; range: $-1 \le y \le 1$

**B.** Domain: all real numbers; range: $-1 \le y \le 1$

**C.** Domain: all real numbers; range: all real numbers

**D.** Domain: $-1 \le t \le 1$; range: all real numbers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • FORWARD 1 • Q2/5

1372

QUESTION ID

**S37-F1-Q2**

TYPE

Multiple choice

CORRECT ANSWER

**B. Domain: all real numbers; range: $-1 \le y \le 1$**

**GRADING NOTE** The unit circle definition works for every real t, and sin t is a y-coordinate on the unit circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • FORWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

cos t and sin t can both equal 0 for the same t.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • FORWARD 1 • Q3/5

1373

QUESTION ID

**S37-F1-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE (0, 0) is not on the unit circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • FORWARD 1 • Q3/5

The unit-circle definition agrees with the triangle definition for

**A.** $t < 0$ only

**B.** $\pi/2 < t < \pi$ only

**C.** all t

**D.** $0 < t < \pi/2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • FORWARD 1 • Q4/5

1374

QUESTION ID

**S37-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**D. $0 < t < \pi/2$**

**GRADING NOTE** Acute angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • FORWARD 1 • Q4/5

Why is cos(−t) = cos t, using the unit circle?

**A.** Because −t and t are the same angle, so they end at the same point on the unit circle.

**B.** Because cos is always positive, so cos(−t) and cos t cannot have different signs.

**C.** The endpoints for t and −t are reflections of each other over the y-axis, which keeps the x-coordinate (the cosine).

**D.** The endpoints for t and −t are reflections of each other over the x-axis, which keeps the x-coordinate (the cosine).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • FORWARD 1 • Q5/5

1375

QUESTION ID

## S37-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

**D. The endpoints for t and −t are reflections of each other over the x-axis, which keeps the x-coordinate (the cosine).**

**GRADING NOTE** Reflection over the x-axis keeps x and changes the sign of y. The others name the wrong reflection (it changes x), state a false fact, or confuse the angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • FORWARD 1 • Q5/5

SET 37 • HSF-TF.A.2

# FORWARD 2

Explain symmetry and periodicity • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4

Which unit-circle argument shows that sin(−t) = −sin t?

**A.** sin(−π/2) = −1 = −sin(π/2), so the rule holds for every t.

**B.** The endpoints for t and −t are reflections of each other over the x-axis, so their y-coordinates (the sines) are opposites.

**C.** The endpoints for t and −t are the same point, so their sines are opposites.

**D.** The endpoints for t and −t are always reflections of each other over the y-axis, so their y-coordinates (the sines) are opposites.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 37 • FORWARD 2 • Q1/5

1377

**QUESTION ID**
**S37-F2-Q1**

**TYPE**
Multiple choice

**CORRECT ANSWER**
**B. The endpoints for t and −t are reflections of each other over the x-axis, so their y-coordinates (the sines) are opposites.**

**GRADING NOTE** Equal arcs in opposite directions from (1, 0) give mirror images over the x-axis. The others name the wrong reflection (it keeps y), check one value, or describe the points wrongly.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 37 • FORWARD 2 • Q1/5

Is sine an even function, an odd function, or neither?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 37 • FORWARD 2 • Q2/5

1378

QUESTION ID

**S37-F2-Q2**

TYPE

Short answer

CORRECT ANSWER

**Odd**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 37 • FORWARD 2 • Q2/5

**True or false?**

tan(t + π) = tan t for every t where it is defined.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 37 • FORWARD 2 • Q3/5

1379

QUESTION ID

## S37-F2-Q3

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** The endpoints of t and t + π are opposite points; the ratio y/x is unchanged.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 37 • FORWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

The period of y = tan t is

**A.** π

**B.** 4π

**C.** π/2

**D.** 2π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 37 • FORWARD 2 • Q4/5

1380

QUESTION ID

**S37-F2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**A. π**

GRADING NOTE Opposite points give equal ratios.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 37 • FORWARD 2 • Q4/5

Why is cos an even function?

**A.** The points for t and −t are reflections over the x-axis, which keeps the x-coordinate, so cos(−t) = cos t.

**B.** The points for t and −t are reflections over the y-axis, which keeps the x-coordinate, so cos(−t) = cos t.

**C.** cos t is positive for every t, so cos(−t) and cos t are always equal.

**D.** The points for t and −t are reflections over the x-axis, which negates the x-coordinate, so cos(−t) = −cos t.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 37 • FORWARD 2 • Q5/5

1381

QUESTION ID
**S37-F2-Q5**

TYPE
Multiple choice

CORRECT ANSWER
**A. The points for t and −t are reflections over the x-axis, which keeps the x-coordinate, so cos(−t) = cos t.**

**GRADING NOTE** Reflecting over the x-axis changes only the sign of the y-coordinate. The others name the wrong reflection, use a false fact (cos is not always positive), or say the x-coordinate changes sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 37 • FORWARD 2 • Q5/5

SET 38 • HSF-TF.B.5 • TRIGONOMETRIC FUNCTIONS

# Write sinusoidal models with given amplitude, period, and midline

| | | |
|---|---|---|
| **MAIN** | Algebra II main question | **HSF-TF.B.5** |
| **BACKWARD 1** | Amplitude and midline from the maximum and minimum | **7.NS.A.3** |
| **BACKWARD 2** | Period and frequency | **6.RP.A.2** |
| **BACKWARD 3** | Vertical stretches, shifts, and reflections | **HSF-BF.B.3** |
| **FORWARD 1** | Write models from data or with a phase shift | **HSF-TF.B.5** |
| **FORWARD 2** | Solve for times in sinusoidal models | **HSF-TF.B.7** |

SET 38 • HSF-TF.B.5

# MAIN

Algebra II main question • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

**ANSWER KEY** Reduced copy of the question page, with its answer

Write a cosine function with amplitude 3, period 2π, midline y = 1, and a maximum at x = 0, in the form y = A·cos(Bx) + D with A > 0 and B > 0.

A = _____

B = _____

D = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • MAIN • Q1/5

1384

QUESTION ID

**S38-M-Q1**

TYPE

Short answer

CORRECT ANSWER

**3; 1; 1**

GRADING NOTE Form required: all three. B = 2π/2π = 1; a cosine with A > 0 has its maximum at x = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A Ferris wheel has diameter 40 m, its center is 25 m above the ground, and it turns once every 4 minutes. A rider starts at the bottom. Write the height h after t minutes in the form $h = D - A\cdot\cos(Bt)$ with $A > 0$ and $B > 0$. Give B exactly, in terms of π.

A = _____

B = _____

D = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • MAIN • Q2/5

1385

**QUESTION ID**

**S38-M-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**20; π/2; 25**

**GRADING NOTE** Form required: all three. Amplitude 20, midline 25, B = 2π/4; the minus sign starts the rider at the bottom.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • MAIN • Q2/5

Water depth at a harbor varies between 1 ft (low tide) and 9 ft (high tide) with a period of 12 hours. High tide is at t = 0. Write the depth d after t hours in the form d = A·cos(Bt) + D with A > 0 and B > 0. Give B exactly, in terms of π.

A = _____

B = _____

D = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • MAIN • Q3/5

1386

**QUESTION ID**

**S38-M-Q3**

**TYPE**

Short answer

**CORRECT ANSWER**

**4; π/6; 5**

**GRADING NOTE** Form required: all three. Midline (9 + 1)/2 = 5, amplitude (9 − 1)/2 = 4, B = 2π/12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • MAIN • Q3/5

A town's average temperature is lowest, 10°C, in month t = 0 and highest, 30°C, at t = 6, with a 12-month period. Write T(t) in the form T = D − A·cos(Bt) with A > 0 and B > 0. Give B exactly, in terms of π.

A = _____

B = _____

D = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • MAIN • Q4/5

1387

**QUESTION ID**

**S38-M-Q4**

**TYPE**

Short answer

**CORRECT ANSWER**

**10; π/6; 20**

**GRADING NOTE** Form required: all three. Midline 20, amplitude 10, B = 2π/12; the minus sign puts the minimum at t = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • MAIN • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Write a sine function with amplitude 2, period π, and midline y = −1 that is at the midline and increasing at x = 0, in the form y = A·sin(Bx) + D with A > 0 and B > 0.

A = _____

B = _____

D = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • MAIN • Q5/5

1388

QUESTION ID

**S38-M-Q5**

TYPE

Short answer

CORRECT ANSWER

**2; 2; −1**

GRADING NOTE Form required: all three. B = 2π/π = 2; a sine with A > 0 is at its midline and increasing at x = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • MAIN • Q5/5

SET 38 • HSF-TF.B.5

# BACKWARD 1

Amplitude and midline from the maximum and minimum • Grade 7 • The Number System • 7.NS.A.3

**ANSWER KEY** Reduced copy of the question page, with its answer

Find (9 + 1)/2 and (9 − 1)/2.

(9 + 1)/2 = _____

(9 − 1)/2 = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 38 • BACKWARD 1 • Q1/5

1390

QUESTION ID

**S38-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**5; 4**

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 38 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A quantity varies between 10 and 30. Find the midline value and the amplitude.

Midline value: _____

Amplitude: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 38 • BACKWARD 1 • Q2/5

1391

QUESTION ID

**S38-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**20; 10**

GRADING NOTE Form required: both. (30 + 10)/2 and (30 − 10)/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 38 • BACKWARD 1 • Q2/5

**True or false?**

The amplitude is the maximum minus the minimum.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 38 • BACKWARD 1 • Q3/5

1392

QUESTION ID

**S38-B1-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE It is half the difference.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 38 • BACKWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

For max 45 and min 5, the midline is

**A.** y = 20

**B.** y = 50

**C.** y = 25

**D.** y = 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 38 • BACKWARD 1 • Q4/5

1393

QUESTION ID

**S38-B1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**C. y = 25**

GRADING NOTE (45 + 5)/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 38 • BACKWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A Ferris wheel has diameter 40 m with its center 25 m above the ground. Give its maximum and minimum heights in meters.

Maximum (m): _____

Minimum (m): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 38 • BACKWARD 1 • Q5/5

1394

**QUESTION ID**

## S38-B1-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**45; 5**

**GRADING NOTE** Form required: both. 25 ± 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 38 • BACKWARD 1 • Q5/5

SET 38 • HSF-TF.B.5

# BACKWARD 2

Period and frequency • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

A wheel turns once every 4 minutes. How many turns per minute?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 38 • BACKWARD 2 • Q1/5

1396

QUESTION ID

**S38-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**1/4**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 38 • BACKWARD 2 • Q1/5

A spring oscillates 3 times per second. What is the period?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 38 • BACKWARD 2 • Q2/5

1397

QUESTION ID

**S38-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**1/3 s**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 38 • BACKWARD 2 • Q2/5

**True or false?**

Frequency is equal to the period.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 38 • BACKWARD 2 • Q3/5

1398

QUESTION ID

**S38-B2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE Frequency is the reciprocal: f = 1/period.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 38 • BACKWARD 2 • Q3/5

A period of 12 hours means a frequency of

**A.** 12 cycles per hour

**B.** 2 cycles per hour

**C.** 1/2 cycle per hour

**D.** 1/12 cycle per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 38 • BACKWARD 2 • Q4/5

1399

**QUESTION ID**

## S38-B2-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. 1/12 cycle per hour**

**GRADING NOTE** Reciprocal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 38 • BACKWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

y = sin(Bx) has period 2π/B. Find B for period 4.

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 38 • BACKWARD 2 • Q5/5

1400

QUESTION ID

**S38-B2-Q5**

TYPE

Short answer

CORRECT ANSWER

**π/2**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 38 • BACKWARD 2 • Q5/5

SET 38 • HSF-TF.B.5

# BACKWARD 3

Vertical stretches, shifts, and reflections • High School — Algebra I • Functions: Building Functions • HSF-BF.B.3

How is y = 3cos x + 1 obtained from y = cos x?

**A.** Stretch vertically by 3, then shift up 1

**B.** Stretch vertically by 3, then shift right 1

**C.** Stretch horizontally by 3, then shift up 1

**D.** Shift up 3, then stretch vertically by 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 38 • BACKWARD 3 • Q1/5

1402

QUESTION ID

**S38-B3-Q1**

TYPE

Multiple choice

CORRECT ANSWER

**A. Stretch vertically by 3, then shift up 1**

**GRADING NOTE** Multiplying the output by 3 stretches vertically; adding 1 shifts up.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 38 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

How does the graph of y = −cos x compare with the graph of y = cos x?

**A.** It is the reflection over the y-axis

**B.** It is shifted down 1

**C.** It is the reflection over the x-axis

**D.** It is the same graph

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 38 • BACKWARD 3 • Q2/5

1403

**QUESTION ID**

**S38-B3-Q2**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. It is the reflection over the x-axis**

**GRADING NOTE** Negating the output reflects over the x-axis (cos is even, so a reflection over the y-axis would not change it). The others reflect over the y-axis (which leaves cos x unchanged), shift down, or claim no change.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 38 • BACKWARD 3 • Q2/5

**True or false?**

y = cos x + 5 has a maximum of 5.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 38 • BACKWARD 3 • Q3/5

1404

QUESTION ID

**S38-B3-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** The maximum of cos x is 1, so the maximum is 6; 5 is the midline.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 38 • BACKWARD 3 • Q3/5

$y = -20\cos x$ starts (at $x = 0$) at

**A.** its minimum

**B.** its maximum

**C.** the midline going up

**D.** the midline going down

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 38 • BACKWARD 3 • Q4/5

1405

**QUESTION ID**

**S38-B3-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. its minimum**

**GRADING NOTE** −20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 38 • BACKWARD 3 • Q4/5

First stretch the graph of y = sin x vertically by 2, then shift it down 1. Which equation results?

**A.** y = sin(2x) − 1

**B.** y = 2sin(x − 1)

**C.** y = 2(sin x − 1)

**D.** y = 2sin x − 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 38 • BACKWARD 3 • Q5/5

1406

QUESTION ID

**S38-B3-Q5**

TYPE

Multiple choice

CORRECT ANSWER

**D. y = 2sin x − 1**

GRADING NOTE Stretch: y = 2sin x; then shift down: y = 2sin x − 1. Shifting first and then stretching would give 2(sin x − 1) = 2sin x − 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B3-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 38 • BACKWARD 3 • Q5/5

SET 38 • HSF-TF.B.5

# FORWARD 1

Write models from data or with a phase shift • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

**ANSWER KEY** Reduced copy of the question page, with its answer

A rider's height is 3 m at t = 0 (the bottom), 27 m at the top after 15 s, and the motion repeats every 30 s. Write h(t) in the form h = D − A·cos(Bt) with A > 0 and B > 0. Give B exactly, in terms of π.

A = _____

B = _____

D = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 1 • Q1/5

1408

QUESTION ID

**S38-F1-Q1**

TYPE

Short answer

CORRECT ANSWER

**12; π/15; 15**

GRADING NOTE Form required: all three. Amplitude (27 − 3)/2 = 12, midline 15, B = 2π/30. The minus sign puts the minimum at t = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Sunrise times vary between 5:30 and 7:30 (written as 5.5 and 7.5 hours), with period 365 days and the latest sunrise at day t = 0. Write S(t) in the form S = D + A·cos(Bt) with A > 0 and B > 0. Give B exactly, in terms of π.

A = _____

B = _____

D = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 1 • Q2/5

1409

QUESTION ID

**S38-F1-Q2**

TYPE

Short answer

CORRECT ANSWER

**1; 2π/365; 6.5**

GRADING NOTE Form required: all three. Amplitude 1, midline 6.5, B = 2π/365; the maximum is at t = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 1 • Q2/5

**True or false?**

$y = 4\sin(x - \pi/2)$ is the same as $y = 4\cos x$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 1 • Q3/5

1410

QUESTION ID

**S38-F1-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** $\sin(x - \pi/2) = -\cos x$, so $4\sin(x - \pi/2) = -4\cos x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 1 • Q3/5

A sine model with period 10, amplitude 3, midline 7, and a maximum at t = 2.5 is

**A.** 7 + 3sin(10t)

**B.** 7 + 3sin(πt/5)

**C.** 3 + 7sin(πt/5)

**D.** 7 + 3cos(πt/5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 1 • Q4/5

1411

QUESTION ID

**S38-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**B. 7 + 3sin(πt/5)**

**GRADING NOTE** Quarter period 2.5 gives the maximum of sine.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 1 • Q4/5

A sinusoidal model has a maximum of 18 at t = 3 and the next minimum of 6 at t = 7. Write it in the form $y = D + A\cdot\cos(B(t - 3))$ with $A > 0$ and $B > 0$. Give B exactly, in terms of π.

A = _____

B = _____

D = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 1 • Q5/5

1412

**QUESTION ID**

**S38-F1-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**6; π/4; 12**

**GRADING NOTE** Form required: all three. Half a period is 7 − 3 = 4, so the period is 8 and B = 2π/8; amplitude 6, midline 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 1 • Q5/5

SET 38 • HSF-TF.B.5

# FORWARD 2

Solve for times in sinusoidal models • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7

**ANSWER KEY** Reduced copy of the question page, with its answer

For h = 25 − 20cos(πt/2), find the first time t > 0 when the rider is at 35 m. Give an exact fraction. A calculator may be used.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 38 • FORWARD 2 • Q1/5

1414

**QUESTION ID**

**S38-F2-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**4/3**

**GRADING NOTE** Form required: the exact value. cos(πt/2) = −1/2 gives πt/2 = 2π/3, so t = 4/3 (about 1.33 min).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 38 • FORWARD 2 • Q1/5

For d = 4cos(πt/6) + 5, find all times in 0 ≤ t < 12 when the depth is 5 ft.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 38 • FORWARD 2 • Q2/5

1415

**QUESTION ID**

**S38-F2-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**3 or 9**

**GRADING NOTE** Form required: both, in any order. cos(πt/6) = 0 gives πt/6 = π/2 or 3π/2.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S38-F2-Q2**

**High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 38 • FORWARD 2 • Q2/5**

**True or false?**

A sinusoidal model usually reaches a given value between its min and max twice per period.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S38-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 38 • FORWARD 2 • Q3/5

1416

QUESTION ID

## S38-F2-Q3

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Once going up and once going down.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S38-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 38 • FORWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

For T = 20 − 10cos(πt/6), T = 25 first when

**A.** t = 2

**B.** t = 4

**C.** t = 3

**D.** t = 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 38 • FORWARD 2 • Q4/5

1417

QUESTION ID

**S38-F2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**B. t = 4**

**GRADING NOTE** cos(πt/6) = −1/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 38 • FORWARD 2 • Q4/5

Why does a sinusoidal model take each value strictly between its minimum and maximum exactly twice in each period?

**A.** Because every function takes each value exactly twice in each interval of the same length as the period.

**B.** In each period the graph rises from the minimum to the maximum and falls back, passing each height once each way.

**C.** Because the period is 2π, and 2π contains each value exactly twice, once in each half of the period.

**D.** Because the graph crosses its midline twice in each period, and each crossing gives one of the two times the graph takes the value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 38 • FORWARD 2 • Q5/5

1418

QUESTION ID
**S38-F2-Q5**

TYPE
Multiple choice

CORRECT ANSWER
**B. In each period the graph rises from the minimum to the maximum and falls back, passing each height once each way.**

**GRADING NOTE** Each period has one rising half and one falling half. The others overgeneralize, give an irrelevant fact, or only describe the midline.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 38 • FORWARD 2 • Q5/5

SET 39 • HSF-TF.B.5 • TRIGONOMETRIC FUNCTIONS

# Interpret the parameters of sinusoidal models

| | | |
|---|---|---|
| **MAIN** | Algebra II main question | **HSF-TF.B.5** |
| **BACKWARD 1** | Build the model from amplitude, period, and midline | **HSF-TF.B.5** |
| **BACKWARD 2** | Evaluate a sinusoidal model | **HSF-TF.A.2** |
| **BACKWARD 3** | Convert rates between units | **6.RP.A.3.d** |
| **FORWARD 1** | Predict how changing a parameter changes the situation | **HSF-TF.B.5** |
| **FORWARD 2** | Interpret periodic features from graphs and tables | **HSF-IF.B.4** |

SET 39 • HSF-TF.B.5

# MAIN

Algebra II main question • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

A Ferris wheel rider's height is h(t) = 12 + 10sin(πt/15) meters after t seconds. Give the maximum height, the minimum height, and the time for one revolution.

Maximum (m): _____

Minimum (m): _____

One revolution (s): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • MAIN • Q1/5

1421

**QUESTION ID**

**S39-M-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**22; 2; 30**

**GRADING NOTE** Form required: all three. 12 ± 10, and the period is 2π/(π/15) = 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Hours of daylight are $D(t) = 12 + 3\sin(2\pi(t - 80)/365)$, t in days. What do 12, 3, and 365 represent?

**A.** 12: average daylight (midline, hours); 3: greatest change from 12 (amplitude, hours); 365: day of greatest daylight.

**B.** 12: average daylight (midline, hours); 3: days in one cycle (period); 365: greatest daylight (amplitude, hours).

**C.** 12: greatest daylight (maximum, hours); 3: average daylight (midline, hours); 365: days in one cycle (period).

**D.** 12: average daylight (midline, hours); 3: greatest change from 12 (amplitude, hours); 365: days in one cycle (period).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • MAIN • Q2/5

1422

**QUESTION ID**

## S39-M-Q2

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. 12: average daylight (midline, hours); 3: greatest change from 12 (amplitude, hours); 365: days in one cycle (period).**

**GRADING NOTE** Midline 12, amplitude 3, period $2\pi/(2\pi/365) = 365$ days.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • MAIN • Q2/5

A model of blood pressure is P(t) = 100 + 20sin(2.5πt), t in seconds. Find the period and the number of heartbeats per minute.

Period (s): _____

Beats per minute: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • MAIN • Q3/5

1423

**QUESTION ID**

**S39-M-Q3**

**TYPE**

Short answer

**CORRECT ANSWER**

**0.8; 75**

**GRADING NOTE** Form required: both. 2π/(2.5π) = 0.8 s; 60/0.8 = 75.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • MAIN • Q3/5

A spring's position is $y = -4\cos(3\pi t)$ cm, t in seconds. Give the amplitude, the period, and the frequency.

Amplitude (cm): _____

Period (s): _____

Frequency (cycles per second): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • MAIN • Q4/5

1424

**QUESTION ID**

**S39-M-Q4**

**TYPE**

Short answer

**CORRECT ANSWER**

**4; 2/3; 1.5**

**GRADING NOTE** Form required: all three. $|-4| = 4$; $2\pi/(3\pi) = 2/3$; $1/(2/3) = 1.5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • MAIN • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Two sounds are y = sin(880πt) and y = sin(440πt), t in seconds. Give the frequency of each in cycles per second.

Frequency of sin(880πt): _____

Frequency of sin(440πt): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • MAIN • Q5/5

1425

QUESTION ID

**S39-M-Q5**

TYPE

Short answer

CORRECT ANSWER

**440; 220**

GRADING NOTE Form required: both. Frequency = B/(2π): 880π/(2π) = 440 and 440π/(2π) = 220, so the first sound is higher.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • MAIN • Q5/5

SET 39 • HSF-TF.B.5

# BACKWARD 1

Build the model from amplitude, period, and midline • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

**ANSWER KEY** Reduced copy of the question page, with its answer

What is the amplitude of $y = 12 + 10\sin(\pi t/15)$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • BACKWARD 1 • Q1/5

1427

**QUESTION ID**

**S39-B1-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**10**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

What is the midline of $y = 12 + 10\sin(\pi t/15)$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • BACKWARD 1 • Q2/5

1428

QUESTION ID

**S39-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**y = 12**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • BACKWARD 1 • Q2/5

**True or false?**

The period of $y = \sin(Bt)$ is $B/(2\pi)$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • BACKWARD 1 • Q3/5

1429

QUESTION ID

**S39-B1-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** The period is $2\pi/B$ (for $B > 0$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • BACKWARD 1 • Q3/5

The period of $y = \sin(\pi t/15)$ is

**A.** 30

**B.** 15

**C.** $\pi/15$

**D.** $2\pi$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • BACKWARD 1 • Q4/5

1430

**QUESTION ID**

**S39-B1-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. 30**

**GRADING NOTE** $2\pi/(\pi/15)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • BACKWARD 1 • Q4/5

Write a sine model with amplitude 4, period 2/3, and midline 0, in the form $y = A \cdot \sin(Bt)$ with $A > 0$ and $B > 0$. Give B exactly, in terms of π.

A = _____

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • BACKWARD 1 • Q5/5

1431

**QUESTION ID**

**S39-B1-Q5**

**TYPE**

Short answer

**CORRECT ANSWER**

**4; 3π**

**GRADING NOTE** Form required: both. B = 2π/(2/3) = 3π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • BACKWARD 1 • Q5/5

SET 39 • HSF-TF.B.5

# BACKWARD 2

Evaluate a sinusoidal model • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

Evaluate 12 + 10sin(π/2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • BACKWARD 2 • Q1/5

1433

QUESTION ID

**S39-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**22**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Evaluate 12 + 10sin(3π/2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • BACKWARD 2 • Q2/5

1434

QUESTION ID

**S39-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**2**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • BACKWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

$-4\cos(0) = 0.$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • BACKWARD 2 • Q3/5

1435

QUESTION ID

**S39-B2-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE cos 0 = 1, so −4cos 0 = −4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • BACKWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

100 + 20sin(π/2) equals

**A.** 100

**B.** 20

**C.** 120

**D.** 80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • BACKWARD 2 • Q4/5

1436

**QUESTION ID**

**S39-B2-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. 120**

**GRADING NOTE** sin(π/2) = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • BACKWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Evaluate 12 + 3sin(0).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • BACKWARD 2 • Q5/5

1437

QUESTION ID

**S39-B2-Q5**

TYPE

Short answer

CORRECT ANSWER

**12**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • BACKWARD 2 • Q5/5

SET 39 • HSF-TF.B.5

# BACKWARD 3

Convert rates between units • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d

**ANSWER KEY** Reduced copy of the question page, with its answer

A cycle lasts 0.8 s. How many cycles are in 60 s?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 39 • BACKWARD 3 • Q1/5

1439

QUESTION ID

**S39-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**75**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 39 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

1.5 cycles per second is how many cycles per minute?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 39 • BACKWARD 3 • Q2/5

1440

QUESTION ID

**S39-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**90**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 39 • BACKWARD 3 • Q2/5

**True or false?**

A period of 30 s means 30 revolutions per minute.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 39 • BACKWARD 3 • Q3/5

1441

QUESTION ID

**S39-B3-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** 60/30 = 2 revolutions per minute.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 39 • BACKWARD 3 • Q3/5

440 cycles per second is how many cycles per millisecond?

**A.** 4.4

**B.** 44

**C.** 440

**D.** 0.44

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 39 • BACKWARD 3 • Q4/5

1442

QUESTION ID

**S39-B3-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**D. 0.44**

**GRADING NOTE** 1000 ms per second.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 39 • BACKWARD 3 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A tide has period 12 hours. How many cycles in 3 days?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 39 • BACKWARD 3 • Q5/5

1443

QUESTION ID

**S39-B3-Q5**

TYPE

Short answer

CORRECT ANSWER

**6**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 39 • BACKWARD 3 • Q5/5

SET 39 • HSF-TF.B.5

# FORWARD 1

Predict how changing a parameter changes the situation • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

**ANSWER KEY** Reduced copy of the question page, with its answer

A Ferris wheel rider's height is $h(t) = 12 + 10\sin(\pi t/15)$ m after t seconds. The wheel is replaced by one of the same size that turns twice as fast. In the new model $h(t) = 12 + 10\sin(Bt)$, what is B? Give B exactly, in terms of π.

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • FORWARD 1 • Q1/5

1445

**QUESTION ID**

**S39-F1-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**2π/15**

**GRADING NOTE** Form required: the exact value. Twice as fast halves the period from 30 s to 15 s, so B = 2π/15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • FORWARD 1 • Q1/5

Daylight in one city is D(t) = 12 + 3sin(2π(t − 80)/365) hours. A city farther from the equator has daylight between 8 and 16 hours with the same period and timing. Which parameter changes, and to what?

**A.** The amplitude changes to 4

**B.** The amplitude changes to 8

**C.** The midline changes to 16

**D.** The period changes to 8 days

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • FORWARD 1 • Q2/5

1446

**QUESTION ID**

**S39-F1-Q2**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. The amplitude changes to 4**

**GRADING NOTE** The midline (16 + 8)/2 = 12 stays the same; the amplitude becomes (16 − 8)/2 = 4. The others confuse the maximum with the midline, change the period, or use the full range.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • FORWARD 1 • Q2/5

**True or false?**

Changing the midline changes the period.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • FORWARD 1 • Q3/5

1447

QUESTION ID

**S39-F1-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE The midline only shifts the graph vertically.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • FORWARD 1 • Q3/5

Multiplying B by 3 makes the period

**A.** 3 units longer

**B.** unchanged

**C.** three times as long

**D.** one third as long

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • FORWARD 1 • Q4/5

1448

QUESTION ID

**S39-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**D. one third as long**

**GRADING NOTE** 2π/(3B).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • FORWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A heart rate rises from 75 to 100 beats per minute. What is the new period, in seconds?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • FORWARD 1 • Q5/5

1449

**QUESTION ID**
**S39-F1-Q5**

**TYPE**
Short answer

**CORRECT ANSWER**
**0.6 s**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • FORWARD 1 • Q5/5

SET 39 • HSF-TF.B.5

# FORWARD 2

Interpret periodic features from graphs and tables • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4

A sinusoidal model of a tide's depth has consecutive maxima at t = 2 and t = 14 hours. What is its period, in hours?

Period (hours): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 39 • FORWARD 2 • Q1/5

1451

**QUESTION ID**

**S39-F2-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**12**

**GRADING NOTE** Form required: the value. For a sinusoidal model, consecutive maxima are one period apart: 14 − 2 = 12.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S39-F2-Q1**

**High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 39 • FORWARD 2 • Q1/5**

A temperature model has a high of 30 and a low of 10. What are its midline value and amplitude?

Midline value: _____

Amplitude: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 39 • FORWARD 2 • Q2/5

1452

**QUESTION ID**

## S39-F2-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**20; 10**

**GRADING NOTE** Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 39 • FORWARD 2 • Q2/5

**True or false?**

For a nonconstant sinusoidal function, consecutive minima are one period apart.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 39 • FORWARD 2 • Q3/5

1453

QUESTION ID

**S39-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** A sinusoid has exactly one minimum per period. (For a general periodic function this can fail: it may have several local minima in one period.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 39 • FORWARD 2 • Q3/5

On which interval does $h(t) = 12 + 10\sin(\pi t/15)$ increase?

**A.** $0 < t < 15$

**B.** $0 < t < 7.5$

**C.** $15 < t < 30$

**D.** $7.5 < t < 22.5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 39 • FORWARD 2 • Q4/5

1454

QUESTION ID

**S39-F2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**B. 0 < t < 7.5**

**GRADING NOTE** Rises to the maximum at t = 7.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 39 • FORWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

How can the frequency be read from the graph of a periodic function?

**A.** Read the maximum height of the graph, which is the number of cycles per unit of the input.

**B.** Find the length of one cycle (the period) and take its reciprocal, the number of cycles per unit of the input.

**C.** Find half the difference between the maximum and the minimum, which is the number of cycles per unit.

**D.** Count the number of times the graph crosses the x-axis in the window shown; that count is the frequency.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 39 • FORWARD 2 • Q5/5

1455

QUESTION ID

**S39-F2-Q5**

TYPE

Multiple choice

CORRECT ANSWER

**B. Find the length of one cycle (the period) and take its reciprocal, the number of cycles per unit of the input.**

**GRADING NOTE** Frequency = 1/period. The others give the maximum, a count that depends on the midline, or the amplitude.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 39 • FORWARD 2 • Q5/5

SET 40 • HSF-TF.C.8 • TRIGONOMETRIC FUNCTIONS

# Prove the Pythagorean identity

| | | |
|---|---|---|
| **MAIN** | Algebra II main question | **HSF-TF.C.8** |
| **BACKWARD 1** | The Pythagorean Theorem | **8.G.B.7** |
| **BACKWARD 2** | Coordinates of the point at angle θ | **HSF-TF.A.2** |
| **BACKWARD 3** | Expand and simplify algebraic squares | **HSA-SSE.A.2** |
| **FORWARD 1** | Prove identities derived from the Pythagorean identity | **HSF-TF.C.8** |
| **FORWARD 2** | Use the angle-addition formulas | **HSF-TF.C.9** |

SET 40 • HSF-TF.C.8

# MAIN

Algebra II main question • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8

Which argument proves $\sin^2\theta + \cos^2\theta = 1$ for every angle $\theta$, using the unit circle?

**A.** On the unit circle $\sin\theta + \cos\theta = 1$ for every angle $\theta$, because the coordinates of a point add to the radius; squaring both sides gives $\sin^2\theta + \cos^2\theta = 1$.

**B.** The point at angle $\theta$ is $(\sin\theta, \cos\theta)$, so $\sin\theta + \cos\theta = 1$.

**C.** For $\theta = 30°$: $(1/2)^2 + (\sqrt{3}/2)^2 = 1$, so the identity holds.

**D.** The point at angle $\theta$ is $(\cos\theta, \sin\theta)$, and every point on the unit circle satisfies $x^2 + y^2 = 1$; substituting gives $\cos^2\theta + \sin^2\theta = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S40-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • MAIN • Q1/5

1458

QUESTION ID

## S40-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

**D. The point at angle $\theta$ is $(\cos\theta, \sin\theta)$, and every point on the unit circle satisfies $x^2 + y^2 = 1$; substituting gives $\cos^2\theta + \sin^2\theta = 1$.**

**GRADING NOTE** The coordinates of the point satisfy the circle equation. The others check one angle or rely on the false statement $\sin\theta + \cos\theta = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S40-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • MAIN • Q1/5

A right triangle has legs a and b, hypotenuse c, and acute angle θ opposite a. Which argument proves $\sin^2\theta + \cos^2\theta = 1$?

**A.** $\sin\theta = a/c$ and $\cos\theta = b/c$, so $\sin^2\theta + \cos^2\theta = (a + b)^2/c^2$, and $(a + b)^2 = c^2$ by the Pythagorean Theorem, so the sum is 1.

**B.** $\sin\theta = b/c$ and $\cos\theta = a/c$, so $\sin\theta + \cos\theta = (a + b)/c = 1$.

**C.** $\sin\theta = a/c$ and $\cos\theta = b/c$, so $\sin^2\theta + \cos^2\theta = (a^2 + b^2)/c^2 = c^2/c^2 = 1$ by the Pythagorean Theorem.

**D.** In a 3-4-5 triangle, $(3/5)^2 + (4/5)^2 = 1$, so the identity holds.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • MAIN • Q2/5

1459

**QUESTION ID**

## S40-M-Q2

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. $\sin\theta = a/c$ and $\cos\theta = b/c$, so $\sin^2\theta + \cos^2\theta = (a^2 + b^2)/c^2 = c^2/c^2 = 1$ by the Pythagorean Theorem.**

**GRADING NOTE** Both ratios and $a^2 + b^2 = c^2$. The others swap the ratios and use $a + b = c$, square a sum incorrectly, or check one triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • MAIN • Q2/5

Which argument proves $1 + \tan^2\theta = 1/\cos^2\theta$ wherever $\cos\theta \neq 0$?

**A.** Divide $\sin^2\theta + \cos^2\theta = 1$ by $\sin^2\theta$ to get $1 + \cos^2\theta/\sin^2\theta = 1/\sin^2\theta$; since $\cos^2\theta/\sin^2\theta = \tan^2\theta$, this gives $1 + \tan^2\theta = 1/\cos^2\theta$.

**B.** For $\theta = 0$: $1 + 0 = 1/1$, so the identity holds.

**C.** Square $\tan\theta = \sin\theta/\cos\theta$ to get $\tan^2\theta = \sin^2\theta/\cos^2\theta = 1/\cos^2\theta$.

**D.** Divide $\sin^2\theta + \cos^2\theta = 1$ by $\cos^2\theta$, which is allowed because $\cos\theta \neq 0$: $\sin^2\theta/\cos^2\theta + 1 = 1/\cos^2\theta$, and $\sin^2\theta/\cos^2\theta = \tan^2\theta$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S40-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • MAIN • Q3/5

1460

**QUESTION ID**

## S40-M-Q3

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. Divide $\sin^2\theta + \cos^2\theta = 1$ by $\cos^2\theta$, which is allowed because $\cos\theta \neq 0$: $\sin^2\theta/\cos^2\theta + 1 = 1/\cos^2\theta$, and $\sin^2\theta/\cos^2\theta = \tan^2\theta$.**

**GRADING NOTE** Divide the Pythagorean identity by $\cos^2\theta$. The others divide by $\sin^2\theta$ and then treat $\cos^2\theta/\sin^2\theta$ as $\tan^2\theta$ (it is $1/\tan^2\theta$), replace $\sin^2\theta$ by 1, or check one angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S40-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • MAIN • Q3/5

θ is in quadrant II, and its point on the unit circle is (x, y) with $x < 0 < y$. Which argument proves $\sin^2\theta + \cos^2\theta = 1$ with a reference triangle?

**A.** The reference triangle has legs x and y, so $x + y = 1$, which means $\cos\theta + \sin\theta = 1$.

**B.** The reference triangle has legs $|x|$ and $|y|$ and hypotenuse 1, so $|x|^2 + |y|^2 = 1$. Squaring removes the signs, so $x^2 + y^2 = \cos^2\theta + \sin^2\theta = 1$.

**C.** The legs of the reference triangle are x and y, and x is negative in quadrant II, so $x^2 + y^2 = -1$ there, which means $\cos^2\theta + \sin^2\theta = -1$ instead of 1.

**D.** The identity was proved for acute angles, so it is true in quadrant II with no further work.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S40-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • MAIN • Q4/5

1461

**QUESTION ID**

## S40-M-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. The reference triangle has legs $|x|$ and $|y|$ and hypotenuse 1, so $|x|^2 + |y|^2 = 1$. Squaring removes the signs, so $x^2 + y^2 = \cos^2\theta + \sin^2\theta = 1$.**

**GRADING NOTE** Side lengths are $|x|$ and $|y|$; squares remove the signs. The others add the legs, mishandle the negative coordinate, or skip the argument.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S40-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • MAIN • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Which argument proves $(\sin\theta + \cos\theta)^2 = 1 + 2\sin\theta\cos\theta$?

**A.** $(\sin\theta + \cos\theta)^2 = \sin^2\theta + 2\sin\theta\cos\theta + \cos^2\theta$, and $\sin^2\theta + \cos^2\theta = 1$.

**B.** $(\sin\theta + \cos\theta)^2 = \sin^2\theta + \cos^2\theta = 1$, so the term $2\sin\theta\cos\theta$ is extra.

**C.** For $\theta = 0$: $(0 + 1)^2 = 1 + 0$, so the identity holds.

**D.** $(\sin\theta + \cos\theta)^2 = 1^2 = 1$ because $\sin\theta + \cos\theta = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • MAIN • Q5/5

1462

QUESTION ID
**S40-M-Q5**

TYPE
Multiple choice

CORRECT ANSWER
**A. $(\sin\theta + \cos\theta)^2 = \sin^2\theta + 2\sin\theta\cos\theta + \cos^2\theta$, and $\sin^2\theta + \cos^2\theta = 1$.**

GRADING NOTE Expand the square, then use the Pythagorean identity. The others drop the middle term, check one angle, or use the false statement $\sin\theta + \cos\theta = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • MAIN • Q5/5

SET 40 • HSF-TF.C.8

# BACKWARD 1

The Pythagorean Theorem • Grade 8 • Geometry • 8.G.B.7

**ANSWER KEY** Reduced copy of the question page, with its answer

A right triangle has legs 3 and 4. Find the hypotenuse.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 40 • BACKWARD 1 • Q1/5

1464

QUESTION ID

**S40-B1-Q1**

TYPE

Short answer

CORRECT ANSWER

**5**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 40 • BACKWARD 1 • Q1/5

A right triangle has hypotenuse 1 and a leg 0.6. Find the other leg.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 40 • BACKWARD 1 • Q2/5

1465

QUESTION ID

**S40-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**0.8**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 40 • BACKWARD 1 • Q2/5

**True or false?**

In a right triangle, $a^2 + b^2 = c^2$ where c is the hypotenuse.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 40 • BACKWARD 1 • Q3/5

1466

QUESTION ID

**S40-B1-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Pythagorean Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 40 • BACKWARD 1 • Q3/5

Legs 5 and 12 give hypotenuse

**A.** 60

**B.** 7

**C.** 17

**D.** 13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 40 • BACKWARD 1 • Q4/5

1467

QUESTION ID

**S40-B1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**D. 13**

**GRADING NOTE** √169.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 40 • BACKWARD 1 • Q4/5

Legs a and b and hypotenuse c: write $c^2$ in terms of a and b.

$c^2$ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 40 • BACKWARD 1 • Q5/5

1468

QUESTION ID

**S40-B1-Q5**

TYPE

Short answer

CORRECT ANSWER

**$a^2 + b^2$**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 40 • BACKWARD 1 • Q5/5

SET 40 • HSF-TF.C.8

# BACKWARD 2

Coordinates of the point at angle θ • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

Which gives the coordinates of the point on the unit circle at angle θ?

**A.** (sin θ, cos θ)

**B.** (θ, sin θ)

**C.** (cos θ, sin θ)

**D.** (cos θ, tan θ)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 40 • BACKWARD 2 • Q1/5

1470

QUESTION ID

**S40-B2-Q1**

TYPE

Multiple choice

CORRECT ANSWER

**C. (cos θ, sin θ)**

**GRADING NOTE** The x-coordinate is cos θ and the y-coordinate is sin θ.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 40 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Give cos(2π/3) and sin(2π/3) exactly.

cos(2π/3) = _____

sin(2π/3) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 40 • BACKWARD 2 • Q2/5

1471

QUESTION ID

**S40-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**−1/2; √3/2**

GRADING NOTE Form required: both exact. The point is (−1/2, √3/2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 40 • BACKWARD 2 • Q2/5

**True or false?**

sin θ is the x-coordinate of the point at angle θ.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 40 • BACKWARD 2 • Q3/5

1472

QUESTION ID

## S40-B2-Q3

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** It is the y-coordinate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 40 • BACKWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

The point at angle π/4 is

**A.** (1/2, √3/2)

**B.** (1, 1)

**C.** (√3/2, 1/2)

**D.** (√2/2, √2/2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 40 • BACKWARD 2 • Q4/5

1473

QUESTION ID

**S40-B2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**D. (√2/2, √2/2)**

GRADING NOTE 45°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 40 • BACKWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

What is the radius of the unit circle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 40 • BACKWARD 2 • Q5/5

1474

QUESTION ID

**S40-B2-Q5**

TYPE

Short answer

CORRECT ANSWER

**1**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 40 • BACKWARD 2 • Q5/5

SET 40 • HSF-TF.C.8

# BACKWARD 3

Expand and simplify algebraic squares • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

Expand $(a + b)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 40 • BACKWARD 3 • Q1/5

1476

QUESTION ID

**S40-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**$a^2 + 2a \cdot b + b^2$**

**GRADING NOTE** Form required: expanded. $(a + b)(a + b) = a^2 + ab + ab + b^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 40 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Simplify $(a/c)^2 + (b/c)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 40 • BACKWARD 3 • Q2/5

1477

QUESTION ID

**S40-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**$(a^2 + b^2)/c^2$**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 40 • BACKWARD 3 • Q2/5

**True or false?**

$\sin^2\theta$ means $\sin(\theta^2)$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 40 • BACKWARD 3 • Q3/5

1478

QUESTION ID

**S40-B3-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** $\sin^2\theta$ means $(\sin\theta)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 40 • BACKWARD 3 • Q3/5

Dividing $x^2 + y^2 = 1$ by $x^2$ gives

**A.** $1 + y^2/x^2 = 1/x^2$

**B.** $1 + y^2 = 1/x$

**C.** $y^2 = 1$

**D.** $x^2 + y^2 = 1/x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 40 • BACKWARD 3 • Q4/5

1479

QUESTION ID

**S40-B3-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**A. $1 + y^2/x^2 = 1/x^2$**

**GRADING NOTE** Divide each term.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 40 • BACKWARD 3 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Expand $(s + c)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 40 • BACKWARD 3 • Q5/5

1480

QUESTION ID

**S40-B3-Q5**

TYPE

Short answer

CORRECT ANSWER

**$s^2 + 2s \cdot c + c^2$**

GRADING NOTE Form required: expanded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B3-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 40 • BACKWARD 3 • Q5/5

SET 40 • HSF-TF.C.8

# FORWARD 1

Prove identities derived from the Pythagorean identity • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8

**ANSWER KEY** Reduced copy of the question page, with its answer

Which argument proves $\sin^4\theta - \cos^4\theta = \sin^2\theta - \cos^2\theta$?

**A.** Divide both sides by $\sin^2\theta - \cos^2\theta$ to get $\sin^2\theta + \cos^2\theta = 1$, which is the Pythagorean identity, so the identity is true.

**B.** $\sin^4\theta - \cos^4\theta = (\sin\theta - \cos\theta)^4$, which equals $\sin^2\theta - \cos^2\theta$.

**C.** For $\theta = \pi/4$ both sides are 0, so the identity holds.

**D.** $\sin^4\theta - \cos^4\theta = (\sin^2\theta - \cos^2\theta)(\sin^2\theta + \cos^2\theta) = (\sin^2\theta - \cos^2\theta)(1) = \sin^2\theta - \cos^2\theta$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • FORWARD 1 • Q1/5

1482

QUESTION ID

**S40-F1-Q1**

TYPE

Multiple choice

CORRECT ANSWER

**D.**
**$\sin^4\theta - \cos^4\theta = (\sin^2\theta - \cos^2\theta)(\sin^2\theta + \cos^2\theta) = (\sin^2\theta - \cos^2\theta)(1) = \sin^2\theta - \cos^2\theta.$**

**GRADING NOTE** Factor as a difference of squares and use the Pythagorean identity. The others assume the identity (and divide by an expression that can be 0), use the false rule $a^4 - b^4 = (a - b)^4$, or check one angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • FORWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Which argument proves $(1 - \cos\theta)(1 + \cos\theta) = \sin^2\theta$?

**A.** For $\theta = \pi/2$: $(1 - 0)(1 + 0) = 1 = \sin^2(\pi/2)$, so the identity holds.

**B.** $(1 - \cos\theta)(1 + \cos\theta) = 1 - 2\cos\theta + \cos^2\theta$, and by the Pythagorean identity this equals $\sin^2\theta$.

**C.** $(1 - \cos\theta)(1 + \cos\theta) = 1 - \cos^2\theta$, and $1 - \cos^2\theta = \sin^2\theta$ by the Pythagorean identity.

**D.** $(1 - \cos\theta)(1 + \cos\theta) = 1 + \cos^2\theta$, which equals $\sin^2\theta$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • FORWARD 1 • Q2/5

1483

**QUESTION ID**

**S40-F1-Q2**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. $(1 - \cos\theta)(1 + \cos\theta) = 1 - \cos^2\theta$, and $1 - \cos^2\theta = \sin^2\theta$ by the Pythagorean identity.**

**GRADING NOTE** Multiply (difference of squares), then use the identity. The others multiply incorrectly or check one angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • FORWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

$\sin \theta + \cos \theta = 1$ for every $\theta$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • FORWARD 1 • Q3/5

1484

QUESTION ID

**S40-F1-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

GRADING NOTE $\theta = \pi/4$ gives $\sqrt{2}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • FORWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Which equals $1 - \sin^2\theta$?

**A.** $\tan^2\theta$

**B.** $1 - \cos^2\theta$

**C.** $\cos\theta$

**D.** $\cos^2\theta$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • FORWARD 1 • Q4/5

1485

QUESTION ID
**S40-F1-Q4**

TYPE
Multiple choice

CORRECT ANSWER
**D. $\cos^2\theta$**

**GRADING NOTE** Rearranged identity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • FORWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Which argument proves $\cos^2\theta - \sin^2\theta = 2\cos^2\theta - 1$?

**A.** Replace $\sin^2\theta$ with $1 - \cos^2\theta$:
$\cos^2\theta - (1 - \cos^2\theta) = \cos^2\theta - 1 + \cos^2\theta = 2\cos^2\theta - 1$.

**B.** Add $\sin^2\theta$ to both sides to get $\cos^2\theta = 2\cos^2\theta - 1 + \sin^2\theta$, which is always true.

**C.** For $\theta = 0$: $1 - 0 = 2 - 1$, so the identity holds.

**D.** Replace $\sin^2\theta$ with $1 - \cos^2\theta$: $\cos^2\theta - 1 - \cos^2\theta = -1$, so the identity is false for every $\theta$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • FORWARD 1 • Q5/5

1486

QUESTION ID
**S40-F1-Q5**

TYPE
Multiple choice

CORRECT ANSWER
**A. Replace $\sin^2\theta$ with $1 - \cos^2\theta$: $\cos^2\theta - (1 - \cos^2\theta) = \cos^2\theta - 1 + \cos^2\theta = 2\cos^2\theta - 1$.**

GRADING NOTE Substitute $\sin^2\theta = 1 - \cos^2\theta$ and subtract the whole expression. The others drop the parentheses, check one angle, or end with an unproved statement.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • FORWARD 1 • Q5/5

SET 40 • HSF-TF.C.8

# FORWARD 2

Use the angle-addition formulas • High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9

Use sin(a + b) = sin a cos b + cos a sin b to find sin(75°) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 40 • FORWARD 2 • Q1/5

1488

**QUESTION ID**

## S40-F2-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**(√6 + √2)/4**

**GRADING NOTE** Any equivalent form. a = 45°, b = 30°.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S40-F2-Q1**

**High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 40 • FORWARD 2 • Q1/5**

Use cos(a + b) = cos a cos b − sin a sin b with a = b = θ. Which expression equals cos 2θ?

**A.** $\cos^2\theta + \sin^2\theta$

**B.** $2\cos\theta$

**C.** $2\sin\theta\cos\theta$

**D.** $\cos^2\theta - \sin^2\theta$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 40 • FORWARD 2 • Q2/5

1489

**QUESTION ID**

## S40-F2-Q2

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. $\cos^2\theta - \sin^2\theta$**

**GRADING NOTE** $\cos(\theta + \theta) = \cos\theta\cos\theta - \sin\theta\sin\theta$. The others double the cosine, change the sign (that equals 1), or give $\sin 2\theta$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 40 • FORWARD 2 • Q2/5

**True or false?**

$\cos 2\theta = 1 - 2\sin^2\theta$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S40-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 40 • FORWARD 2 • Q3/5

1490

QUESTION ID

**S40-F2-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

**GRADING NOTE** Use the Pythagorean identity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S40-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 40 • FORWARD 2 • Q3/5

sin 2θ equals

**A.** $\sin^2\theta - \cos^2\theta$

**B.** 2sin θ

**C.** 2sin θ cos θ

**D.** sin θ + cos θ

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 40 • FORWARD 2 • Q4/5

1491

QUESTION ID

## S40-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

**C. 2sin θ cos θ**

**GRADING NOTE** a = b = θ.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S40-F2-Q4**

**High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 40 • FORWARD 2 • Q4/5**

$\sin 2\theta = 2\sin\theta\cos\theta$. Use $(\sin\theta + \cos\theta)^2 = 1 + 2\sin\theta\cos\theta$ to write $(\sin\theta + \cos\theta)^2$ in terms of $\sin 2\theta$.

**A.** $2 + \sin 2\theta$

**B.** $\sin 2\theta$

**C.** $1 + \sin 2\theta$

**D.** $1 + 2\sin 2\theta$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 40 • FORWARD 2 • Q5/5

1492

**QUESTION ID**

## S40-F2-Q5

**TYPE**

Multiple choice

**CORRECT ANSWER**

## C. $1 + \sin 2\theta$

**GRADING NOTE** Replace $2\sin\theta\cos\theta$ with $\sin 2\theta$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 40 • FORWARD 2 • Q5/5

SET 41 • HSF-TF.C.8 • TRIGONOMETRIC FUNCTIONS

# Use the Pythagorean identity to find trigonometric values

| | | |
|---|---|---|
| **MAIN** | Algebra II main question | **HSF-TF.C.8** |
| **BACKWARD 1** | Signs of sine, cosine, and tangent by quadrant | **HSF-TF.A.2** |
| **BACKWARD 2** | Square roots of fractions | **8.EE.A.2** |
| **BACKWARD 3** | Solve $x^2 = c$ | **HSA-REI.B.4.b** |
| **FORWARD 1** | Find values when the quadrant is not given, and combine values | **HSF-TF.C.8** |
| **FORWARD 2** | Find the angles with inverse trigonometric functions | **HSF-TF.B.7** |

SET 41 • HSF-TF.C.8

# MAIN

Algebra II main question • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8

**ANSWER KEY** Reduced copy of the question page, with its answer

sin θ = 3/5 and θ is in quadrant II. Find cos θ and tan θ exactly.

cos θ = _____

tan θ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • MAIN • Q1/5

1495

**QUESTION ID**

**S41-M-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**−4/5; −3/4**

**GRADING NOTE** Form required: both exact. $\cos^2\theta = 16/25$ and $\cos\theta < 0$ in quadrant II.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

cos θ = −5/13 and θ is in quadrant III. Find sin θ and tan θ exactly.

sin θ = _____

tan θ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • MAIN • Q2/5

1496

**QUESTION ID**

**S41-M-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**−12/13; 12/5**

**GRADING NOTE** Form required: both exact. $\sin^2\theta = 144/169$ and $\sin\theta < 0$ in quadrant III.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • MAIN • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

tan θ = −2 and θ is in quadrant IV. Find sin θ and cos θ exactly.

sin θ = _____

cos θ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • MAIN • Q3/5

1497

**QUESTION ID**

## S41-M-Q3

**TYPE**

Short answer

**CORRECT ANSWER**

**−2√5/5; √5/5**

**GRADING NOTE** Form required: both exact (−2/√5 and 1/√5 are equivalent). The point (1, −2) has distance √5 from the origin.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • MAIN • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

sin θ = −1/3 and θ is in quadrant IV. Find cos θ and tan θ exactly.

cos θ = _____

tan θ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • MAIN • Q4/5

1498

**QUESTION ID**

**S41-M-Q4**

**TYPE**

Short answer

**CORRECT ANSWER**

**2√2/3; −√2/4**

**GRADING NOTE** Form required: both exact (−1/(2√2) is equivalent). $\cos^2\theta = 8/9$ and cos θ > 0 in quadrant IV.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • MAIN • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

$\cos\theta = 0.6$ and $\sin\theta < 0$. Find $\sin\theta$ and $\tan\theta$.

$\sin\theta =$ _____

$\tan\theta =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • MAIN • Q5/5

1499

**QUESTION ID**

## S41-M-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

**$-0.8$; $-4/3$**

**GRADING NOTE** Form required: both. $\sin^2\theta = 0.64$; $\tan\theta = -0.8/0.6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • MAIN • Q5/5

SET 41 • HSF-TF.C.8

# BACKWARD 1

Signs of sine, cosine, and tangent by quadrant • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

In quadrant II, which of sin, cos, and tan are positive?

**A.** sin only

**B.** sin and tan

**C.** cos only

**D.** all three

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 41 • BACKWARD 1 • Q1/5

1501

QUESTION ID

**S41-B1-Q1**

TYPE

Multiple choice

CORRECT ANSWER

**A. sin only**

**GRADING NOTE** In quadrant II, $y > 0$ and $x < 0$, so sin > 0, cos < 0, and tan = $y/x$ < 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 41 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

In quadrant III, what is the sign of tan θ? Answer Positive or Negative.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 41 • BACKWARD 1 • Q2/5

1502

QUESTION ID

**S41-B1-Q2**

TYPE

Short answer

CORRECT ANSWER

**Positive**

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 41 • BACKWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

In quadrant IV, $\cos\theta < 0$ and $\sin\theta > 0$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 41 • BACKWARD 1 • Q3/5

1503

QUESTION ID

**S41-B1-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Quadrant IV has $x > 0$ and $y < 0$, so $\cos\theta > 0$ and $\sin\theta < 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 41 • BACKWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

If $\sin\theta < 0$ and $\cos\theta > 0$, $\theta$ is in quadrant

**A.** I

**B.** II

**C.** IV

**D.** III

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 41 • BACKWARD 1 • Q4/5

1504

**QUESTION ID**

## S41-B1-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

## C. IV

**GRADING NOTE** (+, −).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 41 • BACKWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

If tan θ < 0 and sin θ > 0, which quadrant is θ in?
Answer I, II, III, or IV.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 41 • BACKWARD 1 • Q5/5

1505

**QUESTION ID**
**S41-B1-Q5**

**TYPE**
Short answer

**CORRECT ANSWER**
**II**

**GRADING NOTE** Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 41 • BACKWARD 1 • Q5/5

SET 41 • HSF-TF.C.8

# BACKWARD 2

Square roots of fractions • Grade 8 • Expressions & Equations • 8.EE.A.2

**ANSWER KEY** Reduced copy of the question page, with its answer

Find √(16/25).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 41 • BACKWARD 2 • Q1/5

1507

QUESTION ID

**S41-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**4/5**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 41 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Find √(8/9) in simplest radical form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 41 • BACKWARD 2 • Q2/5

1508

QUESTION ID

**S41-B2-Q2**

TYPE

Short answer

CORRECT ANSWER

**2√2/3**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 41 • BACKWARD 2 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

√(144/169) = 12/169.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 41 • BACKWARD 2 • Q3/5

1509

QUESTION ID

## S41-B2-Q3

TYPE

True or false

CORRECT ANSWER

## B. False

**GRADING NOTE** Take the root of both: √144/√169 = 12/13.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 41 • BACKWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

√(1/5) equals

**A.** 5

**B.** √5

**C.** 1/25

**D.** √5/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 41 • BACKWARD 2 • Q4/5

1510

QUESTION ID

**S41-B2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**D. √5/5**

**GRADING NOTE** Rationalize.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 41 • BACKWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Find √0.64.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 41 • BACKWARD 2 • Q5/5

1511

QUESTION ID

**S41-B2-Q5**

TYPE

Short answer

CORRECT ANSWER

**0.8**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 41 • BACKWARD 2 • Q5/5

SET 41 • HSF-TF.C.8

# BACKWARD 3

Solve $x^2 = c$ • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve $x^2 = 16/25$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 41 • BACKWARD 3 • Q1/5

1513

QUESTION ID

**S41-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**4/5 or −4/5**

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 41 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve $x^2 = 1 - 9/25$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 41 • BACKWARD 3 • Q2/5

1514

QUESTION ID

**S41-B3-Q2**

TYPE

Short answer

CORRECT ANSWER

**±4/5**

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 41 • BACKWARD 3 • Q2/5

**True or false?**

$x^2 = 0.36$ has only the solution 0.6.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 41 • BACKWARD 3 • Q3/5

1515

**QUESTION ID**

## S41-B3-Q3

**TYPE**

True or false

**CORRECT ANSWER**

## B. False

**GRADING NOTE** Also −0.6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 41 • BACKWARD 3 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

$x^2 + (3/5)^2 = 1$ gives x =

**A.** ±2/5

**B.** 2/5

**C.** 4/5

**D.** ±4/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 41 • BACKWARD 3 • Q4/5

1516

QUESTION ID

**S41-B3-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**D. ±4/5**

**GRADING NOTE** $x^2 = 16/25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 41 • BACKWARD 3 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Solve $5x^2 = 1$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 41 • BACKWARD 3 • Q5/5

1517

QUESTION ID

**S41-B3-Q5**

TYPE

Short answer

CORRECT ANSWER

**$\pm\sqrt{5}/5$**

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 41 • BACKWARD 3 • Q5/5

SET 41 • HSF-TF.C.8

# FORWARD 1

Find values when the quadrant is not given, and combine values • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8

$\sin\theta = 3/5$ and $0 \le \theta < 2\pi$. Give $\cos\theta$ when $\theta$ is in quadrant I and when $\theta$ is in quadrant II.

$\cos\theta$ in quadrant I: _____

$\cos\theta$ in quadrant II: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • FORWARD 1 • Q1/5

1519

**QUESTION ID**

**S41-F1-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**4/5; −4/5**

**GRADING NOTE** Form required: both. $\cos^2\theta = 1 - 9/25 = 16/25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • FORWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

$\cos\theta = -5/13$ and $\theta$ is in quadrant III. Find $\sin\theta \cdot \cos\theta$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • FORWARD 1 • Q2/5

1520

**QUESTION ID**

**S41-F1-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**60/169**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • FORWARD 1 • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

**True or false?**

Knowing tan θ = 1 determines sin θ uniquely.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • FORWARD 1 • Q3/5

1521

QUESTION ID

**S41-F1-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** θ can be in quadrant I or III, so sin θ = ±√2/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • FORWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

If $\sin\theta = 0.8$ and $\tan\theta < 0$, then $\cos\theta =$

**A.** $-0.8$

**B.** $0.2$

**C.** $0.6$

**D.** $-0.6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • FORWARD 1 • Q4/5

1522

QUESTION ID

**S41-F1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**D. −0.6**

GRADING NOTE Quadrant II.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • FORWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Use $1 + \tan^2\theta = 1/\cos^2\theta$ to find $\cos\theta$ if $\tan\theta = 3$ and $\theta$ is in quadrant III.

$\cos\theta =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • FORWARD 1 • Q5/5

1523

**QUESTION ID**
**S41-F1-Q5**

**TYPE**
Short answer

**CORRECT ANSWER**
**$-\sqrt{10}/10$**

**GRADING NOTE** Any equivalent form ($-1/\sqrt{10}$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • FORWARD 1 • Q5/5

SET 41 • HSF-TF.C.8

# FORWARD 2

Find the angles with inverse trigonometric functions • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7

**ANSWER KEY** Reduced copy of the question page, with its answer

sin θ = 3/5 with 0 ≤ θ < 2π. Find both values of θ, in radians, to the nearest thousandth. A calculator may be used.

θ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 41 • FORWARD 2 • Q1/5

1525

**QUESTION ID**

## S41-F2-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**0.644 or 2.498**

**GRADING NOTE** Form required: both, in any order, rounded to 0.001. arcsin(0.6) ≈ 0.6435 and π − arcsin(0.6) ≈ 2.4981.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S41-F2-Q1**

**High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 41 • FORWARD 2 • Q1/5**

**ANSWER KEY** Reduced copy of the question page, with its answer

cos θ = −5/13 and θ is in quadrant III (0 ≤ θ < 2π). Find θ, in radians, to the nearest thousandth. A calculator may be used.

θ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 41 • FORWARD 2 • Q2/5

1526

**QUESTION ID**

## S41-F2-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**4.318**

**GRADING NOTE** Rounding to 0.001. θ = 2π − arccos(−5/13) ≈ 4.3176 (arccos(−5/13) ≈ 1.9656 is in quadrant II).

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S41-F2-Q2**

**High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 41 • FORWARD 2 • Q2/5**

**True or false?**

arcsin gives only angles between −π/2 and π/2.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 41 • FORWARD 2 • Q3/5

1527

QUESTION ID

## S41-F2-Q3

TYPE

True or false

CORRECT ANSWER

## A. True

**GRADING NOTE** Restricted range.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 41 • FORWARD 2 • Q3/5

The quadrant II angle with sin θ = 3/5 is

**A.** π + arcsin(3/5)

**B.** π − arcsin(3/5)

**C.** arcsin(3/5)

**D.** −arcsin(3/5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 41 • FORWARD 2 • Q4/5

1528

**QUESTION ID**

## S41-F2-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. π − arcsin(3/5)**

**GRADING NOTE** Reflect over the y-axis.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S41-F2-Q4**

**High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 41 • FORWARD 2 • Q4/5**

sin θ = 3/5 and θ is in quadrant II. Why does the Pythagorean identity give cos θ, while an inverse function is needed to find θ itself?

**A.** cos θ cannot be found from sin θ alone; θ must be found first with an inverse function, and only then can cos θ be computed.

**B.** The identity relates the values sin θ and cos θ to each other; finding the angle from a value means undoing sine, which needs arcsin (and the quadrant).

**C.** The identity gives θ directly: θ = 1 − sin²θ = 16/25, so no inverse function is needed for θ either.

**D.** The inverse function is needed only for angles measured in degrees; in radians the identity gives θ directly.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 41 • FORWARD 2 • Q5/5

1529

**QUESTION ID**

## S41-F2-Q5

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. The identity relates the values sin θ and cos θ to each other; finding the angle from a value means undoing sine, which needs arcsin (and the quadrant).**

**GRADING NOTE** Values versus angles. The others misuse the identity, confuse units, or claim the identity cannot give cos θ (cos θ = −4/5 here).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 41 • FORWARD 2 • Q5/5

SET 42 • HSF-IF.B.4 • INTERPRETING FUNCTIONS

# Interpret key features of polynomial and rational models

| | | |
|---|---|---|
| **MAIN** | Algebra II main question | **HSF-IF.B.4** |
| **BACKWARD 1** | Interpret key features of linear and quadratic models | **HSF-IF.B.4** |
| **BACKWARD 2** | Describe increasing and decreasing behavior | **8.F.B.5** |
| **BACKWARD 3** | Zeros and sign from factors | **HSA-APR.B.3** |
| **FORWARD 1** | Sketch a model from a verbal description of key features | **HSF-IF.B.4** |
| **FORWARD 2** | Compare key features across representations | **HSF-IF.C.9** |

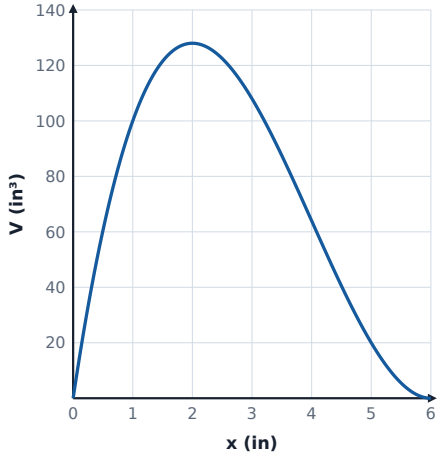
SET 42 • HSF-IF.B.4

# MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4

**ANSWER KEY** Reduced copy of the question page, with its answer

An open box is made from a 12-in square sheet by cutting squares of side x from the corners. Its volume is $V(x) = x(12 - 2x)^2$. Use the graph to give the x that maximizes V, the maximum volume, and the interval where V increases.




x (in) = _____

Maximum volume (in³): _____

V increases for: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • MAIN • Q1/5

1532

**QUESTION ID**

## S42-M-Q1

**TYPE**

Short answer

**CORRECT ANSWER**

**2; 128; 0 < x < 2**

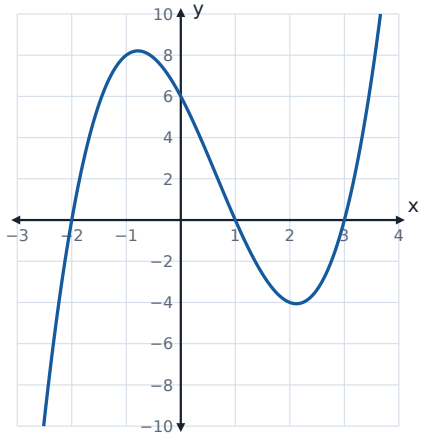
**GRADING NOTE** Form required: all three. $V(2) = 2 \cdot 64 = 128$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

The graph of f(x) = (x + 2)(x − 1)(x − 3) is shown. Give the two intervals where f(x) > 0 and the two intervals where f(x) < 0, each pair from left to right.




f > 0, first interval: _____

f > 0, second interval: _____

f < 0, first interval: _____

f < 0, second interval: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • MAIN • Q2/5

1533

**QUESTION ID**

## S42-M-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**−2 < x < 1; x > 3;**
**x < −2; 1 < x < 3**

**GRADING NOTE** Form required: all four. The zeros −2, 1, 3 split the line; the signs alternate, starting negative on the left (positive leading coefficient, odd degree).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • MAIN • Q2/5

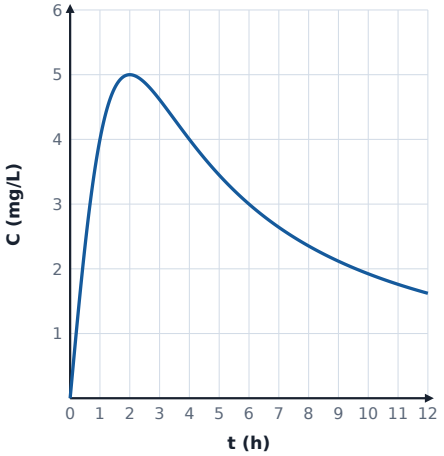
**ANSWER KEY** Reduced copy of the question page, with its answer

A drug concentration is $C(t) = 20t/(t^2 + 4)$ mg/L after t hours. Use the graph to give the maximum concentration, when it occurs, and the value C approaches as t grows.

Maximum (mg/L): _____

t (hours) = _____

C approaches: _____




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • MAIN • Q3/5

1534

QUESTION ID
**S42-M-Q3**

TYPE
Short answer

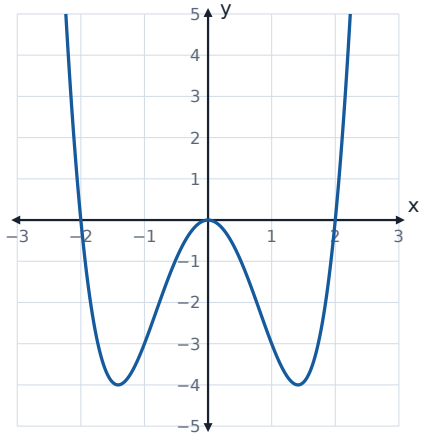
CORRECT ANSWER
**5; 2; 0**

GRADING NOTE Form required: all three. C(2) = 40/8 = 5; for large t the denominator grows faster, so C → 0: the drug is eventually eliminated.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • MAIN • Q3/5

The graph of $f(x) = x^4 - 4x^2$ is shown. Give its x-intercepts, the y-value of its relative maximum, and the y-value of its relative minima.




x-intercepts: _____

Relative maximum value: _____

Relative minimum value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • MAIN • Q4/5

1535

**QUESTION ID**

## S42-M-Q4

**TYPE**

Short answer

**CORRECT ANSWER**

**{−2, 0, 2}; 0; −4**

**GRADING NOTE** Form required: all three. $f = x^2(x^2 - 4)$; relative maximum (0, 0); relative minima at $(\pm\sqrt{2}, -4)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • MAIN • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

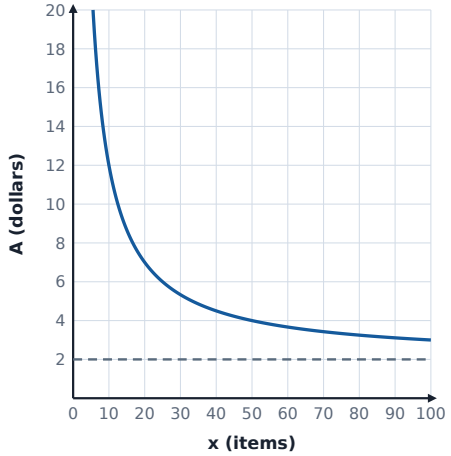
The average cost of making x items is A(x) = (100 + 2x)/x dollars. The graph is shown. Which describes the end behavior and its meaning?

**A.** As x grows, A(x) increases without bound: each extra item adds $2, so the average cost keeps rising.

**B.** As x grows, A(x) approaches $0 (the asymptote y = 0): with enough items the fixed cost and the item cost both vanish.

**C.** As x grows, A(x) approaches $100 (the asymptote y = 100): the $100 fixed cost dominates the average for large x.

**D.** As x grows, A(x) approaches $2 (the asymptote y = 2): the $100 fixed cost is spread over more items, so the average approaches the $2 cost of each item.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • MAIN • Q5/5

1536

**QUESTION ID**

## S42-M-Q5

**TYPE**

Multiple choice

**CORRECT ANSWER**

**D. As x grows, A(x) approaches $2 (the asymptote y = 2): the $100 fixed cost is spread over more items, so the average approaches the $2 cost of each item.**

**GRADING NOTE** A(x) = 2 + 100/x, and 100/x → 0. The others ignore the $2 per item, invert the effect of the fixed cost, or misread the graph.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • MAIN • Q5/5

SET 42 • HSF-IF.B.4

# BACKWARD 1

Interpret key features of linear and quadratic models • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4

A ball's height is $h(t) = -16t^2 + 32t + 4$ feet after t seconds. When is the height greatest, and what is the greatest height?

t (seconds) = _____

Greatest height (ft): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 1 • Q1/5

1538

**QUESTION ID**

**S42-B1-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**1; 20**

**GRADING NOTE** Form required: both. The vertex is at $t = -32/(2 \cdot -16) = 1$, and $h(1) = -16 + 32 + 4 = 20$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

C = 25 + 3x is the cost in dollars of x rides. What does the C-intercept (0, 25) mean?

**A.** A fixed cost of $25 before any rides

**B.** After 25 rides the cost is $3

**C.** Each ride costs $25

**D.** The cost of 3 rides is $25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 1 • Q2/5

1539

**QUESTION ID**

**S42-B1-Q2**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. A fixed cost of $25 before any rides**

**GRADING NOTE** At x = 0 rides, the cost is $25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 1 • Q2/5

**True or false?**

A parabola that opens down has a minimum.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 1 • Q3/5

1540

QUESTION ID

**S42-B1-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Its vertex is the highest point: a maximum.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 1 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

$h(t) = -16t^2 + 32t + 4$ increases for

**A.** all t

**B.** $0 < t < 1$

**C.** $t > 1$

**D.** $1 < t < 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 1 • Q4/5

1541

QUESTION ID

**S42-B1-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**B. 0 < t < 1**

**GRADING NOTE** Before the vertex.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Where is $y = (x - 2)(x - 6)$ negative?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 1 • Q5/5

1542

QUESTION ID

**S42-B1-Q5**

TYPE

Short answer

CORRECT ANSWER

**$2 < x < 6$**

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 1 • Q5/5

SET 42 • HSF-IF.B.4

# BACKWARD 2

Describe increasing and decreasing behavior • Grade 8 • Functions • 8.F.B.5

A graph rises from x = 0 to x = 2 and then falls. Which description is correct?

**A.** Increasing for every x > 0

**B.** Constant for 0 < x < 2

**C.** Increasing for 0 < x < 2 and decreasing for x > 2

**D.** Decreasing for 0 < x < 2 and increasing for x > 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q1

Grade 8 • Functions • 8.F.B.5 • Set 42 • BACKWARD 2 • Q1/5

1544

QUESTION ID

**S42-B2-Q1**

TYPE

Multiple choice

CORRECT ANSWER

**C. Increasing for 0 < x < 2 and decreasing for x > 2**

GRADING NOTE Rising means increasing; falling means decreasing.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q1

Grade 8 • Functions • 8.F.B.5 • Set 42 • BACKWARD 2 • Q1/5

A drug's concentration rises after a dose, reaches a peak, and then levels off toward 0. Which describes its graph against time?

**A.** It stays constant at its peak value.

**B.** It increases to a peak, then decreases and approaches 0.

**C.** It decreases to a minimum, then increases without bound.

**D.** It increases without bound.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q2

Grade 8 • Functions • 8.F.B.5 • Set 42 • BACKWARD 2 • Q2/5

1545

**QUESTION ID**

## S42-B2-Q2

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. It increases to a peak, then decreases and approaches 0.**

**GRADING NOTE** Rise, peak, then decay toward 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q2

Grade 8 • Functions • 8.F.B.5 • Set 42 • BACKWARD 2 • Q2/5

**True or false?**

A highest point between rising and falling parts is a relative maximum.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q3

**Grade 8 • Functions • 8.F.B.5 • Set 42 • BACKWARD 2 • Q3/5**

1546

QUESTION ID

**S42-B2-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q3

**Grade 8 • Functions • 8.F.B.5 • Set 42 • BACKWARD 2 • Q3/5**

A function that falls on $1 < x < 3$ is

**A.** constant there

**B.** decreasing there

**C.** undefined there

**D.** increasing there

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q4

Grade 8 • Functions • 8.F.B.5 • Set 42 • BACKWARD 2 • Q4/5

1547

**QUESTION ID**

**S42-B2-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. decreasing there**

**GRADING NOTE** Falling.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q4

Grade 8 • Functions • 8.F.B.5 • Set 42 • BACKWARD 2 • Q4/5

What is a relative minimum of a graph?

**A.** The lowest point of the whole graph, always

**B.** A point lower than all nearby points of the graph

**C.** A point higher than all nearby points

**D.** A point where the graph crosses the x-axis

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q5

Grade 8 • Functions • 8.F.B.5 • Set 42 • BACKWARD 2 • Q5/5

1548

**QUESTION ID**

## S42-B2-Q5

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. A point lower than all nearby points of the graph**

**GRADING NOTE** Relative means compared with nearby points; it need not be the lowest point overall.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q5

Grade 8 • Functions • 8.F.B.5 • Set 42 • BACKWARD 2 • Q5/5

SET 42 • HSF-IF.B.4

# BACKWARD 3

Zeros and sign from factors • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

**ANSWER KEY** Reduced copy of the question page, with its answer

List the zeros of $(x + 2)(x - 1)(x - 3)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 42 • BACKWARD 3 • Q1/5

1550

QUESTION ID

**S42-B3-Q1**

TYPE

Short answer

CORRECT ANSWER

**−2, 1, 3**

GRADING NOTE Form required: all.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 42 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

List the zeros of $x^4 - 4x^2 = x^2(x - 2)(x + 2)$ with multiplicities.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 42 • BACKWARD 3 • Q2/5

1551

**QUESTION ID**

**S42-B3-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**0 (2), 2 (1), −2 (1)**

**GRADING NOTE** Form required: all.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S42-B3-Q2**

**High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 42 • BACKWARD 3 • Q2/5**

**True or false?**

$V(x) = x(12 - 2x)^2$ has zeros 0 and 12.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 42 • BACKWARD 3 • Q3/5

1552

QUESTION ID
**S42-B3-Q3**

TYPE
True or false

CORRECT ANSWER
**B. False**

**GRADING NOTE** $12 - 2x = 0$ at $x = 6$, so the zeros are 0 and 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 42 • BACKWARD 3 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

$20t/(t^2 + 4) = 0$ when

**A.** never

**B.** $t = -2$

**C.** $t = 2$

**D.** $t = 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 42 • BACKWARD 3 • Q4/5

1553

QUESTION ID

**S42-B3-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**D. t = 0**

**GRADING NOTE** Numerator 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 42 • BACKWARD 3 • Q4/5

Is (x + 2)(x − 1)(x − 3) positive or negative at x = 2?
Answer Positive or Negative.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B3-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 42 • BACKWARD 3 • Q5/5

1554

**QUESTION ID**
**S42-B3-Q5**

**TYPE**
Short answer

**CORRECT ANSWER**
**Negative**

**GRADING NOTE** (4)(1)(−1) = −4.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S42-B3-Q5**

**High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 42 • BACKWARD 3 • Q5/5**

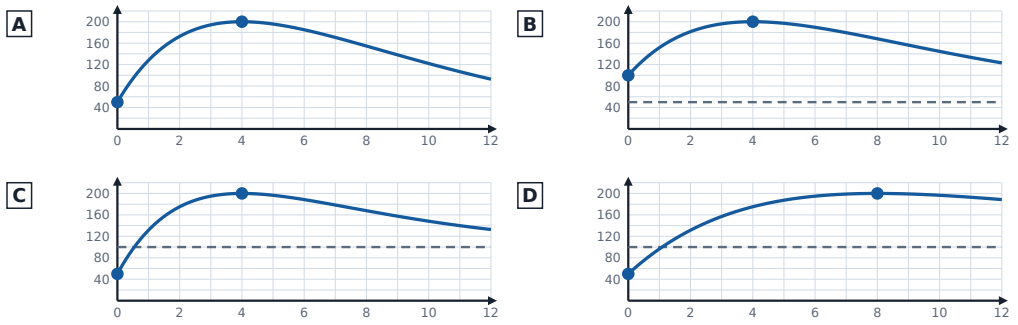
SET 42 • HSF-IF.B.4

# FORWARD 1

Sketch a model from a verbal description of key features • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4

**ANSWER KEY** Reduced copy of the question page, with its answer

Which graph could show a population that is 50 at t = 0, rises to a maximum of 200 at t = 4, and then decreases toward a level of 100 as t grows?




**A.** Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • FORWARD 1 • Q1/5

1556

QUESTION ID

## S42-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

## C. Graph C

**GRADING NOTE** The graph must start at (0, 50), peak at (4, 200), and level off toward y = 100. The others level off toward 0, start at 100 and level off toward 50, or peak at t = 8.

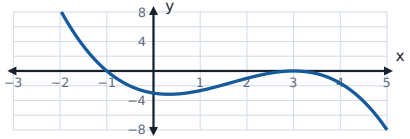
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • FORWARD 1 • Q1/5

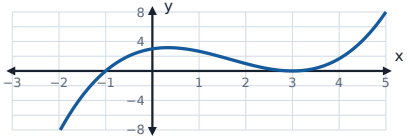
**ANSWER KEY** Reduced copy of the question page, with its answer

Which graph could show a cubic with zeros at −1 and 3 only, a negative y-intercept, falling to the right, and touching the x-axis at 3?

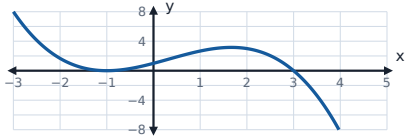
**A**



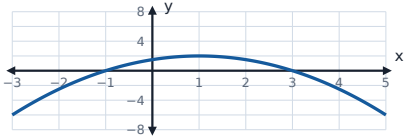

**B**




**C**




**D**




**A.** Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • FORWARD 1 • Q2/5

1557

**QUESTION ID**

## S42-F1-Q2

**TYPE**

Multiple choice

**CORRECT ANSWER**

## A. Graph A

**GRADING NOTE** Touching at 3 and falling to the right means the graph is below the axis on both sides of 3, so it is negative between −1 and 3 and the y-intercept is negative; for example $y = -(x + 1)(x - 3)^2/3$. The others rise to the right, touch at −1 instead of 3, or are a parabola (not a cubic).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • FORWARD 1 • Q2/5

**True or false?**

A graph can have a relative maximum that is not the absolute maximum.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • FORWARD 1 • Q3/5

1558

QUESTION ID
**S42-F1-Q3**

TYPE
True or false

CORRECT ANSWER
**A. True**

**GRADING NOTE** Cubics, for example.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • FORWARD 1 • Q3/5

A model rises to a peak and approaches 0 as t grows. Which formula fits?

**A.** $20t/(t^2 + 4)$

**B.** $20t$

**C.** $t^2$

**D.** $20 - t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • FORWARD 1 • Q4/5

1559

**QUESTION ID**

## S42-F1-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**A. $20t/(t^2 + 4)$**

**GRADING NOTE** Rational with a peak.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S42-F1-Q4**

**High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • FORWARD 1 • Q4/5**

Which quadratic function has zeros 0 and 6 and is positive between them (so it has a relative maximum between the zeros)?

**A.** $f(x) = (x - 6)(x + 6)$

**B.** $f(x) = 6 - x^2$

**C.** $f(x) = x(x - 6)$

**D.** $f(x) = x(6 - x)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • FORWARD 1 • Q5/5

1560

QUESTION ID

**S42-F1-Q5**

TYPE

Multiple choice

CORRECT ANSWER

**D. f(x) = x(6 − x)**

**GRADING NOTE** x(6 − x) is positive for 0 < x < 6 (maximum 9 at x = 3). x(x − 6) is negative there; the others have the wrong zeros.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • FORWARD 1 • Q5/5

SET 42 • HSF-IF.B.4

# FORWARD 2

Compare key features across representations • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9

**ANSWER KEY** Reduced copy of the question page, with its answer

$f(x) = x(12 - 2x)^2$, and g is a quadratic that opens downward with vertex (3, 120). On $0 < x < 6$, f has its maximum at x = 2. Give the maximum value of each function on $0 < x < 6$.

Maximum of f: _____

Maximum of g: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 42 • FORWARD 2 • Q1/5

1562

**QUESTION ID**

**S42-F2-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**128; 120**

**GRADING NOTE** Form required: both. $f(2) = 2 \cdot 8^2 = 128$; a downward-opening parabola has its maximum at its vertex, 120. So f has the greater maximum.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S42-F2-Q1**

**High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 42 • FORWARD 2 • Q1/5**

$C(t) = 20t/(t^2 + 4)$ and a table of $D(t)$ gives $D(1) = 3$, $D(2) = 6$, $D(3) = 4$. Among these values ($t = 1, 2, 3$), which function has the greater peak, and when?

**A.** C, with 6 at t = 2

**B.** D, with 4 at t = 3

**C.** D, with 6 at t = 2

**D.** C, with 5 at t = 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 42 • FORWARD 2 • Q2/5

1563

**QUESTION ID**

## S42-F2-Q2

**TYPE**

Multiple choice

**CORRECT ANSWER**

## C. D, with 6 at t = 2

**GRADING NOTE** $C(1) = 4$, $C(2) = 5$, $C(3) \approx 4.6$, so C peaks at 5 while D reaches 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 42 • FORWARD 2 • Q2/5

**True or false?**

Two functions with the same zeros have the same relative extrema.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 42 • FORWARD 2 • Q3/5

1564

QUESTION ID
**S42-F2-Q3**

TYPE
True or false

CORRECT ANSWER
**B. False**

**GRADING NOTE** Scale changes the extrema values.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S42-F2-Q3**

**High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 42 • FORWARD 2 • Q3/5**

Which has a horizontal asymptote?

**A.** $(x + 2)(x - 1)(x - 3)$

**B.** $x^4 - 4x^2$

**C.** $x(12 - 2x)^2$

**D.** $(100 + 2x)/x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 42 • FORWARD 2 • Q4/5

1565

QUESTION ID

**S42-F2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**D. (100 + 2x)/x**

**GRADING NOTE** Rational with equal degrees.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 42 • FORWARD 2 • Q4/5

A polynomial is given by a formula and a rational function is given by a graph. How can their end behavior be compared?

**A.** Use the polynomial's leading term for large |x|, and read the rational function's asymptote or trend from the graph.

**B.** Compare their y-intercepts, read from the formula and from the graph, since the larger intercept wins for large |x| in both directions.

**C.** Compare their values at x = 1, using the formula and the graph; the larger value at 1 stays larger for large |x|.

**D.** Compare the number of zeros of each: the function with more zeros has the stronger end behavior for large |x|.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 42 • FORWARD 2 • Q5/5

1566

QUESTION ID

## S42-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

**A. Use the polynomial's leading term for large |x|, and read the rational function's asymptote or trend from the graph.**

**GRADING NOTE** End behavior concerns large |x|. The others compare features near x = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 42 • FORWARD 2 • Q5/5

SET 43 • HSF-IF.B.4 • INTERPRETING FUNCTIONS

# Interpret periodic models from graphs and tables

| | | |
|---|---|---|
| **MAIN** | Algebra II main question | **HSF-IF.B.4** |
| **BACKWARD 1** | Interpret maximums and minimums on a graph | **HSF-IF.B.4** |
| **BACKWARD 2** | Read values from a table | **6.EE.C.9** |
| **BACKWARD 3** | Amplitude, period, and midline of a sinusoidal model | **HSF-TF.B.5** |
| **FORWARD 1** | Predict with periodicity | **HSF-IF.B.4** |
| **FORWARD 2** | Write a model from periodic features | **HSF-TF.B.5** |

SET 43 • HSF-IF.B.4

# MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4

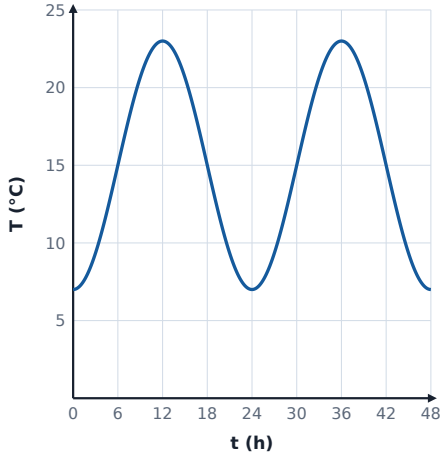
**ANSWER KEY** Reduced copy of the question page, with its answer

The graph shows a city's temperature T (°C) t hours after midnight on two days. Give the period in hours, the maximum temperature in °C, and the first time t (in hours) when it occurs.

Period: _____

Maximum: _____

First time t: _____




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • MAIN • Q1/5

1569

**QUESTION ID**

**S43-M-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**24; 23; 12**

**GRADING NOTE** Form required: all three. One cycle per day; the maximum 23°C first occurs at noon.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • MAIN • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

The table shows the water depth d (ft) at a harbor every 2 hours. Give the period, the depth at low tide, and the first time of low tide.

| t (h) | 0 | 2 | 4 | 6 | 8 | 10 | 12 | 14 | 16 | 18 |
|---|---|---|---|---|---|---|---|---|---|---|
| d (ft) | 9 | 7 | 3 | 1 | 3 | 7 | 9 | 7 | 3 | 1 |

Period (h): _____

Depth at low tide (ft): _____

First low tide at t (h) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • MAIN • Q2/5

1570

QUESTION ID

**S43-M-Q2**

TYPE

Short answer

CORRECT ANSWER

**12; 1; 6**

GRADING NOTE Form required: all three. The values repeat after 12 hours; the least depth is 1 ft, first at t = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • MAIN • Q2/5

**ANSWER KEY** Reduced copy of the question page, with its answer

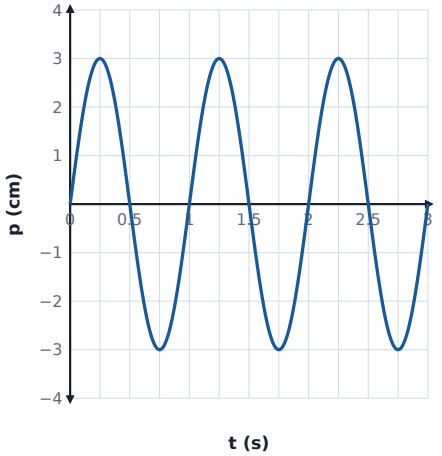
A piston's position p (cm) is graphed for $0 \le t \le 3$ seconds. Give the amplitude, the period, and the two intervals of the first cycle ($0 \le t \le 1$) where p increases, from left to right.

Amplitude (cm): _____

Period (s): _____

First interval: _____

Second interval: _____




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • MAIN • Q3/5

1571

**QUESTION ID**

## S43-M-Q3

**TYPE**

Short answer

**CORRECT ANSWER**

**3; 1; 0 < t < 0.25; 0.75 < t < 1**

**GRADING NOTE** Form required: all four. The wave reaches 3 at t = 0.25 and −3 at t = 0.75.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • MAIN • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

The table shows hours of daylight D in month m (m = 0 is January). Give the maximum, the minimum, the period, and the interval of m (from the table) in which daylight increases.

| m | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| D (h) | 9 | 9.4 | 10.5 | 12 | 13.5 | 14.6 | 15 | 14.6 | 13.5 | 12 | 10.5 | 9.4 | 9 |

Maximum (h): _____

Minimum (h): _____

Period (months): _____

Increasing for: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • MAIN • Q4/5

1572

QUESTION ID

**S43-M-Q4**

TYPE

Short answer

CORRECT ANSWER

**15; 9; 12; 0 < m < 6**

**GRADING NOTE** Form required: all four. The maximum 15 is at m = 6 and the minimum 9 at m = 0 and 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • MAIN • Q4/5

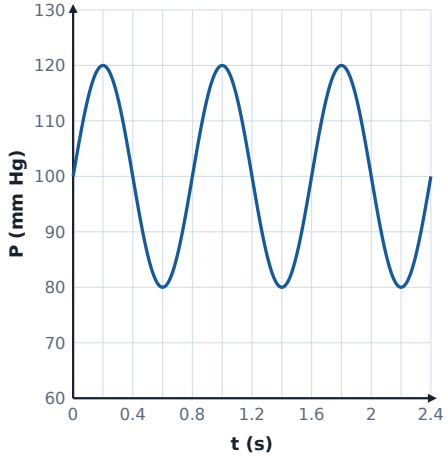
**ANSWER KEY** Reduced copy of the question page, with its answer

The graph shows blood pressure P (mm Hg) over 2.4 seconds. Give the maximum and minimum pressures and the number of heartbeats per minute.

Maximum: _____

Minimum: _____

Beats per minute: _____




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • MAIN • Q5/5

1573

**QUESTION ID**

## S43-M-Q5

**TYPE**

Short answer

**CORRECT ANSWER**

## 120; 80; 75

**GRADING NOTE** Form required: all three. Three cycles in 2.4 s give a period of 0.8 s, and 60/0.8 = 75.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • MAIN • Q5/5

SET 43 • HSF-IF.B.4

# BACKWARD 1

Interpret maximums and minimums on a graph • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4

A graph of a ball's height (m) against time peaks at 45. What does the peak mean?

**A.** The greatest height reached is 45 m

**B.** The ball rises 45 m per second

**C.** The ball lands at 45 m

**D.** The ball is in the air for 45 s

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • BACKWARD 1 • Q1/5

1575

QUESTION ID
**S43-B1-Q1**

TYPE
Multiple choice

CORRECT ANSWER
**A. The greatest height reached is 45 m**

**GRADING NOTE** The maximum output of a height graph is the greatest height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • BACKWARD 1 • Q1/5

A graph of harbor depth d (ft) against time t (h) is lowest at the point (6, 1). Give the least depth and the time it occurs.

Least depth (ft): _____

Time (h): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • BACKWARD 1 • Q2/5

1576

**QUESTION ID**

## S43-B1-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**1; 6**

**GRADING NOTE** Form required: both. The point (t, d) = (6, 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • BACKWARD 1 • Q2/5

**True or false?**

The minimum value of a graph is its lowest y-value.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • BACKWARD 1 • Q3/5

1577

QUESTION ID

**S43-B1-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • BACKWARD 1 • Q3/5

A relative maximum on a graph is

**A.** the last point

**B.** any x-intercept

**C.** a point higher than nearby points

**D.** the y-intercept

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • BACKWARD 1 • Q4/5

1578

**QUESTION ID**

## S43-B1-Q4

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. a point higher than nearby points**

**GRADING NOTE** Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • BACKWARD 1 • Q4/5

A temperature graph rises on $0 < t < 2$. What does this mean?

**A.** The temperature stays constant

**B.** The temperature is above 2 degrees

**C.** The temperature is increasing during that time

**D.** The temperature is decreasing during that time

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • BACKWARD 1 • Q5/5

1579

**QUESTION ID**

**S43-B1-Q5**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**C. The temperature is increasing during that time**

**GRADING NOTE** A rising graph means increasing outputs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • BACKWARD 1 • Q5/5

SET 43 • HSF-IF.B.4

# BACKWARD 2

Read values from a table • Grade 6 • Expressions & Equations • 6.EE.C.9

**ANSWER KEY** Reduced copy of the question page, with its answer

Harbor depths d (ft) at t = 0, 2, 4, 6, 8, 10, 12 hours are 9, 7, 3, 1, 3, 7, 9. What is d when t = 4?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 43 • BACKWARD 2 • Q1/5

1581

QUESTION ID

**S43-B2-Q1**

TYPE

Short answer

CORRECT ANSWER

**3 ft**

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 43 • BACKWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Harbor depths d (ft) at t = 0, 2, 4, 6, 8, 10, 12 hours are 9, 7, 3, 1, 3, 7, 9. At which listed times is d = 9?

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 43 • BACKWARD 2 • Q2/5

1582

**QUESTION ID**

## S43-B2-Q2

**TYPE**

Short answer

**CORRECT ANSWER**

**0 or 12**

**GRADING NOTE** Form required: both, in any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 43 • BACKWARD 2 • Q2/5

**True or false?**

A repeated pattern in a table suggests a periodic relationship.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 43 • BACKWARD 2 • Q3/5

1583

QUESTION ID

**S43-B2-Q3**

TYPE

True or false

CORRECT ANSWER

**A. True**

GRADING NOTE Same values recur.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 43 • BACKWARD 2 • Q3/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Hours of daylight in months m = 0, 3, 6, 12 are 9, 12, 15, 9. Daylight of 15 h occurs at

**A.** m = 12

**B.** m = 6

**C.** m = 0

**D.** m = 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 43 • BACKWARD 2 • Q4/5

1584

QUESTION ID

**S43-B2-Q4**

TYPE

Multiple choice

CORRECT ANSWER

**B. m = 6**

GRADING NOTE Read.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 43 • BACKWARD 2 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Harbor depth is 7 ft at t = 2 h and 3 ft at t = 4 h. By how much does the depth d change from t = 2 to t = 4?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 43 • BACKWARD 2 • Q5/5

1585

QUESTION ID

**S43-B2-Q5**

TYPE

Short answer

CORRECT ANSWER

**−4 ft**

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 43 • BACKWARD 2 • Q5/5

SET 43 • HSF-IF.B.4

# BACKWARD 3

Amplitude, period, and midline of a sinusoidal model • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

**ANSWER KEY** Reduced copy of the question page, with its answer

Give the amplitude and the midline of
$T = 15 - 8\cos(\pi t/12)$.

Amplitude: _____

Midline: y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • BACKWARD 3 • Q1/5

1587

QUESTION ID
**S43-B3-Q1**

TYPE
Short answer

CORRECT ANSWER
**8; 15**

GRADING NOTE Form required: both. The amplitude is $|-8|$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • BACKWARD 3 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Give the period of $p = 3\sin(2\pi t)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • BACKWARD 3 • Q2/5

1588

**QUESTION ID**

**S43-B3-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**1**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • BACKWARD 3 • Q2/5

**True or false?**

A wave between 80 and 120 has amplitude 40.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • BACKWARD 3 • Q3/5

1589

QUESTION ID

**S43-B3-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** Amplitude is half the difference: (120 − 80)/2 = 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • BACKWARD 3 • Q3/5

A wave with maximum 9 and minimum 1 has midline

**A.** y = 8

**B.** y = 5

**C.** y = 10

**D.** y = 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • BACKWARD 3 • Q4/5

1590

**QUESTION ID**

**S43-B3-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. y = 5**

**GRADING NOTE** (9 + 1)/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • BACKWARD 3 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

A period of 0.8 s means how many cycles per minute?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • BACKWARD 3 • Q5/5

1591

QUESTION ID

**S43-B3-Q5**

TYPE

Short answer

CORRECT ANSWER

**75**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B3-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • BACKWARD 3 • Q5/5

SET 43 • HSF-IF.B.4

# FORWARD 1

Predict with periodicity • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4

**ANSWER KEY** Reduced copy of the question page, with its answer

Use the temperature model $T(t) = 15 - 8\cos(\pi t/12)$ to find T at t = 100 hours.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • FORWARD 1 • Q1/5

1593

**QUESTION ID**

**S43-F1-Q1**

**TYPE**

Short answer

**CORRECT ANSWER**

**11°C**

**GRADING NOTE** Any equivalent form. $100 = 4 \cdot 24 + 4$; $T(4) = 15 - 8\cos(\pi/3) = 11$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • FORWARD 1 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Harbor depths repeat every 12 hours, and the depth is 1 ft at t = 6 h. Predict the depth at t = 30 hours.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • FORWARD 1 • Q2/5

1594

**QUESTION ID**

**S43-F1-Q2**

**TYPE**

Short answer

**CORRECT ANSWER**

**1 ft**

**GRADING NOTE** Any equivalent form. 30 = 24 + 6.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S43-F1-Q2**

**High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • FORWARD 1 • Q2/5**

**True or false?**

If f has period 12, then f(50) = f(10).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • FORWARD 1 • Q3/5

1595

QUESTION ID

**S43-F1-Q3**

TYPE

True or false

CORRECT ANSWER

**B. False**

**GRADING NOTE** $50 = 4 \cdot 12 + 2$, so f(50) = f(2); f(10) need not equal it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • FORWARD 1 • Q3/5

A piston's position is p = 3sin(2πt) cm, with period 1 s. At t = 7.25 s, p is

**A.** 0

**B.** 3

**C.** −3

**D.** 1.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • FORWARD 1 • Q4/5

1596

**QUESTION ID**

**S43-F1-Q4**

**TYPE**

Multiple choice

**CORRECT ANSWER**

**B. 3**

**GRADING NOTE** Same as t = 0.25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • FORWARD 1 • Q4/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Daylight follows a 12-month cycle with a maximum of 15 h at m = 6 (m = 0 is January). In which month after m = 12 does daylight next reach 15 h?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • FORWARD 1 • Q5/5

1597

QUESTION ID

**S43-F1-Q5**

TYPE

Short answer

CORRECT ANSWER

**m = 18**

**GRADING NOTE** Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • FORWARD 1 • Q5/5

SET 43 • HSF-IF.B.4

# FORWARD 2

Write a model from periodic features • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

Harbor depth varies between 9 ft (at t = 0) and 1 ft (at t = 6 h) and repeats every 12 hours. Write the depth in the form d = D + A·cos(Bt) with A > 0 and B > 0. Give B exactly, in terms of π.

A = _____

B = _____

D = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • FORWARD 2 • Q1/5

1599

QUESTION ID

**S43-F2-Q1**

TYPE

Short answer

CORRECT ANSWER

**4; π/6; 5**

**GRADING NOTE** Form required: all three. Midline 5, amplitude 4, B = 2π/12; the maximum is at t = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • FORWARD 2 • Q1/5

**ANSWER KEY** Reduced copy of the question page, with its answer

Daylight varies between 9 h (month m = 0) and 15 h (m = 6) and repeats every 12 months. Write daylight in the form k − A·cos(Bm) with A > 0 and B > 0. Give B exactly, in terms of π.

A = _____

B = _____

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • FORWARD 2 • Q2/5

1600

QUESTION ID

**S43-F2-Q2**

TYPE

Short answer

CORRECT ANSWER

**3; π/6; 12**

**GRADING NOTE** Form required: all three. Midline 12, amplitude 3, B = 2π/12; the minus sign puts the minimum at m = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • FORWARD 2 • Q2/5

1600